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From Vision to Value:

The development of a Conceptual Smart Checkout Solution

PAPER WITHIN *Product development*
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The authors take full responsibility for opinions, conclusions and findings presented.

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Abstract

The retail sector is facing operational changes, driven by rising labor costs, staff shortages, and high employee turnover rates. While traditional self-checkout (SCO) systems have been widely implemented to maintain high service availability and reduce customer waiting times, they introduce certain barriers, including perceived system complexity, frequent scanning errors, and technical friction. Computer-vision (CV) technology and Machine Learning (ML) offer an alternative through barcode-less product recognition, commonly known as smart checkouts. However, a defined research gap remains regarding how to design and scale a vision-based product recognition SCO within the Swedish convenience retail landscape, where retailer specific preferences, assortments complexities, and regulatory frameworks apply differently than internationally.

This Master's thesis, conducted in collaboration with ITAB Shop Products AB, aimed to identify the primary challenges within self-checkout customer journeys focusing on less than ten item purchases. Furthermore, the study researched a possible retail environment for a Minimum Viable Product (MVP) deployment on the Swedish market. Moreover, it explored how a vision-based checkout solution can be physically designed to be user friendly and fit retailer needs.

Following the Double Diamond design framework, a qualitative and empirical methodology was employed. Data collection involved thirty-nine field observations and informal customer interactions across high-density retail locations in Stockholm, Gothenburg, and Jönköping. This was supplemented by semi-structured interviews with retail stakeholders, and a CV and ML engineer. The qualitative data was processed using an inductive thematic analysis to translate field visit observations and interactions into actionable design problems.

The thematic analysis points out that poor visibility, cluttered SCO layouts and perceived SCO complexity can directly contribute to checkout queues and SCO underutilization. Staff shortages and high staff turnover rates were two large themes, rising from interviews and staff interactions. Based on observed product assortment, operative environments, convenience retail formats, and current SCO adoption, cinema retail emerged as the most viable environment for potentially developing an MVP.

The final design deliverable presents a robust, elegant, and minimalistic product architecture featuring a stationary arch structure over an item placement area. The arch integrates a downward-facing item illumination bar and plexiglass to cover three cameras. Two of them are angled at 40 degrees toward the center of the placement area, and the third camera is placed in the center of the arch, angled vertically downward. Furthermore, the concept incorporates an interaction light to signal checkout availability, an ongoing transaction, or a staff assistance requirement. To guide system interaction, the design presents a screen placed on the side of the placement area to keep the area clear and ensure accessibility. The proposed design also has an integrated backup barcode scanner in case the vision system fails to recognize unlearned or hidden items. This combined system is designed to enable reduced checkout times and minimized queue friction, especially during peak hours in time-sensitive environments.

Keywords: Product Development, Industrial Design, Smart Checkout, Computer Vision, Self-Checkout, User Experience, Inclusive Design, Convenience Retail.

Sammanfattning

Detaljhandeln står inför operativa förändringar till följd av stigande arbetskraftskostnader, personalbrist och hög personalomsättning. Även om traditionella självutcheckningssystem (SCO) har implementerats i stor utsträckning för att bibehålla hög servicetillgänglighet och minska kundernas väntetider, medför de vissa barriärer, däribland upplevd systemkomplexitet, frekventa skanningsfel och teknisk friktion. Computer-vision-teknik (CV) och maskininlärning (ML) erbjuder ett alternativ genom streckkodslös produktidentifiering, i folkmun kallat smarta utcheckningar. Det definierades ett forskningsglapp gällande hur man kan designa och potentiellt skala en kamerabaserad SCO med produktigenkänning inom den svenska bekvämlighetshandeln, där återförsäljarspecifika preferenser, komplexa sortiment och regelverk tillämpas annorlunda än internationellt.

Detta masterexamensarbete, som utförts i samarbete med ITAB Shop Products AB, syftade till att identifiera de primära utmaningarna inom självutcheckningens kundresor med fokus på köp av färre än tio artiklar. Vidare undersökte studien en möjligtvis lämplig handelsmiljö för implementering av en Minimum Viable Product (MVP) på den svenska marknaden. Dessutom utforskades hur en visionsbaserad utcheckningslösning kan utformas fysiskt för att vara användarvänlig och passa återförsäljarnas behov.

Genom att följa designramverket Double Diamond tillämpades en kvalitativ och empirisk metodik. Datainsamlingen omfattade trettionio fältobservationer och informella kundinteraktioner på högfrekventerade platser inom bekvämlighets handeln i Stockholm, Göteborg och Jönköping. Detta kompletterades med semistrukturerade intervjuer med intressenter inom detaljhandeln och en ingenjör inom CV och ML. De kvalitativa data bearbetades med en induktiv tematisk analys för att omsätta observationer och interaktioner från fältbesöken till hanterbara designproblem.

Den tematiska analysen visar att bristande sikt, röriga SCO layouts och upplevd SCO-komplexitet direkt kan bidra till utcheckningsköer och att självutcheckningen underutnyttjas. Personalbrist och hög personalomsättning var två stora teman som framkom under intervjuer och personalinteraktioner. Baserat på observerat varusortiment, operativa miljöer, convenience-retailformat och nuvarande SCO-användning, framstod biografer som den mest lämpliga miljön för att designa en potentiell MVP.

Den slutgiltiga designen presenterar en robust, elegant och minimalistisk produktarkitektur med en stationär bågstruktur över ett varuplaceringsområde. Bågen integrerar en nedåtriktad belysningsramp för varor samt plexiglas för att täcka tre kameror. Två av dem är vinklade i 40 grader mot varuplaceringsplanet, och den tredje kameran är placerad i mitten av bågen, vinklad vertikalt nedåt. Vidare innehåller konceptet ett interaktionsljus för att signalera tillgänglig utcheckning, pågående transaktion eller behov av personalassistans. För att vägleda systeminteraktionen presenterar designen en skärm placerad på sidan av varuplaceringsområdet för att hålla ytan fri och säkerställa tillgänglighet. Den föreslagna designen har även en integrerad reservstreckkodsläsare i händelse av att visionssystemet misslyckas med att identifiera oinlärda eller dolda varor. Detta kombinerade system är utformat för att möjliggöra förkortade utcheckningstider och minimerad köfriktion, särskilt under rusningstid i tidskritiska miljöer.

Nyckelord: Produktutveckling, Industridesign, Smart utcheckning, Datorseende, Självutcheckning, Användarupplevelse, Inkluderande design, Bekvämlighetsjhandel.

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Table of Contents

1. Background 1
1.1 Retail Industry and self-checkouts 1
1.2 Vision-based product recognition 1
1.3 ITAB background and problem definition 3
1.4 Objective and Goals 4
1.5 Research questions 5
1.6 Delimitations 5
2. Theoretical framework 6
2.1 Value based outcomes 6
2.2 User experience design 6
2.3 Inclusive design 6
2.4 Product architecture 7
3. Methods and Data Collection 8
3.1 Double Diamond 8
3.2 Field visits 8
3.3 Interviews 9
3.3.1 In-person Interactions 9
3.4 Try it yourself 10
3.5 Thematical analysis 10
3.6 How might we 10
3.7 Competitor product analysis 11
3.8 Mock-ups 11
3.9 Concept testing 11
3.10 Appearance model 11
3.11 User scenarios 11
4. Implementation and result 12
4.1 Problem discovery 12
4.1.1 Field visits 12
4.1.2 In-person interactions 12

4.1.3 Interviews 13
4.1.4 Try it yourself 13
4.1.5 Data management 19
4.1.6 Thematic analysis 19
4.1.7 Main themes 20
4.1.1 Competitor product analysis 33
4.1.2 ART-series brand analysis 39
4.2 PROBLEM DEFINITION 42
4.2.1 Problem focus 42
4.2.2 Cinema store context 43
4.3 SOLUTION DISCOVERY 44
4.3.1 Mood board 44
4.3.2 How might we 45
4.3.3 Concept generation 48
4.3.4 Sketches 48
4.3.5 Mockups 50
4.3.6 Concept testing 59
4.4 CONCEPT SELECTION 61
4.4.1 Appearance model 63
4.5 SOLUTION DELIVERY 64
4.5.1 Appearance model 64
4.5.1 Renders 68
4.5.2 User interface 70
4.6 USER SCENARIOS 75
5. Discussion & conclusion 79
5.1.1 First research question 79
5.1.2 Second research question 80
5.1.3 Third research question 80
5.1.4 ITAB's value proposition 82
5.1.5 Future research 83
6. References 84
7. Annexes 86

List of Figures

Figure 1: Shows a smart checkout from Mashgin. 2
Figure 2: Itab MoveFlow Modular S SCO 3
Figure 3: EasyFlow vision-based checkout system 4
Figure 4: Three perspectives stated in ITAB methodology for a value-based outcome. 6
Figure 5: Obstructed side (a) and forward (b) reach limitations 7
Figure 6: ITAB product development process and triple Diamond framework 8
Figure 7: Mashgin smart checkout starting frames (A: item placement, B: item separation) 14
Figure 8: Mashgin UI during item placement (C: item placement, D: system processing) 14
Figure 9: Mashgin UI menu selection (E) and final itemized receipt (F) 15
Figure 10: User interface instructions following payment initiation. 16
Figure 11: Item placement array (A) and manual barcode scanning prompt (B) 17
Figure 12: Shows some of the observed measurements of a Mashgin smart checkout. 18
Figure 13: Swedish-to-English interview translation using Google Gemini 19
Figure 14: Initial data clustering from Stockholm and Gothenburg field visits 20
Figure 15: Shows the first part of the thematical analysis. 20
Figure 16: Shows the second part of the thematical analysis 20
Figure 17: Lunchtime field observation at a Jönköping Ryhov convenience store 21
Figure 18: Thematic data cluster for time-sensitive environments 22
Figure 19: SCO configurations at a Stockholm transit station and Asecs in Jönköping 24
Figure 20: Customer queue behavior and SCO underutilization at Nordstan 24
Figure 21: Clustering of problems related to unclear SCO placements. 25
Figure 22: Field photograph of the self-checkout layout depicted in the floor plan 26
Figure 23: Simplified store floor plan showing layout modifications in SCO placement 26
Figure 24: Comparison of a customer access format and a staff mediated service model 27
Figure 25: Customer queuing and density at a sports arena bar in Gothenburg. 27
Figure 26: Shows an observed UI at a SCO at a Stockholm gas station near Avicii Arena 29
Figure 27: Size comparison of an NCR SCO setup and Mashgin's smart checkout. 32
Figure 28: Shows a cluster of theft and Fraud 32
Figure 29: Product architecture comparison of Mashgin and Supersmart's SCO systems 33
Figure 30: Product architecture comparison of Autocanteen and Deligo's SCO systems 34
Figure 31: Product architecture comparison of Proppos and Pan-Oston's SCO systems 35
Figure 32: Shows Mashgin's new solution called Auto-magic checkout. 36
Figure 33: Observed lobby's within Swedish cinema stores in Jönköping and Gothenburg 37
Figure 34: Shows an integration of ART series design elements in Sigma gate 2 39
Figure 35: Shows a round edge between a surface transition 40
Figure 36: Shows ventilation holes for a computer behind a screen 40
Figure 37: Illustrates a pattern used to arrange hanging shelves 41
Figure 38: Shows a modular front panel of ITAB's checkout CM700 41
Figure 39: Problem focus matrix categorized by four levels of importance 43
Figure 40: shows some of the largest items observed at a cinema store in Jönköping. 44
Figure 41: Mood board illustrating the inspiration for the intended design direction. 45
Figure 42: Shows the brainstorming from the first "how might we" question. 45
Figure 43: Shows the brainstorming from the second "how might we" question. 47
Figure 44: Shows the brainstorming from the third "how might we" question. 48
Figure 45: Shows the initial ideation sketching. 49
Figure 46: Shows the camera used as reference for the design. 50
Figure 47: Shows the first setups of the mockups. 51
Figure 48: Concept mockup 1 52
Figure 49: Concept mockup 2 53
Figure 50: Concept mockup 3 54
Figure 51: Concept mockup 4 55
Figure 52: Concept mockup 5 56
Figure 53: Wireframe sequence for Scenario 1. 57
Figure 54: Wireframe sequence for Scenario 2. 58
Figure 55: Shows photos from user evaluations. 59
Figure 56: Citations from participants during user interaction. 60
Figure 57: Item placement area as captured through a 0.6 fisheye camera lens 61
Figure 58: Shows the form development inside the software Autodesk Alias. 62

Figure 59: Illustrates the individual parts for the appearance model. 63
Figure 60: Illustrates the assembly process of the appearance model. 64
Figure 61: Shows a hero shot of the assembled appearance model 65
Figure 62 : Shows a closeup of the camera and illumination bar 65
Figure 63: Shows a closeup of a rubber detail 66
Figure 64: Shows a closeup of the placement area. 66
Figure 65: Shows the green interaction light. 66
Figure 66: Shows the yellow interaction light. 67
Figure 67: Shows the red interaction light. 68
Figure 68: Design variation illustrating modular scaling of the arch and placement surface 69
Figure 69: Close-up of the illumination bar 69
Figure 70: Shows a hero shot of the final concept 70
Figure 71: First segment of the UI sequence for user Scenario 1. 71
Figure 72: Second segment of the UI sequence for user Scenario 1 72
Figure 73: First segment of the UI sequence for user Scenario 2 73
Figure 74: Second segment of the UI sequence for user Scenario 2 74
Figure 75: last segment of the UI sequence for user Scenario 2 75
Figure 76: Shows the first scenario about a family going to the cinema 76
Figure 77: Shows the first part of the second scenario 77
Figure 78: Shows the second part of the second scenario 78

Use of AI

In this paper, AI has been used for improving the workflow and supporting the quality of the work. Its specific applications and purposes in this study are detailed below:

Literature reviewing

NotebookLM is an AI research assistant that was utilized to review literature by summarizing and analyzing the reference materials. By generating concise overviews of the literature, the tool helped to provide a rapid understanding of core content and enabled extraction of specific source information. NotebookLM uses only information from the uploaded sources, preventing it from using external information from the internet.

Improving sentence structure and spell checking

The study used Gemini, an AI-powered assistant and multimodal model family, to improve sentence structure. Original sentences were inputted into the AI to generate clearer alternatives, which were then evaluated to ensure the core meaning and most important information remained intact. The refinement process was iterative as prompts were adjusted progressively until the AI generated the desired information. Spellchecking and grammar corrections were also conducted using Gemini. The AI analyzed the provided text for errors and generated revised paragraphs with the appropriate corrections. Figure captions were shortened and improved using google Gemini.

Interview and interaction transcriptions

Klang.ai was used to automatically transcribe and summarize the recorded interviews and meetings. The Swedish-based platform is GDPR-compliant, ensuring that all data remained private and secure. Because the audio was recorded in Swedish, Klang.ai was chosen for its strong understanding of the language. After the AI generated the transcripts, they were manually reviewed and edited to correct any misidentified technical terms.

Image generation

To improve the visual content of the report's figures and tables, Gemini was used to generate custom images and symbols. The AI was guided by text prompts and, in some cases, reference images to ensure accurate results. Utilizing Gemini allowed for rapid asset creation while ensuring that all generated visuals were royalty-free.

Improving rendered images

Weavy.ai is a node-based AI design platform that was used to enhance 3D render quality of the final concept by locking CAD geometry and applying hyper-realistic materials, studio environment lighting. Moreover, it was used for high-fidelity upscaling to enhance pixel quality to the images.

Abbreviations

ART – Architectural Retail Technology

CAD – Computer Aided Design

CV - Computer Vision

EAA – European Accessibility Act

EFT - Electronic Funds Transfer

GDP - Gross Domestic Product

ICT - Information and Communication Technology

ML – Machine Learning

MVP – Minimal Viable Product

OCR - Optical Character Recognition

POS – Point of sales

RT – Retail Technology

SCO – Self-checkout

UX - User Experience

1. Background

This section covers fundamental concepts of the retail industry, with a specific focus on convenience stores and the role of self-checkout (SCO) systems. Moreover, it covers basics of vision-based product recognition as the technology is central in the product development for this project. Additionally, it provides an overview of the partnership with ITAB shop products AB, by covering their profile and certain previous check-out solutions to highlight key insights.

1.1 Retail Industry and self-checkouts

The retail sector is one of the most influential forces in the global economy; in the U.S. alone, it serves as a leading employer and a critical driver of the nation's Gross Domestic Product (GDP). In both global and local markets, retailers need to analyze and redesign their operations and marketing strategies to improve and strive for quality shopping experiences (Pettersson, 2026). Many shops today offer self-service options, such as self-scanning and SCOs, to supplement regular manned checkouts and increase average purchase amounts. Implementing SCOs helps retailers maintain high service levels despite rising labor costs and staff shortages. Instead of replacing workers, this technology transitions cashiers into dynamic roles like tech support, customer experience management, or SCO station monitoring (Softpay.io, 2025). Retailers face a number of challenges when it comes to staff retention, training and operational cost. These problems and the fact that consumers today expect high service availability drives implementation of self-service solutions in the retail sector (E.Cannert, Personal communication, 12/1/2026)

According to the payment report from (Sverigesriksbank, 2025), slightly more than 50% of all consumers prefers to pay at SCOs or use self-scanning when shopping in a regular grocery store. The benefits of SCOs are usually shorter waiting times in line to the manned checkout and increased convenience sales. Moreover, it provides customers with greater control over their transactions and accuracy in scanning their items correctly, thereby, enhancing customer privacy (Finalpos, 2025). According to (Softpay.io, 2025), the main motives for using SCOs are time savings, control and autonomy and practical convenience. However, they mention the main barriers preventing their usage from being used as much as retailers want. According to a study regarding by (Softpay.io, 2025), the primary reasons are perceived complexity (40%), a fear of making mistakes among (25%), perceived social isolation (48%), and technical problems (30%) based on the source data.

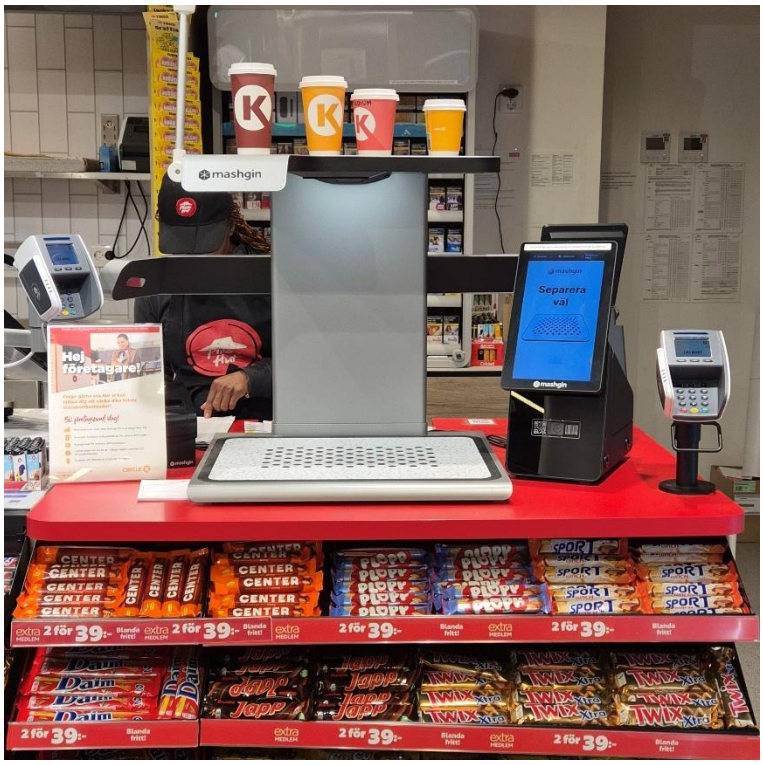
An increasing number of retailers are considering abandoning SCOs due to rising theft rates, particularly involving high-value items such as meat products (Barcoder, 2024). In fact, according to data from (Svenskhandel, 2024), nearly 70% of all shoplifting incidents in grocery stores occurs at self-scanning or SCO kiosks. One reason for this trend might be the lack of human oversight. When a physical cashier is not present, the perceived risk of being caught decreases, making it easier for individuals to commit theft. Although retail crime costs the industry an estimated 10.4 billion SEK annually, further measures are required to enhance security and mitigate the risks of shoplifting, especially at Self-service checkouts.

1.2 Vision-based product recognition

Automated product recognition is driven by computer vision (CV) systems and Machine Learning (ML) that translate raw camera data into actionable classifications and detections. In recent years, a wide range of technologies has developed to enhance retail automation, spanning from barcode scanning to other barcode-less recognition systems. Vision-based product recognition is a technology that uses CV to analyze visual features like shape, color, and texture to identify specific products. Some systems supplement CV with Optical Character Recognition (OCR) to extract textual descriptors, such as brand names or product labels, to accurately identify and categorize items (Pettersson, 2026). With the technical advancements

of the past decade, opportunities are growing to utilize CV, ML and OCR-text in checkouts with the aim of reducing friction in customer journeys. One such product is smart checkouts, which allow customers to simply place items under a camera system that automatically recognizes them, in the ideal case, eliminating the need for traditional barcode scanning. Several companies have developed smart checkouts, such as Mashgin, FUNPAD, Toshiba, Autocanteen, and Pan Oston. Mashgin has deployed its solutions in a variety of stores on the global market, including convenience stores, cafeterias, airport vendors, and sports arenas. See an observed smart check-out from Mashgin in Figure 1.

Figure 1: Shows a smart checkout from Mashgin.



Comment. The photo was taken at Circle K, Gårdatorget in Gothenburg the 10/2/2026.

A case study at Denver's Mile High Stadium showed that Mashgin's smart checkouts doubled overall throughput speed and boosted concession sales by 34%. Additionally, the median transaction time was reduced to 15 seconds compared to their previous checkout option (insight.tech, 2023). The stadium transitioned from traditional, staff-operated counters to automated smart checkouts (Kapustka, 2023). Citing Toby Awalt, Vice President of Marketing for Mashgin from their case study "Our machines are two to four times faster than the typical cashier" (insight.tech, 2023).

At Candler Hospital, installing Mashgin's smart checkouts drove a 110% increase in sales compared to pre-pandemic peaks. The hospital has around 3,000 employees and is open 24/7. Because the shops were previously closed at night due to a lack of 24/7 staffing, Mashgin unlocked new sales and reduced transaction times down to 11 seconds. By eliminating barcode scanning, their smart checkouts can operate up to 400 % faster than conventional checkouts and 800% faster than conventional self-checkouts (Mashgin, 2025).

In a smaller convenience store context, such as the Circle K chain, Mashgin's smart checkout helps to mitigate the impact of labor turnover. The automation of the checkout process allow staff members to move away from the counter and focus on other-value tasks, including customer service, shelf stocking, and store maintenance (insight.tech, 2023).

Notably, Circle K is yet the single store chain on the Swedish market utilizing smart checkouts. However, the rapid emergence of new competitors alongside Mashgin signals a clearly expanding market for smart checkouts. Additionally, it is possible that more companies than

Circle K on the Swedish market implement these kinds of products in the future. Consequently, there is a research gap regarding why other companies have not yet implemented smart checkouts in the Swedish market. This thesis aims to investigate the reasons behind this, while also identifying and addressing pain points in a new smart checkout product concept.

1.3 ITAB background and problem definition

At ITAB, they help customers turn consumer brand experience into physical reality with their know-how, solutions and ecosystem of partners. By analyzing the way consumer expectations and buying patterns are changing, they are able to support physical stores with knowledge of how to satisfy these needs. The company's value propositions are to create the desired consumer brand experience, increase conversion & sales, improve operational efficiency & services, and reduce operational cost. Based on addressing the massive shifts currently taking place within the retail industry, they focus on solving retailers' most significant challenges (ITAB, 2026).

ITAB offers retail solutions and consulting across four main areas: Interior, Retail Technology (RT), Lighting & Sound, and Services. RT offers a global portfolio with a strong focus on the next generation Entrance & Checkout Arena solutions. Examples from the portfolio are conventional checkouts, self-checkouts, loss prevention solutions and a digital platform that supports data aggregation, analytics and orchestration. RT use design principles that support development of user centric & value-based solutions. Moreover, the principles act as a framework that ensures a strong focus on a holistic value proposition and deep contextual understanding of the retail environment (E.Cannert, Personal communication April 20, 2026).

Within the RT checkout segment, they have a SCO family called MoveFlow, which uses barcode scanning for item registration. It can serve both as main self-checkout in a convenience store or as complementary express alternative in a supermarket (ITAB, 2026). See an illustration of MoveFlow in Figure 2.

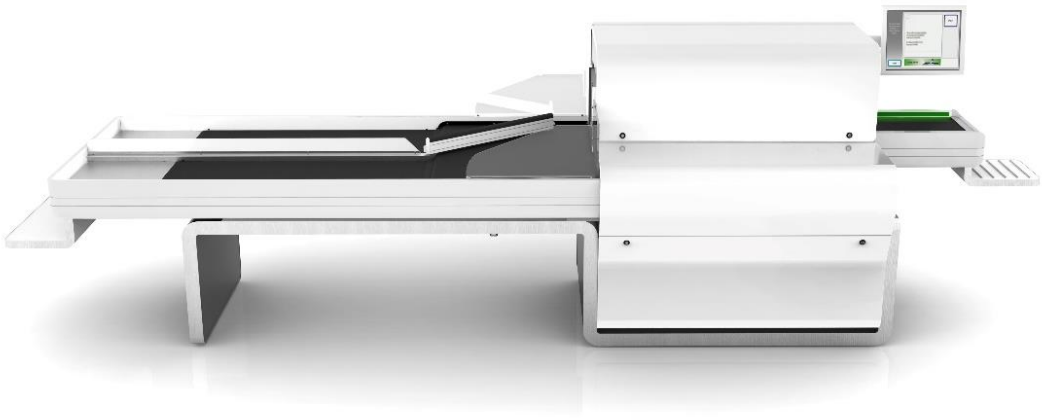
Figure 2: Itab MoveFlow Modular S SCO



Comment. Retrieved from "MoveFlow Modular S" at itab.com (https://itab.com/products-and-services/products/retail-technology/self-checkouts/moveflow-modular-s) Used with permission.

Previous developments at ITAB RT used CV and ML in combination with barcode scanning for a solution called EasyFlow. It was developed in 2011 and continued until 2017 with the aim to reduce checkout times and increase the overall customer throughput in grocery retail environments. The system differed significantly from modern smart checkouts, as it aimed to mirror a traditional checkout flow. By replacing manual staff-scanning with an automated registration tunnel, the system allowed customers to place items on a conveyor belt that fed into the unit, where products were registered automatically before emerging on the other side. See an illustration of EasyFlow in Figure 3.

Figure 3: EasyFlow vision-based checkout system



EasyFlow's was used in a large retail chain during regular operation, enabling data collection of product recognition accuracy and real time performance. It had high accuracy in product recognition, however, the system was semi-automated, meaning that it required a store attendant to register products manually, especially for bakery bread and weight based fresh produce. The system faced challenges due to constantly evolving assortments and the difficulty of updating product variations before they reached the checkout. Furthermore, environmental complexities and the demanding nature of the retail landscape necessitated frequent staff interventions, creating excessive friction that hindered the product's market appeal. Moreover, factors like the need for extensive data collection and a complex, expensive sensor setup eventually hindered its widespread launch on the market (Pettersson, 2026).

While ITAB has developed further solutions using CV and ML, they have recognized that the complexity of implementing this technology in large-scale retail environments is substantial. Consequently, the company are looking more broadly for other opportunities to use vision technology. One of the possible implementation area is a smart checkout for convenience retail. Utilizing the expertise gained from previous projects, this thesis provides an opportunity to conduct research and propose a new smart checkout concept, focusing on its environment, functionality, and target users.

1.4 Objective and Goals

The objective was to address the main challenges in customer journeys focusing on "less than 10 Item purchases", as convenience shopping typically consists of small baskets. Furthermore, the project aims to identify a specific store type suitable for developing a Minimum Viable Product (MVP) on the Swedish market using vision-based product recognition. Adopting an MVP approach allows for a strategic focus on core value propositions, rather than attempting a high-complexity solution without clearly defined technical limits. This strategy represents a practical approach, prioritizing feasibility and value creation.

Moreover, the goal was to clearly describe the roles, jobs and interaction points of customers, store employees and others when it comes to purchase, deployment, support and training of a new product concept. Clearly state how the concept solves a problem that is valuable to the retailer and beneficial for the consumer. Ultimately, the goal was to create a prototype that incorporates solutions to the main challenges and show how the concept provides value-based outcomes. To address these objectives, three research questions were formulated, written in the next sub chapter.

1.5 Research questions

RQ 1: What are the main challenges of a self-checkout customer journey that supports a less than 10 items purchase?

RQ 2: What store type could be more suitable for developing an MVP smart checkout?

RQ 3: How can a vision-based SCO be designed to support user problems and retailer needs?

1.6 Delimitations

Previous studies and large-scale product developments at ITAB have demonstrated that achieving consistent product recognition accuracy in expansive grocery assortment remains a significant challenge. Consequently, large-scale grocery stores are excluded from the scope of this MVP development. Proven technical challenges in these environments have shown them to be unsuitable for initial implementation; therefore, this thesis focuses on smaller-scale convenience stores.

To provide a multidisciplinary approach, this thesis is supported by an extended project team from ITAB. These professionals are included to provide specialized expertise across various domains to better guide decisions throughout the project. The professions included are two Industrial designers (one with many years or prior experience), a User Experience (UX) Designer, an ML and CV engineer, and a Product manager. However, this thesis is conducted within the fields of product development and industrial design, and its primary focus will remain within these disciplines.

While vision-based systems involve numerous technical complexities, this project maintains a primary focus on user and retailer research to generate an product concept focusing on the industrial design. Specific technical details fall outside the project scope; however, due to the ML and CV engineer in the extended project team, certain expertise in these areas was provided. Key insights enabled technical feasibility to support design decisions.

While loss prevention represents a potential value layer for the concept, comprehensive research into fraud detection is excluded from this scope due to the vast complexity of the field. Despite acknowledging the importance of these problems, a comprehensive solution would necessitate a separate, large-scale project and extensive iterative design to be addressed properly.

From a technical standpoint, vision technology could facilitate automated age verification, further reducing friction for customers purchasing restricted items for a smart checkout concept. However, the Swedish market landscape is heavily shaped by Laws and Strict GDPR regulations, which are critical to consider if developing such solution, thereby, it is chosen to be excluded from the project scope. (T.Petterson, personal communication, February 20, 2026) & (E.Cannert, Personal communication Mars 6, 2026).

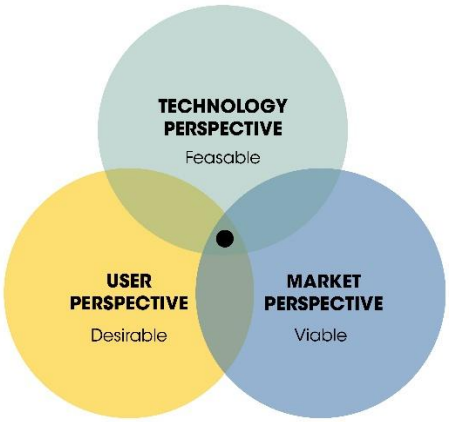
2. Theoretical framework

This section is about company methodology and theories that apply to this thesis.

2.1 Value based outcomes

From the ITAB methodology derives three perspectives that should be balanced with each other to enable a value-based outcome. The three critical perspectives are: User Desirability, Technical Feasibility and Market Viability. The user perspective is about understanding the consumers for their problems and needs. The technical perspective is about understanding the challenges and limitations to create value for the user problems. The market perspective evaluates the viability of the solution in terms of cost-efficiency and long-term profitability. See an illustration of how the perspective aims to be balanced for a value-based product solution. in **.**
Figure 4.

Figure 4: Three perspectives stated in ITAB methodology for a value-based outcome.



2.2 User experience design

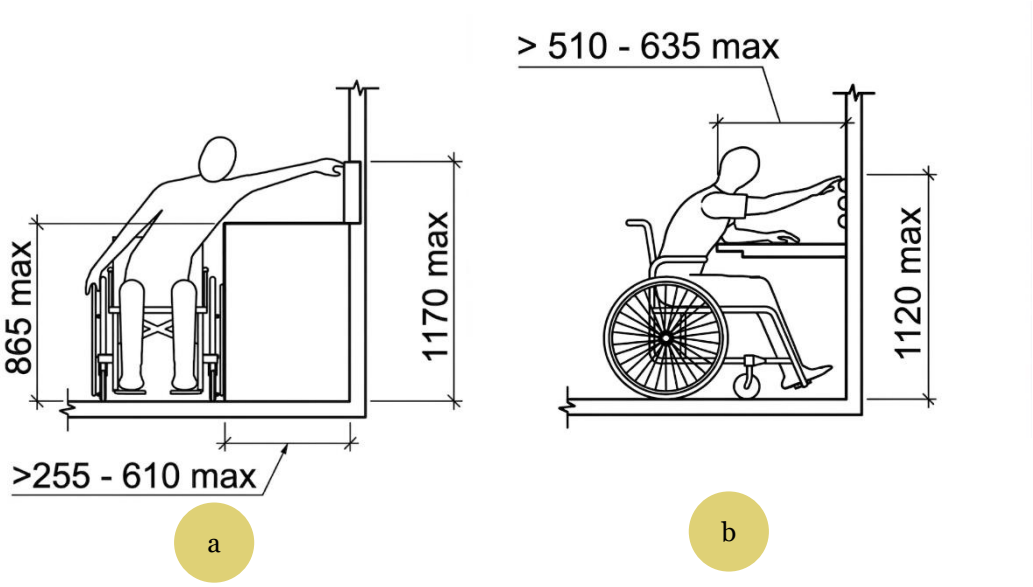
User Experience (UX) Design examines how end users perceive, interact with, and interpret a product or service. Its primary objective is to bridge the gap between developer assumptions and the empirical reality of user behavior, ensuring that solutions are grounded in actual user needs rather than internal preconceptions. A good UX can be defined through functionality, efficiency, and desirability. Functionality means that the product must do the things that users expect it to do to help them accomplish their goals. Efficiency regards the speed and ease of operations, helping people spend more time on things they enjoy and less on what they don't. Desirability addresses the user's emotional reaction, which is shaped by the product's visual and physical attributes, as well as the values conveyed through brand messaging and cultural context (Kuniavsky et al., 2012).

2.3 Inclusive design

Inclusive design can be defined as a methodology for creating products, services, and environments that accommodate the needs of the widest number of people possible without the need for specialized adaptation. This approach strives to relate the diverse capabilities of a population directly to the design process, ensuring that the physical or cognitive demands placed on a person by a design do not exceed their actual capability to respond. (Langdon et al., 2018). In this paper, inclusive design is considered by taking account for the accessibility requirements for fixed Information and Communication Technology (ICT) installations with built-in obstructions. This is relevant because if the product is designed with a table, that table becomes an obstruction. In Sweden, the requirements are guided by a standard that derives from the European Accessibility Act (EAA). The EAA is an EU directive for products and service regarding accessibility for persons with disabilities ((ETSI), 2021).

The EN 301 549 standard outlines two types of requirements for built-in obstructions: obstructed side reach and obstructed forward reach. For a side reach, the maximum allowed obstruction depth is 610 mm, the maximum height is 865 mm, and the maximum reach height to the interface is 1170 mm. For a forward reach, the maximum depth is 635 mm and the maximum reach height is 1120 mm. Following both sets of measurements would ensure a design that is accessible and compliant for both approaches. These requirements are illustrated in Figure 5.

Figure 5: Obstructed side (a) and forward (b) reach limitations



Comment:

Adapted from "*Accessibility requirements for ICT products and services*" (EN 301 549 V3.2.1), by ((ETSI), 2021) (https://www.etsi.org/deliver/etsi_en/301500_301599/301549/03.02.01_60/en_301549v030201p.pdf). Image generated using Google Gemini (Nano Banana 2) on April 29, 2026.

2.4 Product architecture

Understanding product architecture is essential for identifying a product's core technologies and working principles. Furthermore, a well-defined architecture is a primary lever for delivering product variety and managing complexity (Ulrich et al., 2019). To contextualize the design, a competitive benchmarking was done of an existing Vision based SCO product architecture was conducted. This facilitated an analysis of functional allocation, identifying where systems successfully streamlined the user experience and where architectural constraints introduced cognitive friction into the checkout journey. Primary research commenced with a field observation at a retail-integrated refueling station in Jönköping. This site-specific analysis provided a foundational understanding of the system's operational logic and established a baseline for the product architecture of a CV-based (SCO) kiosk. By documenting the physical interplay between the user and the current hardware, it became possible to identify the current functional components while highlighting opportunities for architectural differentiation in a new design.

3. Methods and Data Collection

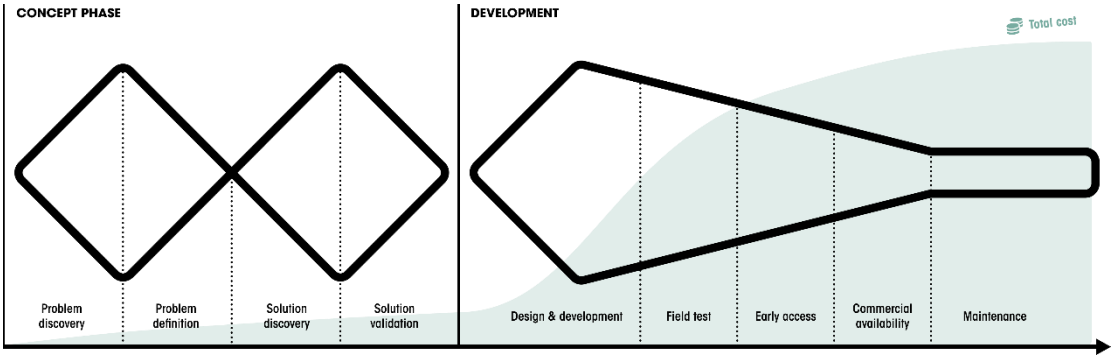
This section contains the methods chosen for the study, explains how they work and how they have been employed.

3.1 Double Diamond

The Double Diamond is a strategic design framework divided into four key phases: Discover, Define, Develop, and Deliver. The first diamond begins with Discovery, a divergent phase focused on researching problems, mapping systems, and identifying stakeholder needs. By gathering a broad range of insights, the project's scope expands to ensure a holistic understanding. In the Definition phase, the diamond converges; insights are synthesized into concepts that are tested via low-fidelity mockups. Through concept screening, the scope is narrowed to a single direction. The second diamond opens with Development, a divergent stage where concepts are combined and refined through detailed testing. Finally, the Delivery phase converges on the final solution, focusing on high-fidelity modeling, prototyping, and finalizing all project deliverables (Designcouncil, 2026).

In alignment with the Industrial Design process at ITAB Shop Products, this strategic framework served as the foundation for the project planning, as it is a core component of the company's methodology. Within ITAB's product development timeline, the Double Diamond represents the Concept Phase, which defines the scope of this project. While the framework is based on the original Design Council model, it has been tailored to reflect ITAB's specific internal workflows. Notably, the design and development phase is excluded from this project's scope, as it falls outside the concept phase. See the product development process in Figure 6.

Figure 6: ITAB product development process and triple Diamond framework



3.2 Field visits

Observations are a widely used data collection method for understanding processes and the relationships between individuals, objects and systems. It is often combined with other research methods, for instance, interviews or interactions, to increase reliability, nuance contextualization and broaden the analysis. Observations can vary in form, such as being direct or indirect, structured or unstructured, and are influenced by factors including time, location, and the presence of the observer (Säfsten et al., 2020). In this study, the observations were direct as the data was collected from people performing actions during field visits at convenience stores.

The purpose of the field visits were to analyse user behaviour within convenience retail environments. This included mapping customer journeys from the moment of entry to product selection, as well as observing reactions to the store environment and interactions with staff at manned, self-service or smart checkouts. Furthermore, the study examined how different product architectures impact user interaction, with the goal of identifying which touchpoints result in perceived friction.

Field visits typically involve multiple sites to identify recurring patterns. To ensure the reliability of these observations, researchers inquire about both typical and atypical events,

validating whether the observed behaviors represent common practices (Kuniavsky et al., 2012). Therefore, it was important to extend the field visits beyond just a few convenience stores in Jönköping, to see how user patterns reoccur or differ.

Locations were selected based on expected high traffic volume. While Jönköping provided access to some high-density convenience stores, the research was expanded to Stockholm and Gothenburg to capture more high-density convenience retail environments with significantly higher throughput demands.

3.3 Interviews

To capture the insights and experiences of various stakeholders, qualitative interviews are an effective approach for gaining a deep understanding of a subject. The data collection technique can be defined as a professional conversation with a specific purpose that usually follows a specific structure. The researcher should describe and understand conditions and the meaning of the conditions by the respondents. The interviews for this thesis involved semi-structured dialogues with stakeholder professionals, utilizing a predetermined set of open-ended questions to guide each conversation. See the interview guide for the interview with one of the interviews in Annex A and its interview summary in Annex B.

The interviews targeted professionals in different fields, including an expert in IT, a Sales Manager, a ML and CV engineer, and a Market Coordinator. The primary objective was to investigate parts of the market perspective and technology framework. Interviewing individuals from diverse professions provides a broader perspective for the study. However, this variety introduces potential bias, as the lack of comparable data from similar stakeholders necessitates a more nuanced and cautious analysis.

However, for the market perspective, the interviews explored whether these organizations have considered adopting smart-checkout technologies and, if so, the specific reasons why they ultimately chose not to implement them. The discussions also aimed to identify strategic priorities regarding employee value creation, operational efficiency, and customer experience. Finally, the study sought to gauge the industry's appetite for specific emerging technologies, providing insight into how these retail chains view innovation, revenue growth, and the future of convenience retail.

For the technology perspective, an internal ML and CV engineer at ITAB was interviewed. The goals were to build understanding of complexities with vision technology when it comes to using it in different environments and with different assortments. Also, to understand what is important to think about for designing a vision based Smart checkout, such as camera, lens and chip sizes, how performance of the camera is impacted based on different parameters, camera angles, product sizes and quantity.

3.3.1 In-person Interactions

To supplement the observations during the field visits, in-person interactions was chosen to gather perspectives of customers and staff. The in-person interactions were conducted as unstructured interviews; they were very open and involved a few overall areas that were discussed freely, as described by (Säfsten et al., 2020). Because in-person interactions require the conductor to handle several tasks at once, the resulting data is inherently influenced by the interviewer's subjective skill and multitasking capabilities. To mitigate the risks of data loss and subjective bias, the study employed standardized data collection techniques, including audio recordings and field notes, to ensure an accurate and verifiable record of the interactions (Säfsten et al., 2020).

This format allows the interviewer to extract specialized knowledge and explore complex topics that are essential for directing the project in the right direction (Säfsten et al., 2020).

3.4 Try it yourself

To gain insights into how users interact with a design, practitioners often adopt the 'Try it yourself' method. This process involves testing a product or service in real-world conditions to bridge the gap between theoretical design and actual usage (Milton & Rodgers, 2013). By experiencing the user's journey firsthand, designers develop a more robust awareness of the emotional and practical complexities involved, guiding more informed design evaluations. The 'learning-by-doing' approach allows for a direct comparison of subtle variations across design solutions. These insights are essential for uncovering new creative avenues and identifying strategic opportunities for innovation.

3.5 Thematical analysis

As a qualitative study, data analysis is more of a reflective craft rather than a rigid formula. The techniques for gathering and interpreting data are driven by the research objectives. By applying a thematical analysis, the study followed an iterative cycle of coding and refinement to ensure emergent themes addressed the research questions. This process transformed raw observational data into insights by identifying recurring patterns in potential artifacts, workarounds, and strategies from the end users. While observations were essential for documenting behavior, interactions were essential to capture people's perspectives and motivations.

To minimize personal bias, it is crucial that the documentation clearly distinguishes between raw data and interpretation. Conducting regular reviews with colleagues to challenge each other's assumptions and ensure that interpretations remain neutral and objective is also crucial in this analysis method (Kuniavsky et al., 2012). For this specific project, the extended project team served as a board for continuous discussions and peer review throughout the process. Following this, the data was re-evaluated, and adjustments were made to enhance both the clarity of the analysis and its presentation. These steps were taken to ensure a good level of transparency throughout the study.

In the result section, the thematical analysis is presented with main themes, following an inverted pyramid structure, prioritizing the most general conclusions at the beginning. It is presented with a descriptive headline, a summary of the observed phenomenon including evidence with citations from interactions and interviews and observed photos to support the findings.The goal of the thematical analysis was to identify patterns and themes that can be translated into actionable design problems.

3.6 How might we

Within Design Thinking the "How Might We" is a framework that serves as a vital bridge between the definition and ideation stages. Following the synthesis of research during the discovery phase, HMW statements are utilized to translate core problems into generative questions. This process, frequently supported by visual aids and sketches, helps prioritize key problems and catalyzes the development of innovative solutions (Dekker, 2020).

- **How:** Suggests that solutions exist; our task is simply to discover them.
- **Might:** Emphasizes exploration and experimentation without the pressure of an immediate, "correct" answer.
- **We:** Highlights the necessity of collaboration, acknowledging that the best solutions are developed together.

In alignment with ITAB methodology, The HMW framework was used to explore potential solutions during the solution discovery phase.

3.7 Competitor product analysis

Conducting a competitor or benchmarking analysis allows designers to systematically evaluate a product against existing market alternatives using established criteria. This comparative method identifies the advantages and limitations of competing or analogous systems, providing a framework to assess the developing design alongside its rivals. Furthermore, this process helps establish qualitative metrics regarding subjective performance (Milton & Rodgers, 2013).

3.8 Mock-ups

To evaluate the scale and visual proportions of a product concept, designers frequently rely on full-scale mock-ups crafted from low-fidelity, easily workable materials like heavy cardboard glued together. These physical models are typically implemented during the initial and intermediate phases of the development phase to analyze user-product interactions and verify a product's form or design before it is finalized (Milton & Rodgers, 2013).

3.9 Concept testing

Concept testing can be described as an experimental activity that can evaluate a design. It has a purpose that benefits of having a defined purpose to understand whether the participants like a concept, if it addresses user needs, or how the development should be continued. Choosing a population that could be potential users for the end product is important to mirror the target market in as many ways as possible. There are many ways the surveying can be executed, such as sending a digital survey, focus groups or face to face interactions. In this study, face to face interaction has been the chosen format as the final design aims to be interactive, it would be necessary to have interactive concept testing (Ulrich et al., 2019).

3.10 Appearance model

Appearance models serve primarily as high-fidelity visual tools for executive reviews, stakeholder approvals, promotional photography, and trade exhibitions. Typically machined, hand-made from accessible materials like foam, timber, or plastic, these prototypes are accurately painted and finished to replicate the intended product's aesthetics. However, because of their delicate composition, they lack long-term durability and are generally unsuitable for long-time functional use or consumer handling (Milton & Rodgers, 2013). Despite their fragile nature, the appearance models in this project were intended to serve as interactive tools for user interaction. Because the developed product aimed to be stationary, requiring no lifting, carrying, or structural load bearing, the risk of physical damage was minimal. Consequently, it was deemed safe enough to utilize these models for hands-on user evaluation.

3.11 User scenarios

Scenarios are used to tell a story about the intended user's experiences in a specific situation. Based on the data collection, scenarios were developed to highlight recurring customer checkout journeys, a method consistent with the framework proposed by (Boeijen et al., 2020). The scenarios serve to highlight the primary pain points that create friction within the checkout user experience and address those in a potential future scenario that could transform situation and resolve friction.

4. Implementation and result

This section represents the implementation of the methods and their results

4.1 Problem discovery

4.1.1 Field visits

The execution was done by traveling to stores in Jönköping Ryhov, asecs and in the city during multiple occations. In Stockholm, the field visits took place at different types of stores for two days. Specifically, observations were conducted at T-Centralen, the Avicii Arena district, Hammarby Sjöstad, Sergels torg,, the Barkarby Gateway retail area and Sergels torg were visited. The other study visits to Gothenburg lasted three days, enabling visits to a large sports arena during a hockey game at Scandinavium. Additionally, visits were made at other stores mainly at the central transit station and inside a large shopping mall called Nordstan. A large store in the Heden district of Gothenburg were also visited. The store types visited during this study are presented below:

- **Large fuel stations**
- **Franchise convenience stores**
- **Service stores in grocery retail shops**
- **Cinema stores**
- **Sports arena**
- **Pharmacies**
- **Lunch restaurants**

During the field visits, notes, photos, videos and recordings were documented on a mobile phone to keep track of events, reactions and interactions.(Kuniavsky et al., 2012), emphasize this as a critical step for establishing credibility; furthermore, it enables the observer to systematically track findings for subsequent analysis. Moreover, the environmental details should be kept in mind during the observation as that many times have an impact on the outcome of the situation.

In total, 39 different store visits were made during the field visits. Some stores were visited only for a short amount of time due to low customer rates and low SCO adoption, which made the focus shift to other more crowded places. Other stores were visited for a longer time, up to one hour and thirty minutes. For instance, Jönköping Cinema store was visited three times. Once to interact with the staff about their working habits, customer rates, most commonly sold items and what situations they find more friction. The other two visits were during friday evenings. The first observation at the cinema in Jönköping was the 20:th of mars at 20.15 during one hour. The second cinema visit was the 10:th April at 19.00 for one hour and thirty minutes.

4.1.2 In-person interactions

To understand the user perspective, in-person interactions were conducted during field visits with a selection of store managers, frontline staff, and customers. Following the completion of their transactions, customers were approached to discuss their user experience and the rationale behind their choice of a manned checkout, SCO or smart checkout. These discussions aimed to empathize with their decisions, uncovering motivations, perceived ease of use, and any specific friction points encountered. Moreover, participants were asked open-ended questions to realize the frequency of certain events without leading them toward a specific response. As described by (Kuniavsky et al., 2012), this form of indirect interviewing minimizes researcher bias and encourages participants to share an authentic worldview. Questions focused on the participants' immediate experience to avoid speculative predictions, as users tend to idealize and simplify their future behavior. Subsequently, participants were encouraged to elaborate on their responses through simple 'why' follow-up questions.

The interactions with store managers and frontline staff aimed to evaluate store organization, core operational tasks, sales optimization strategies, theft risks and customer satisfaction. By engaging both user groups, the study sought to identify the primary drivers of checkout preferences, the influence of specific product categories on behavior, and general attitudes toward vision-based smart-checkouts.

4.1.2.1 Participants

Prior to the discussions, the interviewee introduced himself as a student doing a thesis work. The respondents were briefed on the aim of the discussion and they granted informed consent for their perspectives to be anonymously included in the study. The sessions were recorded using a mobile device to minimize the risk of inaccurate citations and to ensure the participants' intended meanings were preserved. Additionally, notes were taken during and after the interactions to highlight key themes, facilitating easier access during the synthesis stage of the data analysis.

Participants were selected based on availability and a randomized 'first-come, first-served' approach. While there is an inherent risk of subconscious personal selection, the study prioritized a diverse demographic profile to strengthen the validity of the results. The final sample included a broad representation of participants across gender, age groups (ranging from adolescents to middle-aged individuals), and diverse nationalities (although mostly Swedish), providing a wide perspective on the observed behaviors.

4.1.3 Interviews

Digital interviews were the most efficient way to engage external stakeholders, accommodating their flexible and remote working patterns. The sessions were held via Microsoft Teams or Google Meet, depending on the interviewee's preference, ensuring a simple and accessible environment for the dialogue. The first interview with the internal ML and CV engineer was conducted digitally for efficiency, as an in-person meeting was not required. In contrast, two interviews were held in person: the second interview with the ML and CV engineer regarding camera specifications and another with a product manager at ITAB.

Participants were primarily recruited through calling company switchboards or sending emails to professionals within related fields and asking for interviews. Contact information was gathered via corporate websites, general service inquiries, and direct outreach. Additionally, attempts were made to receive contact details through in-store staff during field visits. However, adherence to General Data Protection Regulation (GDPR) protocols meant that many employees were unable to disclose internal contact information. These privacy constraints posed a significant challenge for the recruitment process, ultimately limiting the number of interviews that could be conducted for the study.

4.1.4 Try it yourself

In this project, the 'Try it yourself' method was employed to evaluate interactions with self-service systems across various retail environments, from the initial impression of the system's location to the final receipt of purchased items. This approach served to triangulate the findings from the thematic analysis. Additionally, the method was used to pre-evaluate mock-ups, assessing their suitability and the emotional responses elicited by different design alternatives.

The first field visit was at a gas station in Jönköping to understand how another vision-based smart checkout system is working from a UX-perspective. It was tested by picking a lots of products and trying to register them at the checkout and finding interaction points. In the first test, a coffee was the Item of choice. See illustrations of the purchase process in Figure 7 - Figure 11 below.

Figure 7: Mashgin smart checkout starting frames (A: item placement, B: item separation)

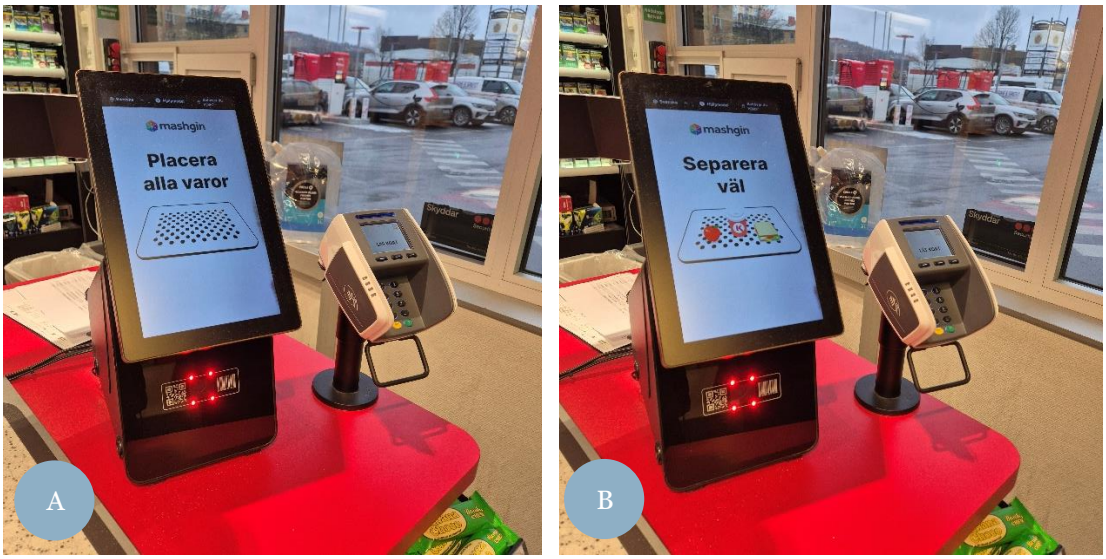


Figure 8: Mashgin UI during item placement (C: item placement, D: system processing)



Figure 9: Mashgin UI menu selection (E) and final itemized receipt (F)

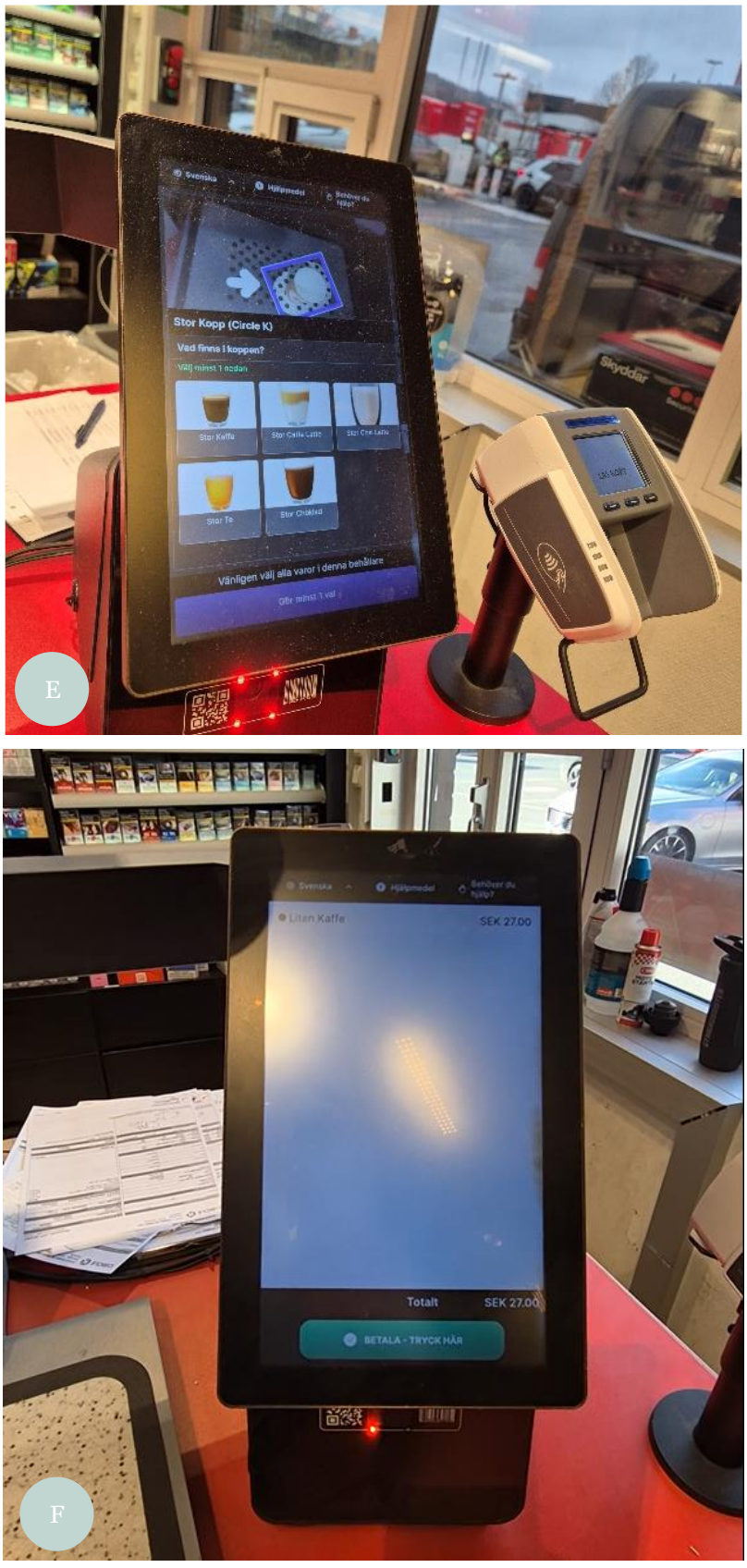
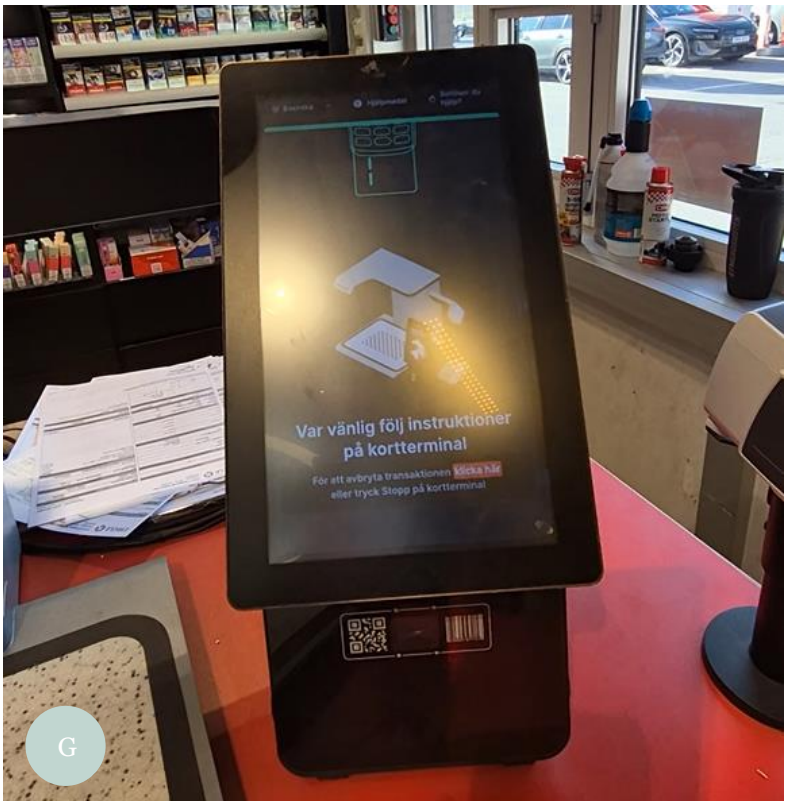


Figure 10: User interface instructions following payment initiation.



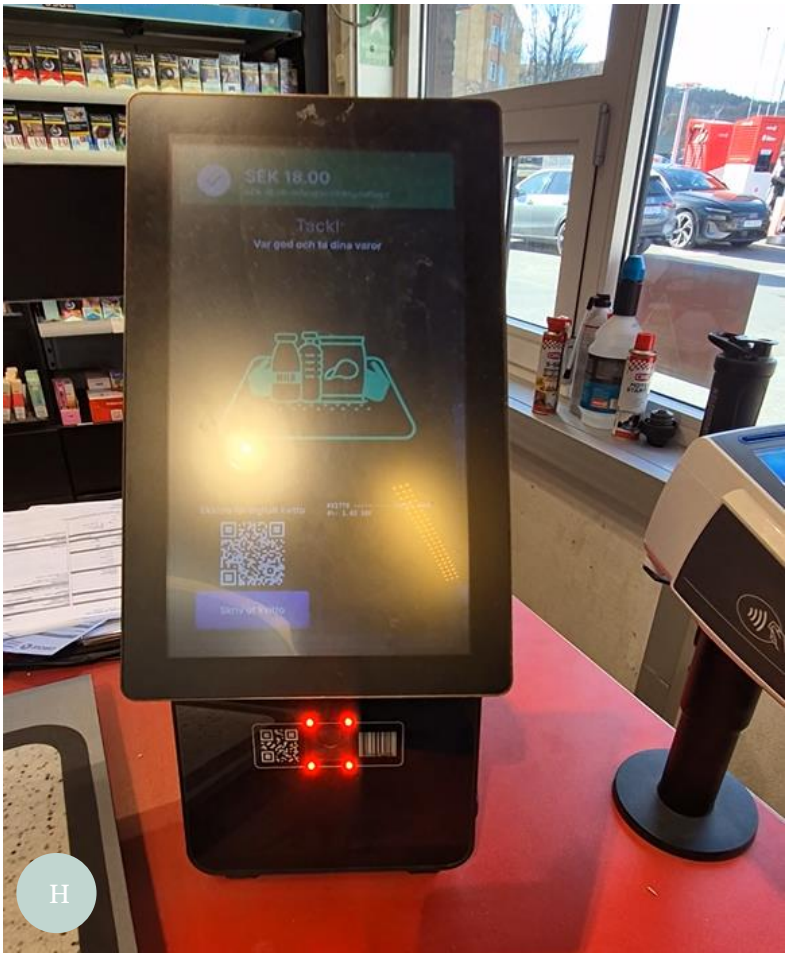
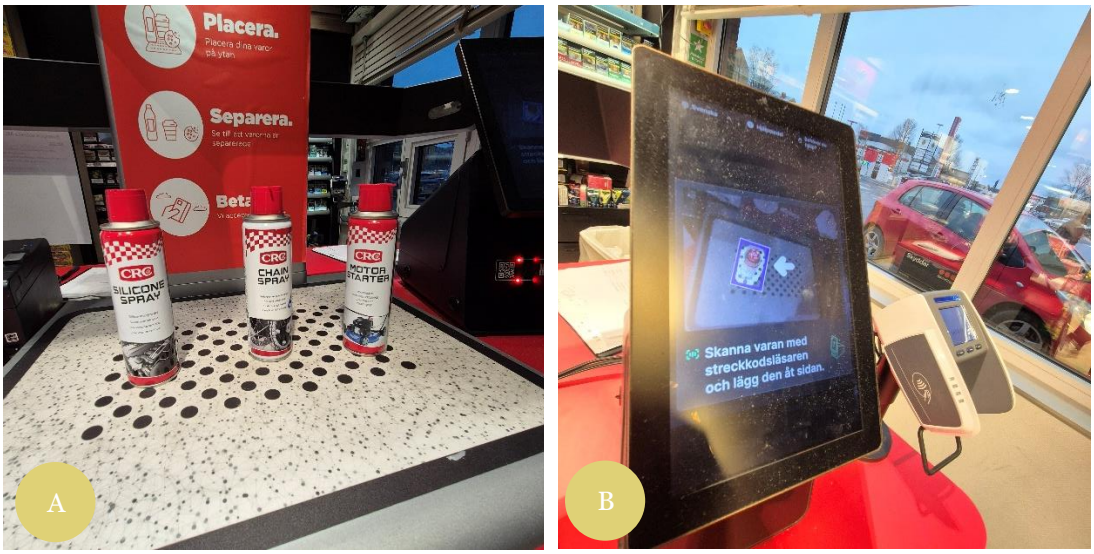


Figure 11: Item placement array (A) and manual barcode scanning prompt (B)



Subsequently, many other Items were tested to understand what happens in the CO when they are selected to test. Items tested were:

- Bakery Item (cinnamon roll)
- Flower
- Newspaper
- Energy drinks (Power king and NOCCO)
- Vehicle Oils
- Slush
- Sunglasses
- Beers in multi-pack
- Bag candy
- Chips
- Sandwiches
- USB-C charger

From these tested Items it was noted that slush became detected as a coffee by the system as the coffee menu popped up. During staff interaction it was mentioned that it was not currently possible to buy slush in the machine. Same with Newspapers, sunglasses and flowers, as they were neither recognized by the system or were possible to register with the barcode scanner.

Noted was also that the energy drinks does not have any age control in this smart checkout, and when asking staff why, they said that it is because they have very many customers buying that and if the staff would need to control them all the time they would have to constantly confirm them, and the convenience is as good as gone for the users. They also mentioned that they always keep an eye on people using the smart checkout so if there are any underaged customers trying to purchase energy drinks they interrupt them.

To have some reference of design element measurements that could be useful for components in smart checkout, the Mashgin design was studied in detail. Measurements was taken on the component that encloses the cameras, sizes of the table, computer box, scanner and light source. See the illustrations of the measuring done in Figure 12.

Figure 12: Shows some of the observed measurements of a Mashgin smart checkout.



4.1.5 Data management

Primary data, consisting of audio recordings, field notes, images and videos were gathered using a mobile phone. The interaction and interview data were imported to Klang.ai, a Swedish-developed AI service for automatic transcription, summarization, and analysis of conversations. This platform was selected specifically for its adherence to GDPR and EU data sovereignty regulations, ensuring that sensitive information remains protected and is not utilized for external model training. The use of AI-assisted transcription significantly accelerated the data processing phase compared to traditional manual transcribes.

To ensure accuracy of the transcriptions, they underwent a manual review and correction process by listening to the original audio while reading the transcription. Since all interviews and interactions were made in Swedish, they needed to be translated into English. See an example of how they were translated in Figure 13.

Figure 13: Swedish-to-English interview translation using Google Gemini



Subsequently, all individuals and company names were anonymized for ethical reasons, ensuring the information is presented transparently. The data were then imported into MIRO, a digital workspace where the thematical analysis were conducted. Given the high number of interactions, the study presents the highlights in a “main themes” section from a thematic analysis. This approach allows for the synthesis of key findings across the entire dataset, identifying recurring patterns and insights. The selection of quotes was based on their importance in relation to the research questions. Individual quotes were selected not just for their own meaning, but for how they support the broader themes of the study.

4.1.6 Thematic analysis

The thematical analysis was conducted through an inductive-bottom-up approach, which means letting the data suggest groups, rather than starting with fixed categories. Moreover, it included examining text from transcriptions, listening through the recordings, quotations, field notes, as well as looking at images and videos and try to understand what the material really means in context with everything else. This led to generation of codes based on clustered data. To give the reader an insight into the vast amount of data analyzed, the comprehensive thematic analysis is illustrated below. See Figure 14-16.

Figure 14: Initial data clustering from Stockholm and Gothenburg field visits

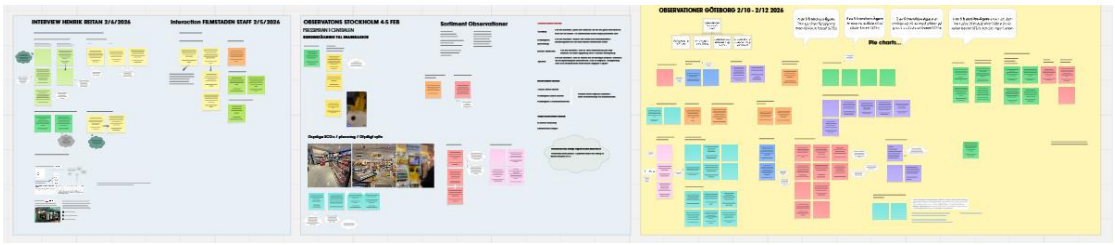


Figure 15: Shows the first part of the thematical analysis.

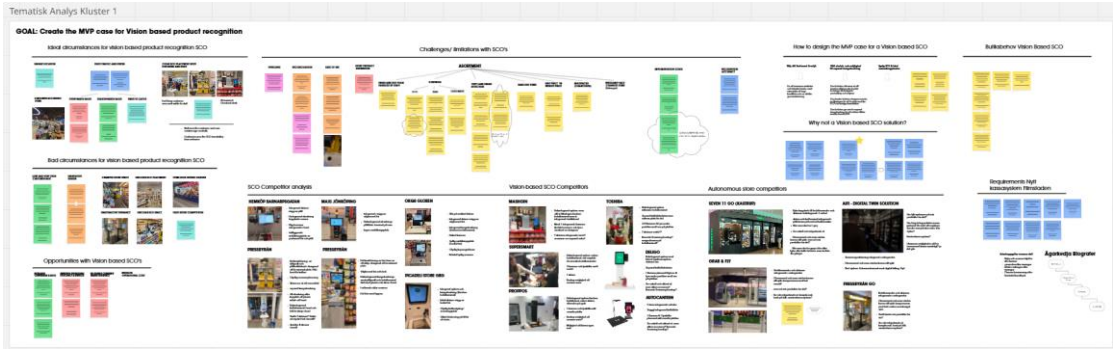
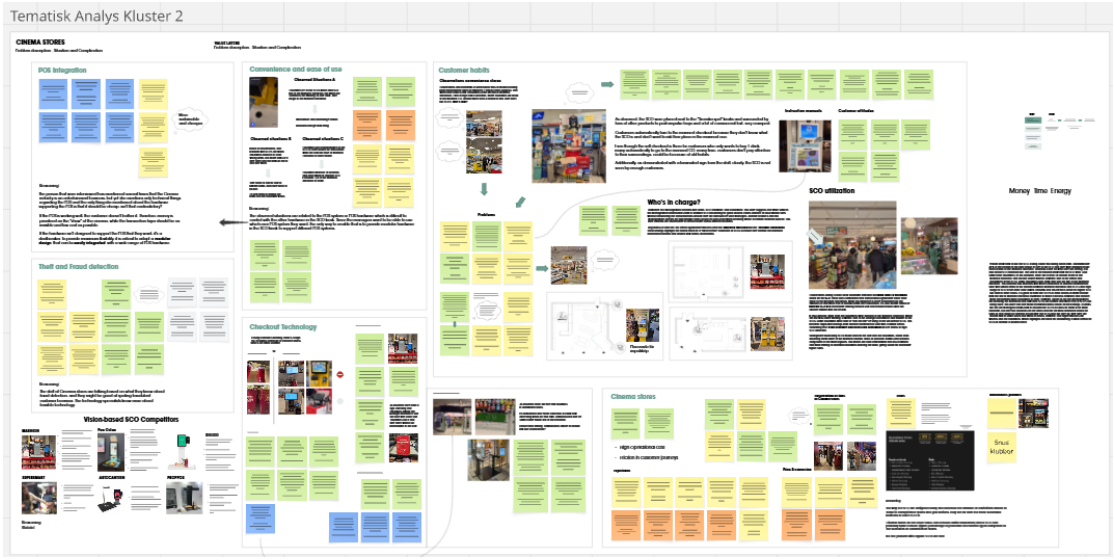


Figure 16: Shows the second part of the thematical analysis



4.1.7 Main themes

To navigate the complexities of identifying a store type that potentially could be suitable for developing a conceptual MVP, the main themes are presented with a funneling process. This approach evaluates stakeholder needs, perspectives, regulatory requirements, and operational challenges across diverse store environments. After revealing what store type that could be suitable according to the research that has been conducted in this specific study, the following themes are presented regarding identified problems in customer journeys for less than 10 item purchases.

4.1.7.1 Time-sensitive environments

Observations were made in both low-traffic and high-traffic environments. Based on 17 convenience store visits and five different gas stations it was shown that convenience stores with low customer traffic rarely have SCO's. When they do, they are hardly ever used as

observed and according to what store managers have said. Instead, most customers head towards the manned counter during times with low customer traffic at the observed stores. According to the observations in this specific study, the viability of SCO seems to be dependent on high-traffic environments characterized by when customers have a specific time to catch. During customer interactions, it was mentioned that SCOs doesn't appear to be the fastest checkout option if the traditional checkout is manned, which could be a reason why many customers prefer to head toward manned counters in low crowded convenience stores. Conversely, another customer mentioned that *"Self-scanning is preferred during peak hours to save time",* which could be connected to a higher usage of SCO when the store traffic is high and people have a time to catch. Observations from a convenience store at Ryhov during lunchtime in Jönköping showed high usage of SCOs at a time when many customers were queueing at the manned counter, see Figure 17.

Figure 17: Lunchtime field observation at a Jönköping Ryhov convenience store



A similar pattern was mentioned by a vice store manager of a large gas station with a unique location at the Avicii arena area in Stockholm. It was mentioned that the SCO's are a necessary solution during high customer rates in the store because they often get sales peaks before and after concerts and games. The store has contact with the arenas near the area and receives notifications two months ahead to be able to staff up properly, still the SCO's are said to be a necessary solution to help checkout customers during extreme peaks.

At a couple of convenience stores, including one at Gothenburg transit station, and one at Asecs Jönköping it was mentioned employees that *"When there is a line to the manned checkout, the SCO are used much more frequently", and "the higher the pace within the store, the greater impact of self service solutions".* During an interview with a CIO and it-specialist from a convenience retail management group it was mentioned that "*Smart checkouts has the biggest potential in environments where a rapid checkout process is critical, and where there are many frequent users"*. This could suggest that time-sensitive environments are a cornerstone for the adoption of smart checkouts. See a cluster of the thematical analysis that seems to lean toward a main theme that could be called time sensitive environments in Figure 18.

Figure 18: Thematic data cluster for time-sensitive environments



During the interview with the IT manager over the large convenience store, it was mentioned that they have high staff turnover rate. One reason why SCO's are implemented are to help sale peaks which can be particularly fragile when dealing with high staff turnover rate. When SCO's do not help as much as needed, they are ineffective for the stores that needs it the most. Therefore it could be of high importance to have Self service alternatives that help as effective as possible. And from a

In summary, the initial screening process identifies several retail environments that seems to be badly suited for a smart checkout based on time sensitive environments.

- **Pharmacies** present a unique challenge for automated retail based on interactions with pharmacy staff from Jönköping. It was mentioned that the staff core purpose is centered on patient safety and precise dosing rather than throughput. Moreover, it was mentioned during the interview with the CIO over the over convenience retail management group that the pharmacy industry is controlled by many regulations.

Because of strict pharmaceutical regulations and the necessity for professional oversight, smart checkouts which could be beneficial for rapid customer journeys, conflict with the industry's legal and ethical requirements. For these reasons, pharmacies are excluded as a viable environment for the initial MVP.

- **Service stores in grocery retail shops** are as observed, heavily reliant on staff for mail services, returns, and age-restricted sales. Despite automation of mail management, and tobacco management as observed, it's believed that an automated SCO kiosks can't add significant value here.

- **Gym kiosks** typically experience low daily customer sales according to staff interactions at 3 large gyms in Stockholm. Additionally, they mentioned that food and age-restricted products, such as energy drinks, are the items sold most frequently. Sometimes protein shakes requires manual preparation by staff, making fully autonomous sales impossible for these products. Gym receptions at the observed stations are only unmanned during low-traffic periods which could speak for low sales potential and low demand for a swift customer journey. Therefore, gym kiosks has been deemed unsuitable for specifically developing a smart checkout MVP.

4.1.7.2 Customer Habits & SCO placement

Visibility and strategic placement are critical factors in the adoption of more successfully implemented SCOs. Interactions with a franchise owner of a convenience store in Jönköping indicated that SCO placement typically depends on store layout, prioritizing staff oversight and customer accessibility, objectives that align with field observations at convenience stores. During observations at Nordstan in Gothenburg, Stockholm central station and Jönköping Asecs, it was noted that many customers pass by the SCO's and head to the manned checkout despite high store traffic. In these stores, the SCO units are positioned either in the center of the store, or at the manned counter. See Figure 19 and Figure 20.

During a customer observation at a transit station convenience store in Stockholm, it was noted that a customer failed to perceive the SCO alternative. Another couple in the same store explicitly stated that they always prefer checking out at the manned counter; both behaviors suggest the strong influence of established customer habits. Similarly, at a store in Nordstan, five people were observed waiting in line at the manned counter. Once the customer at the back of the queue finally completed their purchase, they were approached to determine why they chose the manned option. The customer explained that they were unaware of any alternatives, noting that they were not from the area. These interactions and observations suggest that if SCO terminals do not visibly engage customers from a strategic location, that can prevent users from using them. See a clustering of the problem regarding Customer habits and SCO placement in Figure 21.

Figure 19: SCO configurations at a Stockholm transit station and Asecs in Jönköping

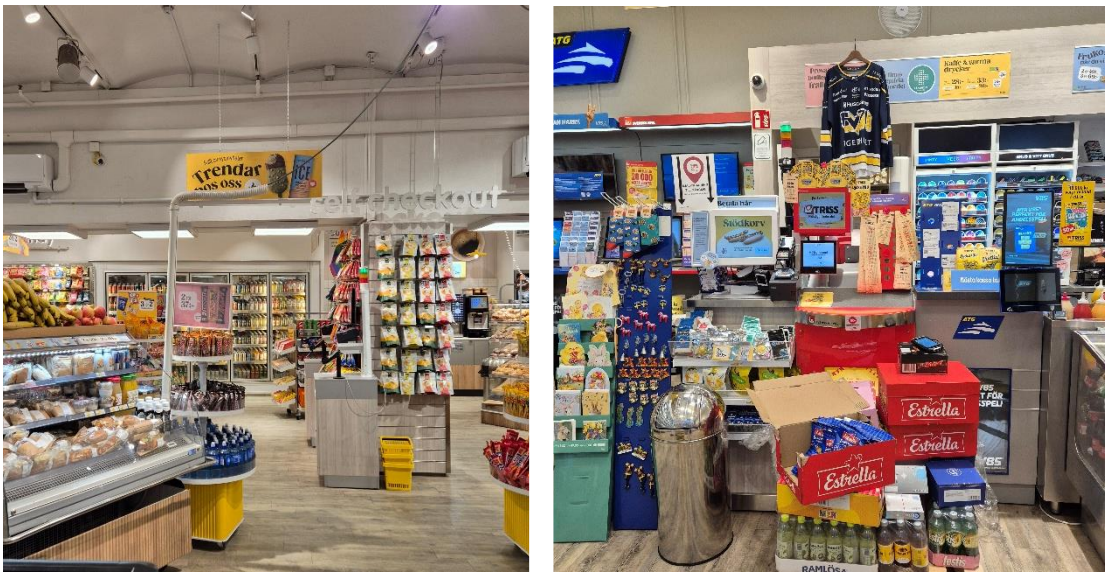


Figure 20: Customer queue behavior and SCO underutilization at Nordstan



Figure 21: Clustering of problems related to unclear SCO placements.



From employees at a couple of convenience stores, it was noted that they frequently inform customers of the SCO alternatives, highlighting that the current SCO setup does not lighten as much workload as intended. Some employees mentioned experiencing high stress during peak hours, believing that if the SCO area were more clearly positioned and less cluttered, it would help relieve pressure on the staff.

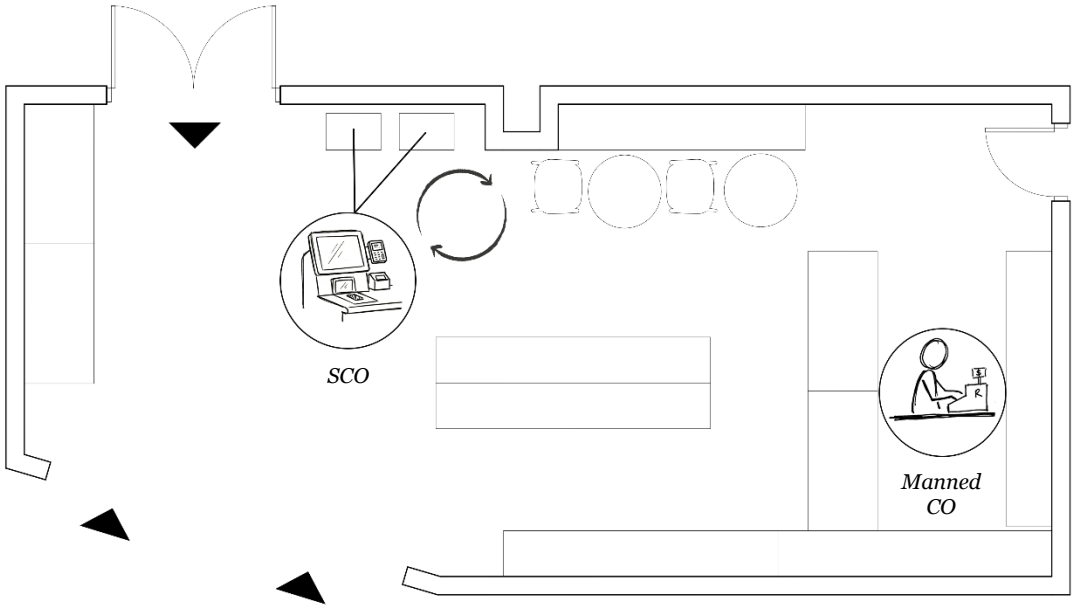
In another Store in Stockholm, an employee interaction suggests that a SCO's former, more hidden location resulted in significant underuse, as many customers did not spot the kiosks. Repositioning the SCO kiosks near the starting point of the queue-line to the manned checkout, optimized the customer flow and led to a significant boost in usage. See the placement of the SCO in the store where the employee works, in Figure 23 & Figure 23. Moreover, this placement still ensures a clear line-of-sight for staff, facilitating the possibility to confirm or decline age-restricted items before transactions. By maintaining visual contact with the SCO area, personnel at the manned stations can efficiently authorize or decline purchases, while making the SCOs more accessible.

Figure 22: Field photograph of the self-checkout layout depicted in the floor plan



Comment. The photo was taken with a mobile camera the 4/2/2026.

Figure 23: Simplified store floor plan showing layout modifications in SCO placement



Comment. The Floor plan was created in Autodesk Revit, and the symbols were generated with Gemini Nano Banana pro, and the composition was done in Photoshop.

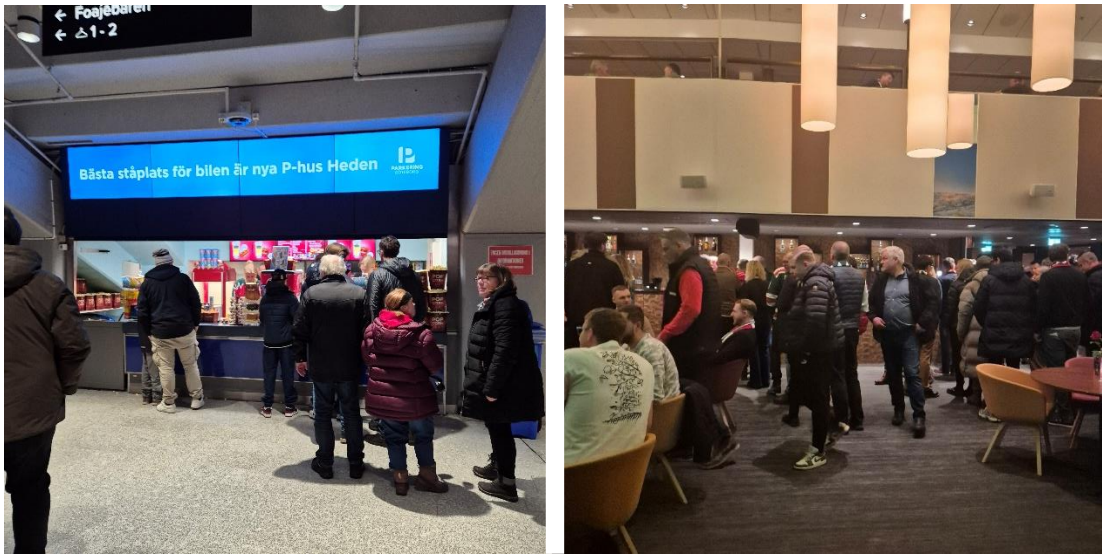
4.1.7.3 Item accessibility

Convenience stores have different architectural layouts which impacts the possibility for customers to utilize self-service alternatives. Observations at convenience stores within a large sports arena in Gothenburg showed that most items are sold behind manned counters. See an illustration of the distinction between open shelf accessibility and restricted staff-assisted availability in Figure 24. This appeared to create friction, as long lines frequently formed and crowding at concession stalls. Consequently, this layout could present a major obstacle for smart checkout systems, as it prevents a "grab-and-go" experience. Based on this observation, it is believed that store layouts that allow customers access to items from shelves should be a better alternative for smart checkouts to provide good a user experience. See an an illustration of the observation of lines at the sports arena in Figure 25.

Figure 24: Comparison of a customer access format and a staff mediated service model



Figure 25: Customer queuing and density at a sports arena bar in Gothenburg.



During staff interactions, it was mentioned that the lack of a clear checkout line make it difficult to manage order priority, leading to customer service delays. Observations of the lines make it understandable that the lines have poor structure.

Other observations at convenience stores outside the sports arena revealed that tobacco and unprescribed medications frequently are positioned behind manned counters. Interactions with staff highlighted that this placement is a strategic measure to mitigate high theft risks and ensure compliance with age verifications. Opposing these problems, smart checkouts would possess friction for customers to ask staff for these items prior to using the checkout, leaving customers to face the same issue as with traditional SCO's, item restrictions. Moreover, age controls, possibly providing equally as much friction or more than current solutions does in the Swedish retail landscape.

4.1.7.4 Assortments

Certain product categories can generate more friction for customers than others. Based on findings, staff-mediated items often represent friction as they limit the customer's ability to utilize self-service. Hot food and age-restricted goods are often staff mediated. Observations at convenience stores, gas stations, and sports arenas showed that they frequently sell these products; consequently, they face significant self-service challenges with them.

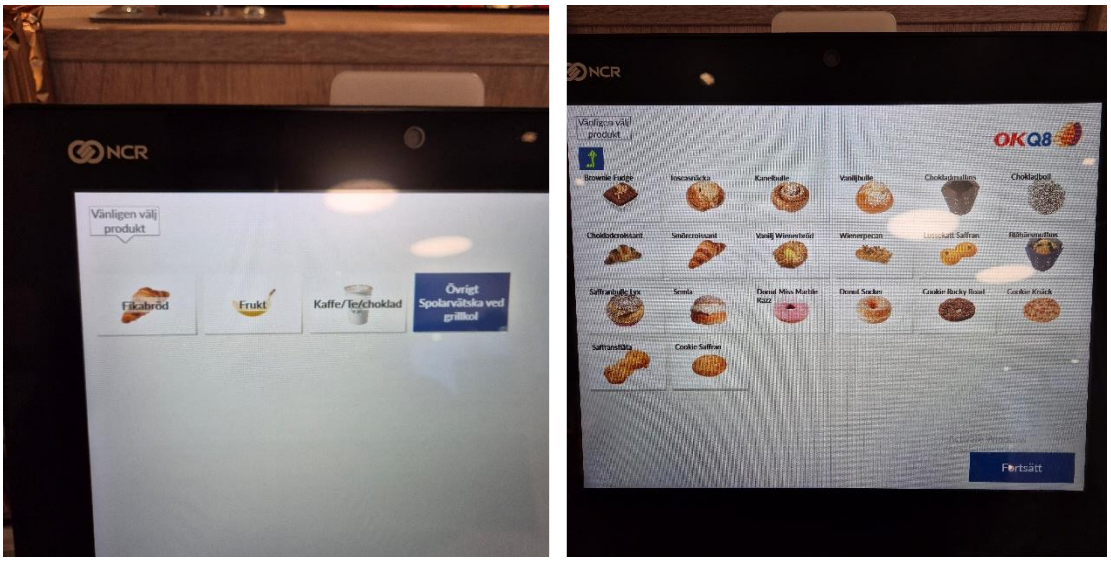
Customer interactions at convenience stores and cinema stores indicate that age-restricted items often influence their decision on whether to utilize a self-checkout. Citing a customer: 'Self-checkouts don't work when you're going to buy snus or energy drinks.' However, the customer expressed that he uses self-service during regular grocery shopping whenever he does not purchase these types of items.

Discussions with the extended project team regarding sales at a sports arena highlighted a potential problem for smart checkouts: selling alcohol. During the discussions, it was noted that legal regulations prohibit the sale of alcohol to intoxicated individuals. The ML and CV engineer highlighted that it could be challenging for a vision-based system to accurately assess a customer's level of sobriety. This could be a potential reason why smart checkouts have not been implemented in Swedish sports arenas, such as the one observed.

Another potential problem regarding grab and go experiences amongst convenience store assortments could be prepared food. Observations of the SCO UI menu at franchise convenience showed that it is impossible to order prepared food. During some customer interactions at convenience stores it was revealed that they stood in line to the manned checkout specifically to buy hotdogs. However, some mentioned that they would prefer to order food at SCOs rather than standing in line to the manned checkout. While integrating food ordering into a smart checkout could be an interesting problem to solve, but it might be out of scope for this thesis as a solution probably would necessitate a more comprehensive solution than just adding food to a UI. During staff interactions at the same convenience stores, it was mentioned that no personnel is specifically assigned to hot food preparation, which highlights that integrating such solution might require organizational changes to function effectively.

Non-barcode items usually possess more friction than barcode items for traditional SCOs as users often must navigate menus to add them. See an observation of UI with non-barcode items in Figure 26.

Figure 26: Shows an observed UI at a SCO at a Stockholm gas station near Avicii Arena



In convenience stores, fruits are often sold per item, meaning that if a user wants to buy more fruits they might have to click several times on the screen to add them. This is an activity that can generate friction as it takes longer time to make several clicks on a screen rather than just scanning an item. Moreover, in retail stores with a wide selection of bakery products, menus can be long and difficult to navigate depending on how they are designed. In the observed UI, item text were supplemented with images which could be helpful, but that was not always the case for all observed SCO's. As observed, customers sometimes manually search for specific bakery items or coffee types which also could be a friction point since the customers do not always know the name of the bakery they are purchasing. A potential workaround could be using a smart checkout to recognize non-barcode items in a more frictionless way it it works properly.

According to an interview with a ML and CV engineer, many vision-based recognition systems struggle with multi-pack products because they feature multiple barcodes. It was explained that the system cannot determine which barcode to prioritize, leading to errors. While manual scanning or staff assistance is often required as a solution, this ultimately creates an additional point of friction for the customer. During observations of one of Mashgin's smart checkouts, their system occasionally struggled to identify multi-barcode products. When recognition failed, the system informed the user via the display to proceed with manual scanning which aligns with the statement of the ML and computer vision engineer. Hence, the problem with multi-packs would need a clever solution to work more frictionless. Additionally, frequently updated packaging was mentioned as a potentially bothersome process. Requiring staff to teach the machine a new product every time its visual appearance changes, or when new items are added, would make system maintenance a highly time-consuming task. Examples of these types of items are newspapers and magazines and seasonaly updated items. Consequently, retail stores that carry many products with frequently updated packaging may face challenges in the ML process if using a smart checkout.

Selecting a viable MVP store for a smart checkout system depends heavily on its product assortment. To evaluate this, a screening of different convenience store formats was conducted based on five specific criteria. See Table 1. Below is a more thorough reasoning of the stated criteria in bullet points.

- **Absence of Service-Heavy items:** Unlike many convenience retailers, cinemas typically prohibit hot food and do not offer in-store preparation, according to interactions with shift leaders at two major locations. This service model effectively eliminates the operational bottlenecks associated with food preparation.

- **Low Friction from Age-Restricted Goods:** While cinemas stock energy drinks, observations reveal that few customers actually purchase them. This is likely due to heavy price competition from nearby retailers with lower price points, a factor noted during customer interactions. Consequently, the low frequency of energy drink transactions minimizes the need for manual age verification. Furthermore, since there is no legal requirement for age stickers on energy drinks in Sweden—rather, it is a voluntary industry agreement—an automated age verification process is not strictly required at SCOs. In fact, staff confirmed that Mashgin's smart checkouts currently do not implement age controls for energy drinks, opening this as a way to reduce friction in a new smart checkout as well.

- **Support for Non-Barcoded Items:** While most items carry barcodes, cinemas often sell specialty merchandise, such as promotional popcorn bowls and collectible beverage containers. A vision-based system can identify these items via image data, eliminating the friction typically caused by manual UI navigation.

- **Seldom updated packaging:** Noted during interactions with staff was that few products frequently change packaging except popcorn or drink merchandise containers, potentially reducing the friction for constantly teaching the system a lot of new items.

- **Existing Weight-Based Infrastructure:** The weighted items sold at cinema stores are usually "pick and mix" candy. Since most cinemas already provide standalone scales, the technical challenge of integrating a scale directly into the SCO recognition area is bypassed.

Based on these criteria, cinema retail emerges as the most suitable environment for implementing a smart checkout MVP.

Table 1: Evaluation of store assortment categories and MVP concept feasibility for cinema retail.

Assortments	Convenience stores	Gas stations	Stadium kiosks	Stadium Bars	Cinema stores
Staff handled food	Yes	Yes	Yes	Yes	No
Age regulated Items	Yes	Yes	Yes	Yes	No
Non-Barcode Items	Yes	Yes	No	Yes	Yes
Multi-barcode Items	Yes	Yes	No	Yes	No
Frequently updated packaging	Yes	Yes	Few	No	Few

4.1.7.5 Upselling

Field visits revealed that convenience retailers are particularly concerned that SCOs do not facilitate upselling, an outcome upon which smaller stores often depend. In lower-traffic environments, implementing SCOs may not be a priority since staff are less occupied than in crowded stores. As one convenience store manager previously noted, "the higher the pace within the store, the greater the impact of self-service solutions." However, this does not diminish the fact that current SCOs are often poorly designed for upselling. Staff discussions at one convenience store confirmed that SCOs conflict with their sales model. Consequently, they suggested that upselling opportunities could be improved if the system recognized the scanned items and recommended complementary products. For instance, the UI could prompt the customer to add a cinnamon bun when coffee is registered, or suggest a "buy two for a discounted price" if it is on promotion. This insight presents an interesting opportunity to explore within the UI design phase of this project. However, the main focus of the thesis is the physical product design, and UI design is a secondary priority.

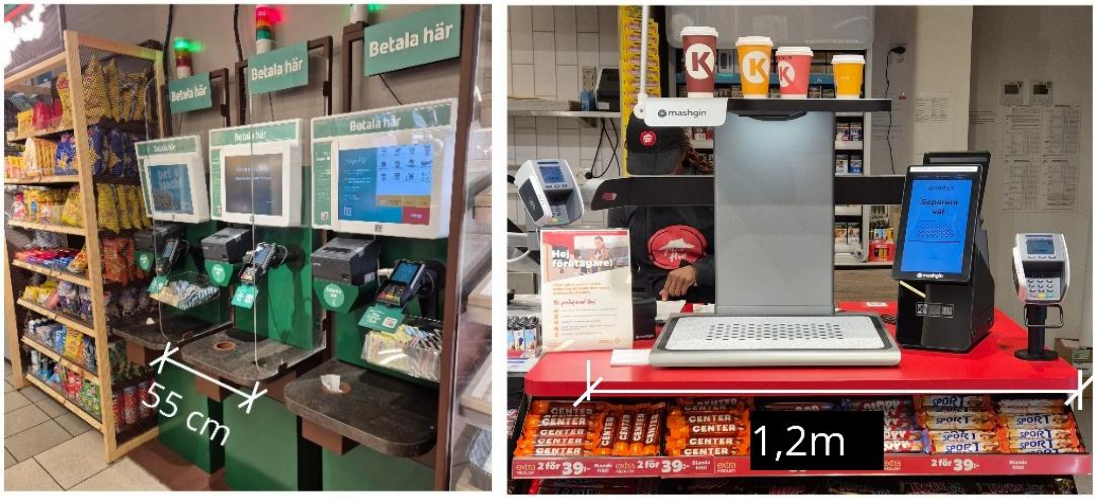
4.1.7.6 POS integration

During interviews, both the IT manager of the convenience store management company and the cinema sales manager emphasized the need to maintain their preferred Point of Sale (POS) system, regardless of the underlying checkout technology. Their preference is to utilize modular POS components, such as a standalone receipt printer, a separate Electronic Funds Transfer (EFT) terminal, and a specific screen type of their choice. The primary reasoning is operational consistency: using the same POS interface at both self-service stations and manned checkouts simplifies staff training, eliminating the need to learn a new system when technology updates. Additionally, a modular setup simplifies maintenance, as a single broken device can be replaced without purchasing an entirely new checkout unit. Therefore, insights from these interviews suggest that designing a smart checkout that integrates with an external, modular POS system would be a highly viable approach.

4.1.7.7 Smart checkout size

The IT manager mentioned that they have considered using Mashgin’s smart checkout, however, there were a couple of reasons why they chose not to. The first and foremost reason was that the physical size of the Mashgin machine was considered too large for their convenience stores. During observations of the SCOs that the convenience store currently uses, it was noted that their size is approximately half the size of Mashgin's. See Figure 27. If unit size is a dealbreaker for implementing Mashgin's vision-based system, it would likely pose a challenge for other smart checkout solutions as well. However, since the proposed MVP store type is a cinema store rather than a franchise convenience store, footprint size may not be a critical issue for this specific project. Nevertheless, space constraints would become a vital factor if the design were scaled for convenience stores, which is why scalability should be kept in mind during the design process.

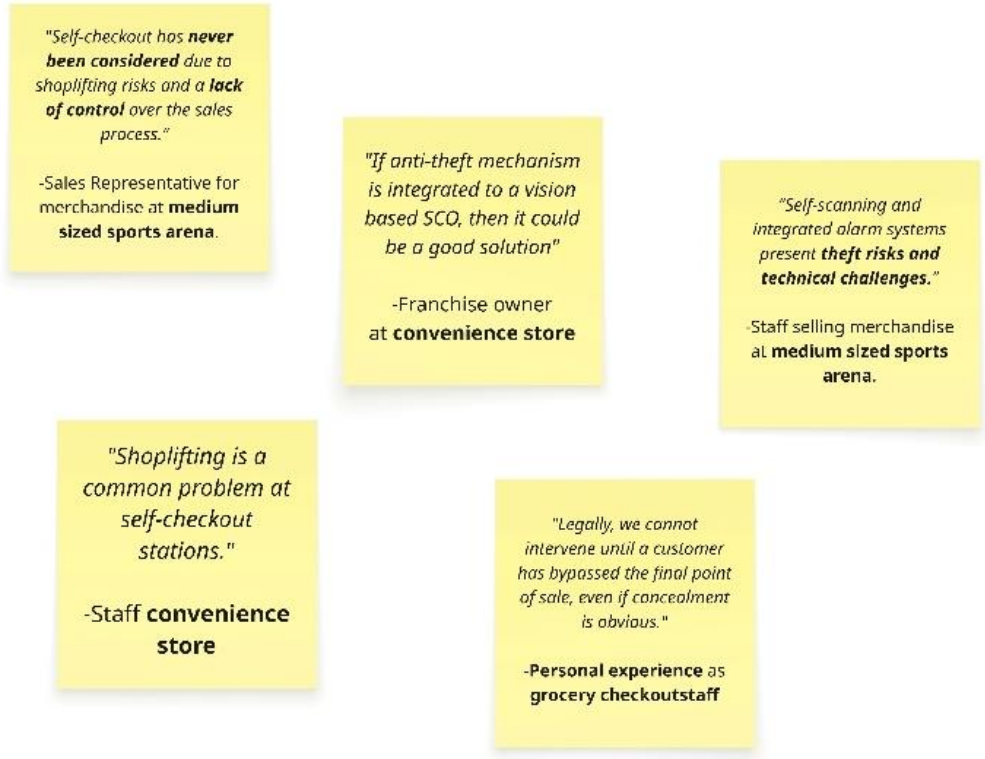
Figure 27: Size comparison of an NCR SCO setup and Mashgin's smart checkout.



4.1.7.8 Theft and fraud

Theft and fraud were mentioned during most interactions with store managers and sales personnel, highlighting a major concern. This is not surprising given the context established in the background section. However, it was mentioned in the delimitations that this study would not focus on theft or fraud solutions, which is why they were excluded from the project scope. Nevertheless, a small cluster is presented to highlight the specific issues mentioned during the field visits. See Figure 28.

Figure 28: Shows a cluster of theft and Fraud

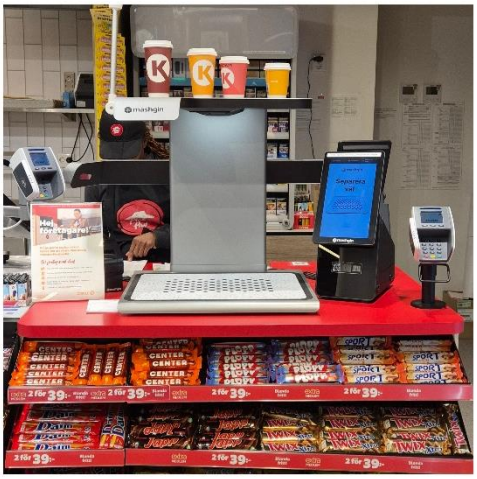


4.1.1 Competitor product analysis

The selected criterion for this competitor product analysis was to analyze by product architecture, as the possibility to evaluate on a deeper level was limited due to unavailability in Sweden. The Mashgin smart checkout was observed and analyzed on a deeper level with the "try-it-yourself" method, but the other brands were unavailable to physically visit. To keep consistency in the level of analysis among the different brands, they were all analyzed by their cameras, illumination, screens, barcode scanners, and the item placement area. See the competitor analysis in Figure 29-31.

Figure 29: Product architecture comparison of Mashgin and Supersmart's SCO systems

Mashgin



- The setup uses an integrated scanner and Screen.
- The screen size is 18 x 29 cm (10 inches)
- There is an external Electronic Funds Transfer (EFT) and receipt printer.
- Features a minimum of 7 integrated cameras placed on the sides and vertically downward angled.
- Features a top-mounted lighting of 18 x 20 mm.

Supersmart



- The setup uses an integrated scanner and EFT.
- No receipt printer was detected.
- The current screen size is estimated as 28 x 30 cm (15 inches) based on common SCO screen sizes.
- Features at least 2 integrated cameras, potentially 3 on the extruded structure.
- The system doesn't appear to utilize integrated lightsource on the plate.
- The placement area appears significantly smaller than Mashgin's.

Comment: The left image was captured via personal mobile camera while the right was captured by an individual in the extended project team. Used with permission.

Figure 30: Product architecture comparison of Autocanteen and Deligo's SCO systems

Autocanteen



- The design appears very minimalistic to the competitors.
- The screen is assumingly the same size as Mashgin's based on its appearance and proportions.
- The receipt printer is semi integrated into a structure together with the screen and EFT.
- Features 1 camera and possibly a light source faced downward to the placement area.
- The screen size seems fixed, limiting customers to have other screen sizes.

Deligo



- Seems to have modular design for multiple EFTs.
- The screen is assumingly the same size as Mashgin's based on its appearance and proportions.
- The screen also seems to be placed very high compared to other competitors.
- The screen size seems fixed, unless the EFT is removed.
- No printer was detected.
- Features probably 1 integrated cameras behind the screen.

Comment: The image of the Autocanteen chechout is adopted from https://www.autocanteen.com/retrieved 2026/05/17. The image of the Deligo checkout is adopted from https://www.deligovision.com/en-gb retrieved 2026/05/17.

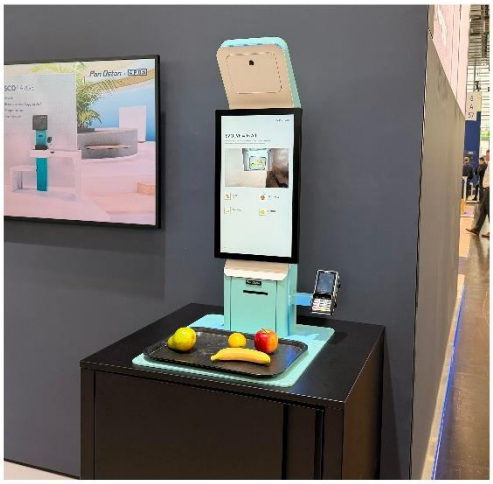
Figure 31: Product architecture comparison of Proppos and Pan-Oston's SCO systems

Proppos



- Seems to be a fully integrated system, with the exception of the EFT.
- The screen is on a separate kiosk and appears to be much larger compared to other competitors.
- The receipt printer appears to be integrated into the screen kiosk.
- Features at least 1 integrated camera faced downward on the placement area.
- Features a top-mounted light

Pan Oston



- Seems to have modular design for multiple EFTs.
- The display is significantly larger than those of many competitors, with a screen size comparable to that of Proppos.
- The receipt printer appears to be integrated into the kiosk.
- Features probably only 1 integrated camera.
- Features no light source for placement area

Comment: Both images were captured via an individual from the extended project team at a retail fair. Used with permission.

4.1.1.1 Summary

Cameras

Out of all the observed competitor products, Mashgin uses the most cameras. While this could provide the highest product recognition accuracy, it may also make the system more expensive. Mashgin's high camera count could be due to the observed product's age; as the oldest design of the observed products, it relies on the hardware and technical knowledge available at the time of its release. In one of Mashgin's more recent design they have 2 cameras on the side. See an illustration in Figure 32.

Figure 32: Shows Mashgin's new solution called Auto-magic checkout.



Comment: Image is adopted from https://www.mashgin.com/ retrieved 2026/05/17.

The other competitors seem to have between one and three cameras, which could be due to knowledge of camera placement that provides good-enough recognition accuracy for the purpose they aim to serve. Having fewer cameras could save money, but it might compromise recognition accuracy. By taking inspiration from the different camera counts and positions of the competitors, the design in this project aimed to find relevant camera positions that could work for a new smart checkout. This was aimed to be tested through mock-ups and by taking images with a mobile camera from the planned position of the cameras.

Illumination

As observed from the competitors, three out of six have some kind of illumination on the item placement area. As mentioned by the interviewed ML and CV engineer at ITAB, the illumination interplays with the camera in different ways depending on which properties the camera has. The drawback of having a smaller camera sensor aperture, the more illumination is required on the item placement area for the camera to receive light. The bigger the camera lens, the less item illumination is required for the camera to receive light. Based on this information, it's possible to think that the competitors that do not illuminate their placement area has a big camera hole.

For the scope of this thesis, it is important to consider the store environment where the product aims to be designed for, specifically, a cinema lobby checkout area. At two observed cinema stores, the environment features low, diffused ambient overhead spotlights to maintain a moody, high-contrast atmosphere. In contrast, intense, warm-toned LED strips are integrated directly under the service counter shelves that illuminate items. See illustrations of the observed cinema environments in Figure 33. Because these areas are dimly lit, additional illumination may be necessary over the item placement area to ensure sufficient light reaches the cameras for accurate recognition.

Figure 33: Observed lobby's within Swedish cinema stores in Jönköping and Gothenburg





Screens

The competitor analysis reveals that all six smart checkouts feature an integrated touchscreen. Adopting that in a new smart checkout design could therefore be a logical step to align with these smart checkout designs. However, screen sizes vary considerably. Recent images on Proppos and Pan Oston from the 2026 EuroShop fair shows larger displays compared to the other competitors with older designs from Mashgin, Autocanteen and Deligo. During discussions with the extended project team, it was mentioned that it could be a trend to market self-checkouts with larger screens to provide a more premium user experience. Also that the screen orientation could be vertical instead of horizontal to make the product appear taller and bigger. Another discussed topic was that many users could be used to the vertical orientation of smart phones which could be one way to align with common user screen habits.

Another individual analysis based on the six competitors is that the screen size could be dictated by what type of items are being sold. For instance, in the Propos solution, they have a screen kiosk next to the placement area, which could be because they are showing how to sell plated food. Menus might need a larger screen for the user to better navigate and more clearly see the options. In the Pan Oston solution, the placed items have a low-to-the-ground volume, which does not pose a risk of covering a large screen compared to what taller items could. As this thesis aimed to develop a solution for an assortment with tall-volume items, Items covering the screen needs to be considered in the design of the screen size and placement. A smaller display (such as Mashgin's 10 x 29 cm format) could be a more practical alternative for this project depending on the screen placement. But linking this to the thematical analysis, modularity of the POS system seemed to be a desirable feature amongst the interviewed participants. Therefore, it could be reasonable to design a screen placement that could allow both for a large and small screen.

Barcode scanners

Both Mashgin and Supersmart includes a secondary barcode scanner as a backup in case the CV system fails. Even with high recognition accuracy, this backup technology is a safeguard against system failures. With Mashgin, users can manually scan or remove items themselves, which could provide a sense of autonomy and resolves the friction of waiting for staff assistance. Conversely, competitors without a backup scanner likely risk disrupting the customer journey when cameras fail to recognize an item. While a system could ask users to reorient or separate unrecognized items through a UI, it still might not be enough to recognize an Item if the system has not learned a specific item or an updated package. During field testing of Mashgin's smart checkout, the UI asks to scan items manually that have not been learned. Implementing a backup barcode reader in a new design could resolve that friction point.

Item placement areas

The size and layout of the item placement area vary among the analyzed smart checkouts. While the majority of competitors do not feature a coordinate system on their placement surfaces, Mashgin is the unique exception. According to the interviewed ML and computer vision engineer at ITAB, integrating a coordinate system is intended to enhance recognition accuracy. However, since this thesis focuses on large items such as popcorn, snacks and drinks that would likely obstruct the markings anyway, such a feature could easily become an obtrusive design element. Consequently, the decision was made to exclude a coordinate system from the ideation of concepts development phase, leaving it as a potential area of interest for future iteration. At this point of the design stage, it is difficult to decide whether it would be necessary to have a coordinate system to avoid friction in the recognition accuracy, that would have to be tested in the future.

4.1.2 ART-series brand analysis

ITAB RT develops the global technology portfolio and brand identity using its own design principles and design language. A segment of their brand called the Architectural Retail Technology (ART) series, which aims to express a cohesive ecosystem of solutions that clearly belongs together. Moreover, the design-language aims to be self-explanatory and uncomplicated with a clear purpose of all elements. The design also aims to be long-lasting since their products are an investment for their clients. Therefore, the design elements should be robust, elegant and minimal to let the retailer brand speak rather than product design have an obtrusive and expressive design language.

The product design within the ART series should have clean surfaces without screws if possible. Surface edges should have soft radiuses or chamfers while it also can be integrated with an LED curvature. See an Illustration of how ITAB RT have integrated a clean sheet metal surface with a chamfer and soft radius in the SIGMA gate 2 in Figure 34.**Figure 1**

Figure 34: Shows an integration of ART series design elements in Sigma gate 2



They also have a radius principle for surface transitions, which also can be seen in the sigma gate very clearly. The rounds should be smooth and clean. See an illustration in Figure 35.

Figure 35: Shows a round edge between a surface transition



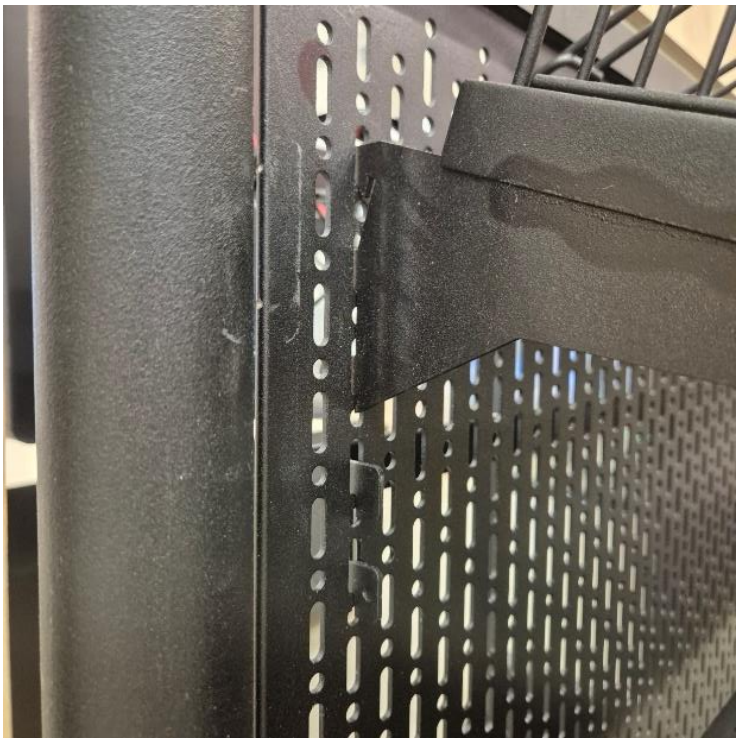
When it comes to patterns, they should serve a clear purpose while being unobtrusive to the retail environment. The purpose of holes could be to ventilate electronics or to attach shelves that provide a design element for upselling opportunities. ITAB's next-generation SCOs feature small, round ventilation holes to cool the electronics while preventing small objects from penetrating the chassis. See an illustration of the hole pattern in Figure 36.

Figure 36: Shows ventilation holes for a computer behind a screen



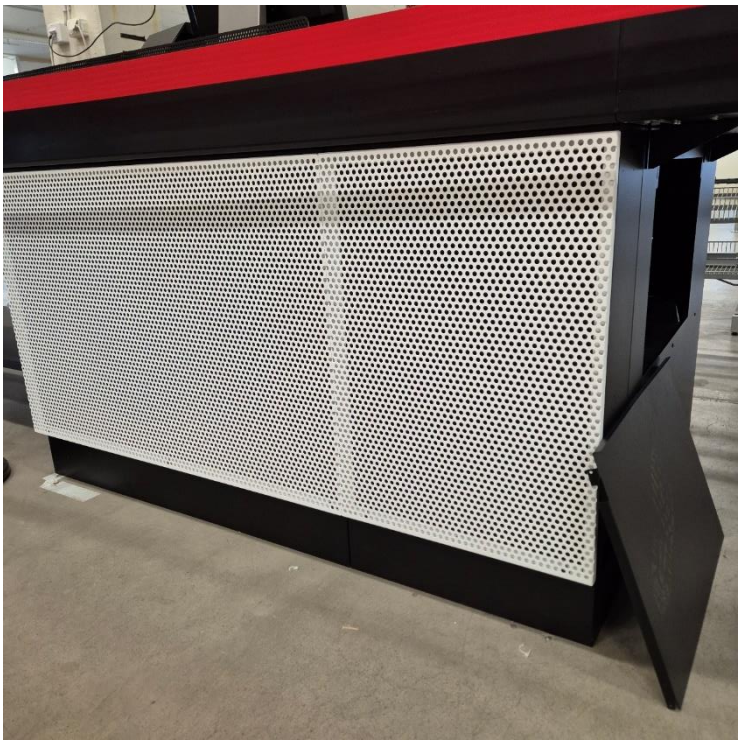
In ITAB's new barrier system, they have a another hole pattern to allow for attaching shelves that provide upselling opportunities. See an illustration in Figure 37.

Figure 37: Illustrates a pattern used to arrange hanging shelves



Materials and colors should follow a “less is more” principle to be as clean and unobtrusive for the retail environment as possible. The colors should also be possible for retailers to customize to provide the desired consumer brand experience. An example of a modular design element that can contribute to the desired consumer brand experience is the front of ITAB’s new conventional checkout, CM700. It can be changed in different colors and patterns. See an illustration in Figure 38. (E.Cannert, Personal communication April 20, 2026)

Figure 38: Shows a modular front panel of ITAB’s checkout CM700



4.2 Problem definition

This chapter reveals the work that has been conducted during the second part of the double diamond methodology.

4.2.1 Problem focus

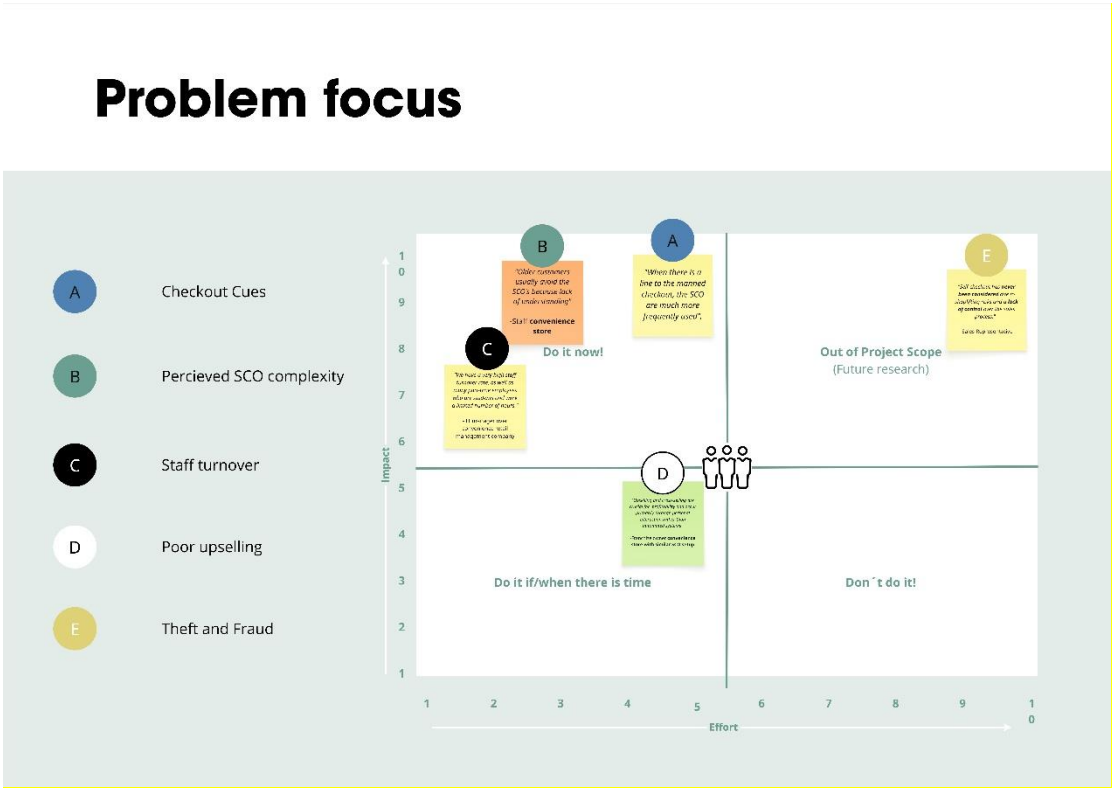
To conclude the thematic analysis, a problem focus matrix was created, which consists of four different areas: *Do it now*, *Do it if/when there is time*, *Don't do it*, and *Out of project scope*. The matrix derives from a screening framework used at ITAB (E.Cannert, Personal communication May 11, 2026). The main problem that this project focuses on is checkout queues. As highlighted in the first theme, 'time-sensitive environments', and the second main theme, 'customer habits and SCO placement,' many problems seems to lead to checkout queues. When many customers want to buy items simultaneously, queues appear, which can generate stress among both customers and workers. Ideally, more staff would be available to open another checkout, but that is not always an option. Employees are not always available, and there are days when stores are short-staffed, yet the same number of customers want to buy their items. Some stores do not have the space to install more conventional checkouts.

Some users that were interacted with perceived SCOs as undesirable to use because they prefer the manned checkouts. According to observations, some people avoided the SCO stations or stopped using them midway through the process. As highlighted by ..... it could be due to perceived complexity of the SCOs. During interactions with checkout staff, some customers avoid SCOs because their appearance is confusing or difficult to navigate at first glance.

High staff turnover means retail employees are constantly leaving and being replaced. Considering this rate is important for self-checkout SCO design for a few key reasons. It could reduce the need for staff training as cashiers quit frequently, leaving stores with new, inexperienced staff. If the smart checkout gets designed highly intuitive, new employees won't need extensive training to manage it or help stuck customers. High turnover leads to sudden labor shortages. A well-designed, smart checkout could handle crowds and process payments smoothly even on days when very few employees are working.

As previously noted, poor upselling capability was identified as a key limitation of current self-service terminals during the field visits. Developing a solution for this issue would be highly interesting, particularly for retailers who rely heavily on impulse purchases, including those in the cinema industry. While upselling is not a top priority for this thesis, developing a modular solution within the MVP framework remains a secondary objective if time permits. However, because a MVP inherently focuses on core functionality rather than modularity, this feature remains secondary to the primary design goals. Similarly, as established in the delimitations, developing solutions for theft and fraud falls entirely outside the scope of this project. The prioritization of these various design challenges is illustrated in the Problem Focus Matrix in Figure 39.

Figure 39: Problem focus matrix categorized by four levels of importance



4.2.2 Cinema store context

During the interview with the sales manager from a large cinema company, it was mentioned that they have never installed any SCOs. However, they are looking into options to enable some kind of self-service solution in the future. It was also discussed that the cinema industry is an entertainment business, focusing on the user experience much more than a convenience store. Therefore, it could be more appealing for this industry to implement a smart checkout from the aspect of providing a better user experience. Despite the fact that cinemas traditionally have been more keen on theatre technology, the manager noted that they do not exclude new technology options in the checkout lobby. The interview also highlighted that these locations can be time-sensitive environments; however, since the pandemic, they have experienced significantly shorter queues.

Staff roles were discussed during both during the interview with the sales manager and during interactions with staff working at the cinemas. Both the employees and the manager mentioned that since the industry is a service-oriented business, it relies heavily on human interaction to provide a good customer experience. Autonomous stores were discussed in contrast to cinemas, and the manager emphasized that technology should not replace human interaction in their business. Instead, it could support and supplement it when there are staff shortages or when customer demand is high. The tasks that staff perform inside the cinema were also discussed. They prepare snacks, stock shelves, scan tickets, welcome customers, start and introduce movies, and clean the auditoriums, to mention just a few of their responsibilities. Between movie sessions, staff were observed stocking shelves and standing by dynamically to assist customers arriving for the next show. They emphasized that their customers are their first priority and that the other tasks are secondary. This highlights that there are other tasks they need to do besides just checking out customers, which indicates that a smart checkout solution could support staff when they need to stock shelves as well. See the interview summary in ANNEX B.

When defining the proportions of a smart checkout design, item sizes are important to consider. Therefore, some of the largest and most frequently sold items were measured based on feedback from the store staff. According to personnel, medium popcorn and medium paper cup drinks are the most popular items, while 'Millions' candy pipes and double snack boxes are also commonly sold. Illustrations of these items and their measurements can be seen in Figure 40.

Figure 40: shows some of the largest items observed at a cinema store in Jönköping.



Large popcorn: 18 x 18 x 22 cm



Large paper can drink: ∅8 cm



Millions candy pipe: 30 cm



The double snack box: 10 x 18 x 22 cm

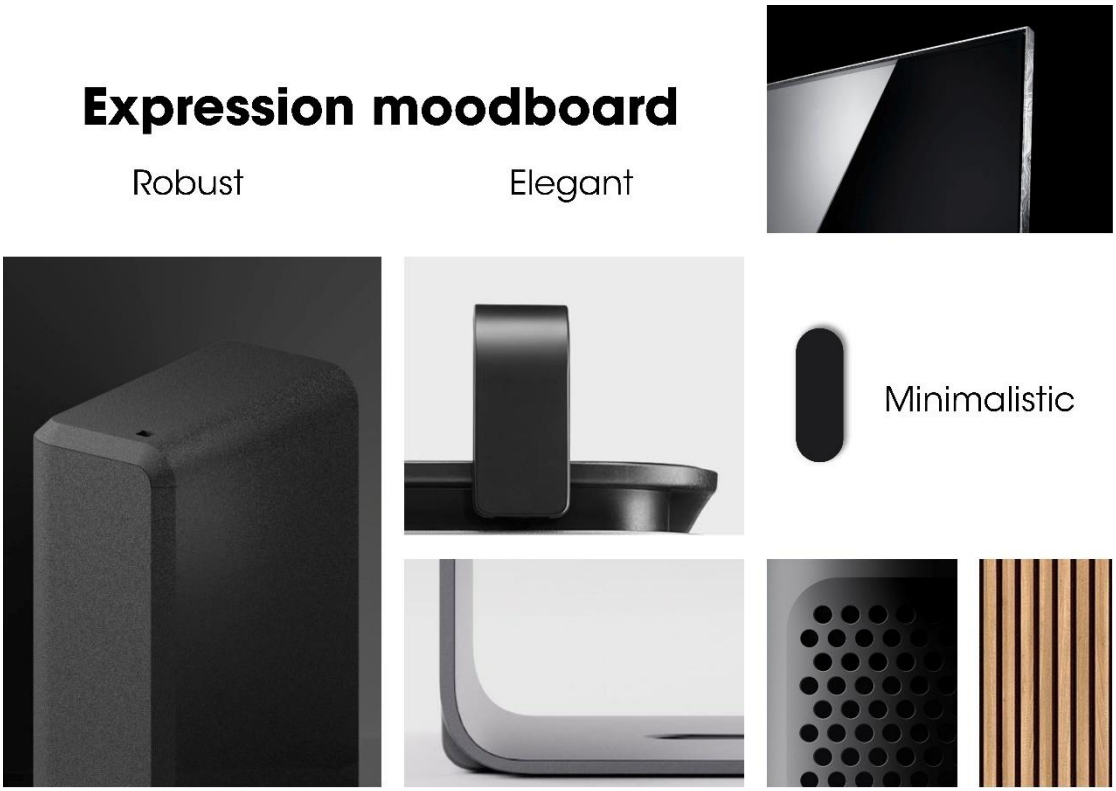
4.3 Solution discovery

This part shows the solution discovery, which is the first part of the second diamond in the double diamond framework.

4.3.1 Mood board

The brand analysis established three key attributes that define the brand's identity: robust, elegant, and minimalistic. These keywords guide the form development of the smart checkout's physical components. Additionally, reference images are integrated alongside these terms to in a mood board to clearly communicate the product concept's design direction. The word “robust” aims to be be reflected by large clean rectangular sheet metal shapes. The word” elegant” aims to be reflected by the smooth and round surface transitions, chamfer integrations for surface transitions and by the reflections from a UI screen. The word “minimalistic” aims to be reflected by neutral and clean shapes, also by the round patterns and wood panels. Even if the patterns are not integrated in the product, it should match them. See an illustration of the mood board in Figure 41.

Figure 41: Mood board illustrating the inspiration for the intended design direction.



4.3.2 How might we

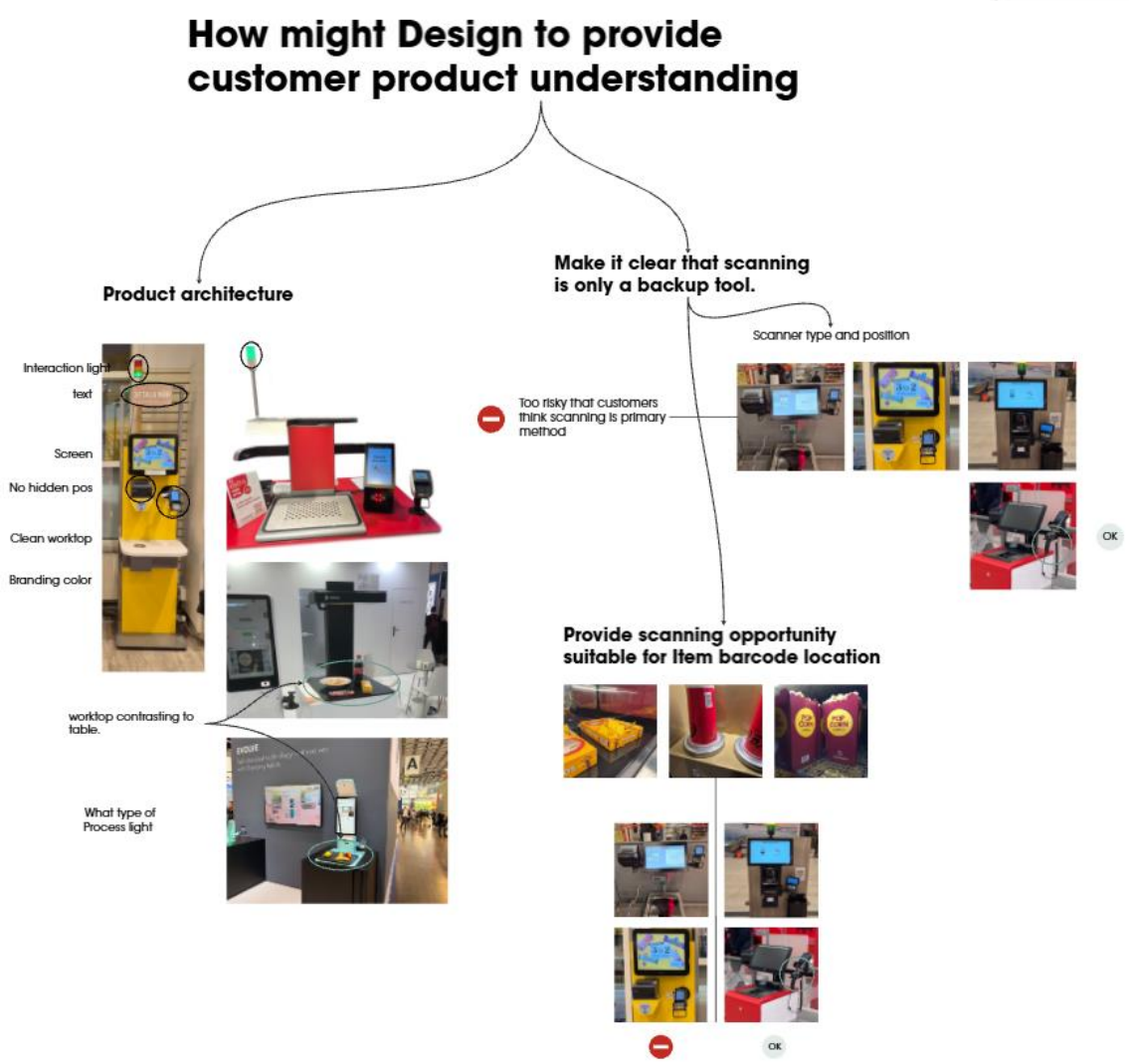
Following the definition of the problem focus, several HMW questions were generated through individual brainstorming and partly in collaboration with the extended project team. The questions served as a foundation for brainstorming possible solutions. For each primary question, the objective was to either propose direct solutions or uncover deeper sub-questions. This methodology was visualized using a hierarchical tree structure, where the initial problem statement branched out into subsequent answers or new inquiries. The questions were inserted to Miro, a digital collaborative whiteboard platform. The first phrase was formulated in the following way:

How might we design to provide customer product understanding?

Providing a suitable product architecture based on the design's purpose was a major focus during ideation for the first question. Based on the field visits, different components were evaluated to see which ones could best show the users the product's intended purpose. For example, most observed SCOs utilized an interaction light and a central screen placement, which could be great opportunities to integrate because customers probably already associate them with an SCO. Having a clear dedicated item placement area was also an idea, since observations showed that when SCOs lacked a defined placement area, they seemed to be significantly less used.

Another key decision was to signal that manual barcode scanning is only a backup tool. Centering the scanner was avoided because it might risk making users think they should use it as a primary solution. Furthermore, since the MVP is targeted for cinema stores, an upward-facing flat-bed scanner embedded in the table was excluded because it would cause items like open popcorn containers to tilt and spill during scanning. Based on these constraints, it was decided that a handheld scanner or a user-facing fixed scanner is the most suitable solution. See the brainstorming for the first "How Might We" question in Figure 42.

Figure 42: Shows the brainstorming from the first “how might we” question.

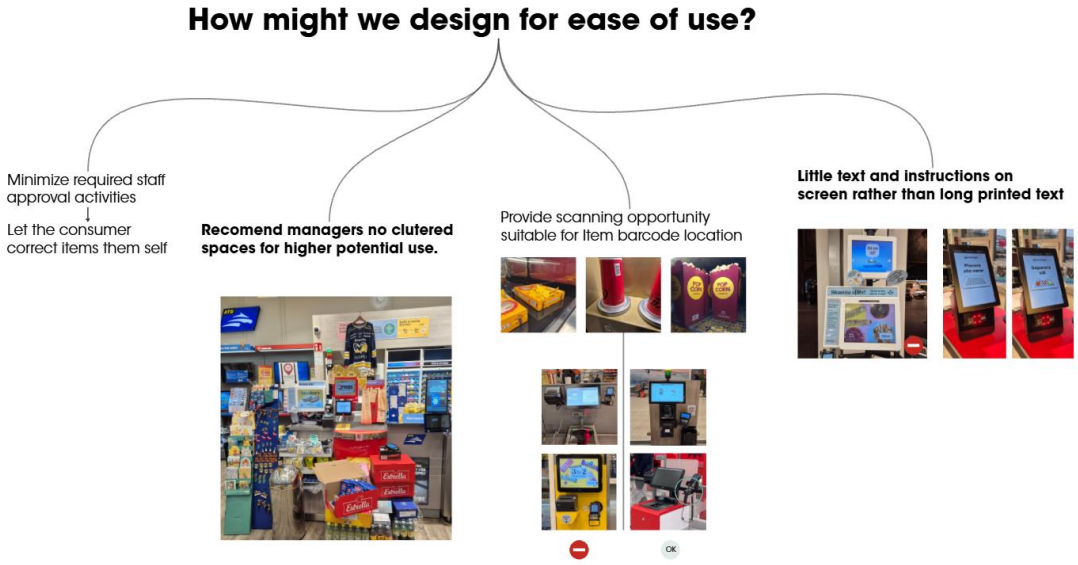


Another formulated HMW question was:

How might we design for ease of use?

Key highlights from this ideation related to this question focused on minimizing staff interventions by allowing consumers to correct scanning errors themselves. The self-correction feature observed during testing with the Mashgin smart could potentially reduce user friction compared to asking for staff assistance. During observations, it was also noted that minimal on-screen instructions seemed appealing than dense blocks of text, a concept that brought to the ideation of user interface wireframes. To improve SCO visibility in cluttered environments, store designers and management could be suggested to maintain clear layouts around checkouts to increase customer usage. See the mapping of ideas for this question in Figure 43.

Figure 43: Shows the brainstorming from the second "how might we" question.

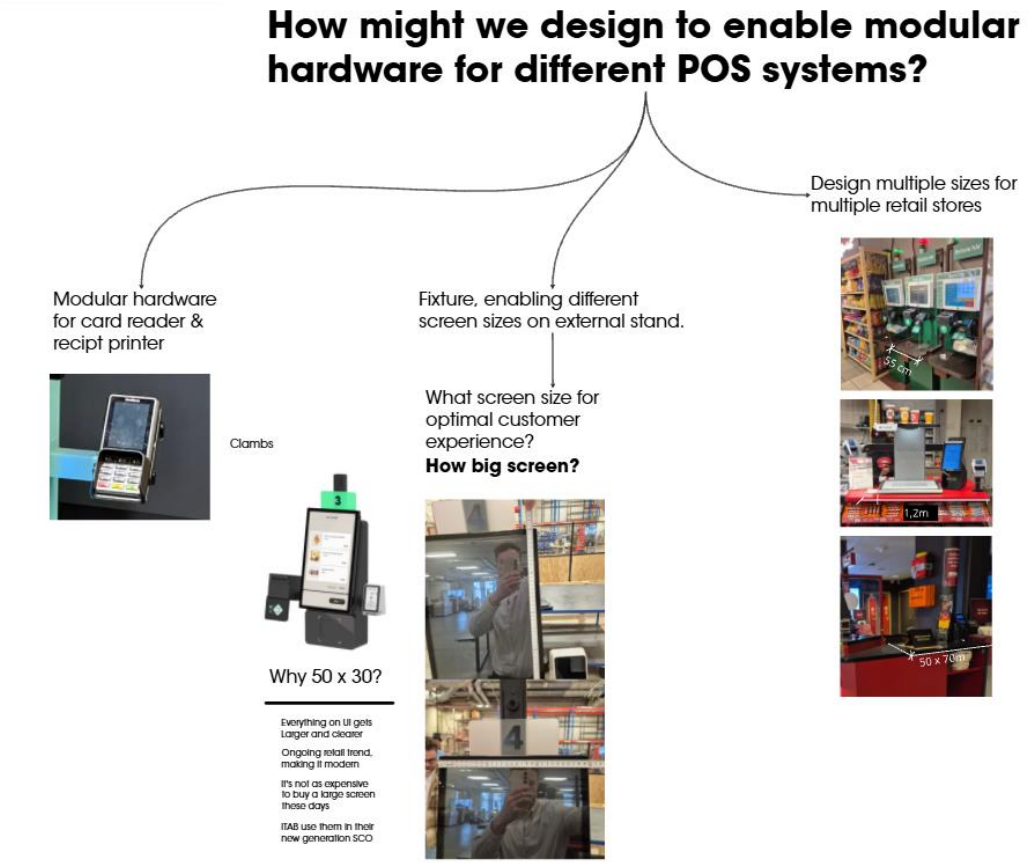


A third HMW question was formulated as follows:

How might we design to enable modular hardware for different POS systems?

Utilizing an unintegrated, modular hardware architecture could be an effective way to offer retailers maximum flexibility when selecting checkout components. While some competitors utilize fully or semi-integrated EFT terminals and printers, keeping the hardware modular provides a clear opportunity for market differentiation. During discussions with the extended project team, it was noted that some retailers seem to want their own specific screen type, which might mean integrating a fixed screen is not the best alternative. Furthermore, as highlighted during the interview with the CIO over the large convenience store management company, excessive size was a primary reason they avoided smart checkouts. Therefore, designing a scalable system that accommodates different component sizes without requiring major redesigns represents an opportunity for future product market scaling. See an illustration of the related HMW question in Figure 44.

Figure 44: Shows the brainstorming from the third “how might we” question.



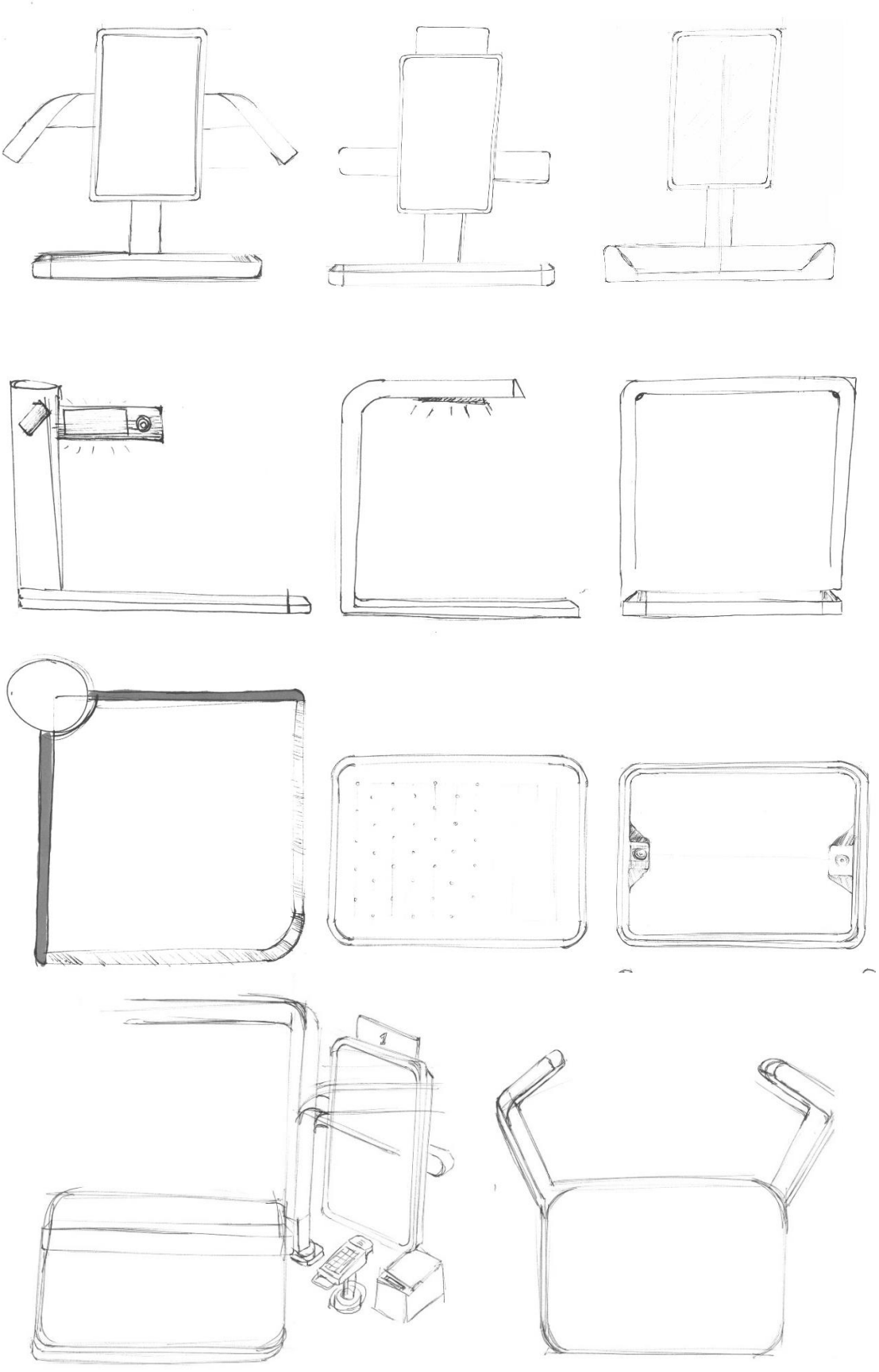
4.3.3 Concept generation

The initial stage of the concept phase focused on exploring the placement of various components to evaluate how their positioning affects user perception. It was also crucial to evaluate the physical dimensions of the potential components to ensure seamless integration within the concept designs. Therefore, the first step was to generate physical models that reflect the intended geometry of components and their purposes. Since the brand analysis indicated a design language of elegance, robustness, and simplicity, cardboard was selected as the primary mock-up material due to its suitability for creating this type of geometry. Utilizing available components from ITAB, such as screens and scanners, allowed for a more efficient and accelerated mock-up process.

4.3.4 Sketches

Sketching was an initial ideation method used to generate potential shapes and configurations for the smart checkout. A central screen layout was explored to focus the user's attention on the middle of the product, which aligns with standard SCO designs. Incorporating an interaction light on top of the screen was also investigated. Based on the competitor analysis, the early concepts featured a structure housing the cameras connected directly to the screen. Integrating a camera into the base was also considered to achieve a more concealed look; however, it was noted that this placement would likely obstruct the camera's view of the items. Additionally, different lighting configurations were evaluated for both the baseplate and the interaction light. One concept placed the interaction light within the baseplate, but this was excluded due to poor visibility, particularly for store staff. Things excluded in the initial sketching was the focus of the scanner device, as it was deemed for secondary component, it could be explored later. The shapes are inspired from the mood board. See the initial sketching process in Figure 45.

Figure 45: Shows the initial ideation sketching.



4.3.5 Mockups

The mockup material used were cardboard and glue as it was considered easy to shape for the forms of the ART-series. To begin with, the base components that was aimed to be created were a camera component, a placement area and a screen, as these serve as foundational pieces in a smart checkout design according to the competitor product analysis. As the camera placement is a central part of the idea generation, it was logical to start with that component. It was created as a rectangular cuboid with rounded edges to simulate the ART series design language. The screen sizes that was simulated were one larger of 300 x 500 mm and one of 180 x 290 mm to enable comparison between different screens together with different mockup setups. Additionally, an item placement area was created based on that should fit less than 10 items in a cinema store. The items used to determine the size were four medium-sized popcorns, four medium-sized drinks, and two “millions candy pipes”. Therefore, the placement area was decided to be 520 x 580 mm.

From the try-it-yourself session with the Mashgin checkout, several measurements were taken as a reference for the camera container. Additionally, the ML and CV engineers suggested that a camera size of 30 x30 x 40 mm could be suitable for a smart checkout solution if ITAB would continue to develop a smart checkout today, which became the baseline size for building the mockups. See Figure 46.

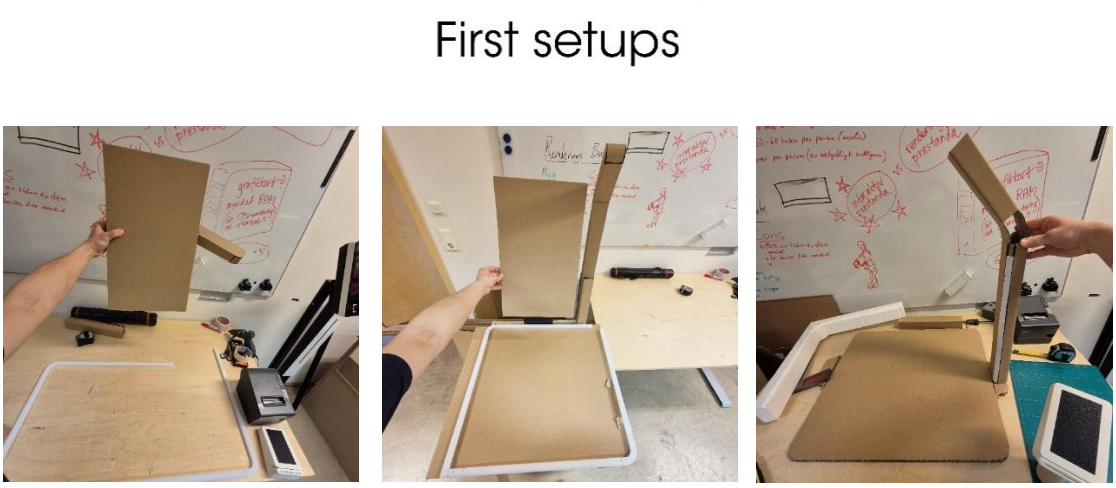
Figure 46: Shows the camera used as reference for the design.



According to the EAA, the maximum depth for an obstructed side reach is 610 mm to an ECT, which means that the placement area or the screen is limited to maximum that depth. The screen must not be any taller than 1120 mm from ground level according to the regulations from EAA regarding obstructed forward reach. For a clearer illustration See Figure 5.

The proportions of the first mockup setup, featuring a central screen layout, were evaluated against the specified EAA parameters. In this setup there were cloudy acrylic plastic shapes representing the borders of the intended item placement area. The idea was to try and integrate the central screen placement with the rectangular cuboid to provide some unified design stationed around the screen. These components were the baseline for the initial mockups. See illustrations of the first setups in Figure 47.

Figure 47: Shows the first setups of the mockups.



Takeaways from the initial setups indicated that a central screen placement would obstruct users trying to place large or tall items, as the images taken represents the maximum height of a large screen with a table height of 850 mm. A popcorn container of 220 mm tall would prevent the users from integrating with the UI, making a central screen position impractical.

The second setup of mockups feature new designs testing other camera and screen positions. One idea was to have two towers in the back that are placed at the edges of the placement area, slightly angled towards the center of the placement area to provide a to provide a cross-sectional camera view. Two similar shapes for camera towers were constructed. The interface screen, which was previously only simulated with cardboard, was replaced with a physical monitor of the same dimensions to provide a more realistic context during mockup evaluation. Initially, an alternative centralized position for the screen was tested: instead of being situated behind the product placement area, it was placed in front of it. However, the large screen size felt impractical for a frontal layout, presenting a risk of beverage spills and creating a physical obstruction that hindered users from placing items on the scanning surface. Furthermore, this placement would be unacceptable under EAA regulations. While this configuration could have been further evaluated using a smaller screen, it was decided to explore alternative positions using the newly acquired monitor instead. This setup was labeled concept 1. See Figure 48.

While the first concept might have a central aim toward the placement area, it probably would face difficulties recognizing items in the front. Especially if there were multiple large items near the back, blocking items in the front. Since it did not feature any illumination of the placement area, there were other factors that would have to be considered as well for next steps. To capture a better camera angle, another concept was created with two diagonal rectangular cuboids to provide vision of the front part of the placement area. This could possibly provide an easier recognition of items placed in the front compared to the previous concept. See an illustration in Figure 49.

While Concept 2 did not include any illumination, Concept 3 integrated lighting aimed down toward the item placement area from a high central position. This concept was thought to contain a centrally placed camera in the same fixture as well. However, this concept would utilize only a single camera, which likely would reduce its recognition accuracy compared to the previous design. While central camera placements were observed in some competitor smart checkouts, relying on a single camera angle poses a challenge for product recognition, as taller items like large popcorn boxes could easily obstruct smaller objects from view. Inspired by other SCO designs from ITAB RT, it featured an interaction light on top of the structure. See Figure 50.

An arch design was created with focus on including a structure to provide illumination for the placement area. The idea was to have a beam that crosses the item placement area and provides illumination to the items, while having cameras on each side and one in the middle. The mock-up design featured 2 different corners to quicker test different variations of the same concept.

Implemented in the arch were also a chamfer with an intended interaction light. As mentioned by the ML and CV engineer, the more cameras, the higher chance for higher recognition accuracy. Therefore, this concept featured three cameras, which was the highest amount yest ideated. Testing the concept with both a larger and a smaller screen were tested. The smaller one seemed to look best with the design based on proportions, as the large one felt bulky together with the rest of the design. With this setup, different positions of the receipt printer and EFT were also evaluated. See Figure 51.

With a similar setup as concept 1, a fifth concept was created featuring a light connecting the two towers, providing space for a third centrally placed camera to potentially increase the recognition accuracy. Inside the beam, could also feature illumination of the item placement area. However as noticed during concept one, this could still struggle with item recognition in the front, meaning that it would have to be further developed to enable that. See Figure 52

Figure 48: Concept mockup 1

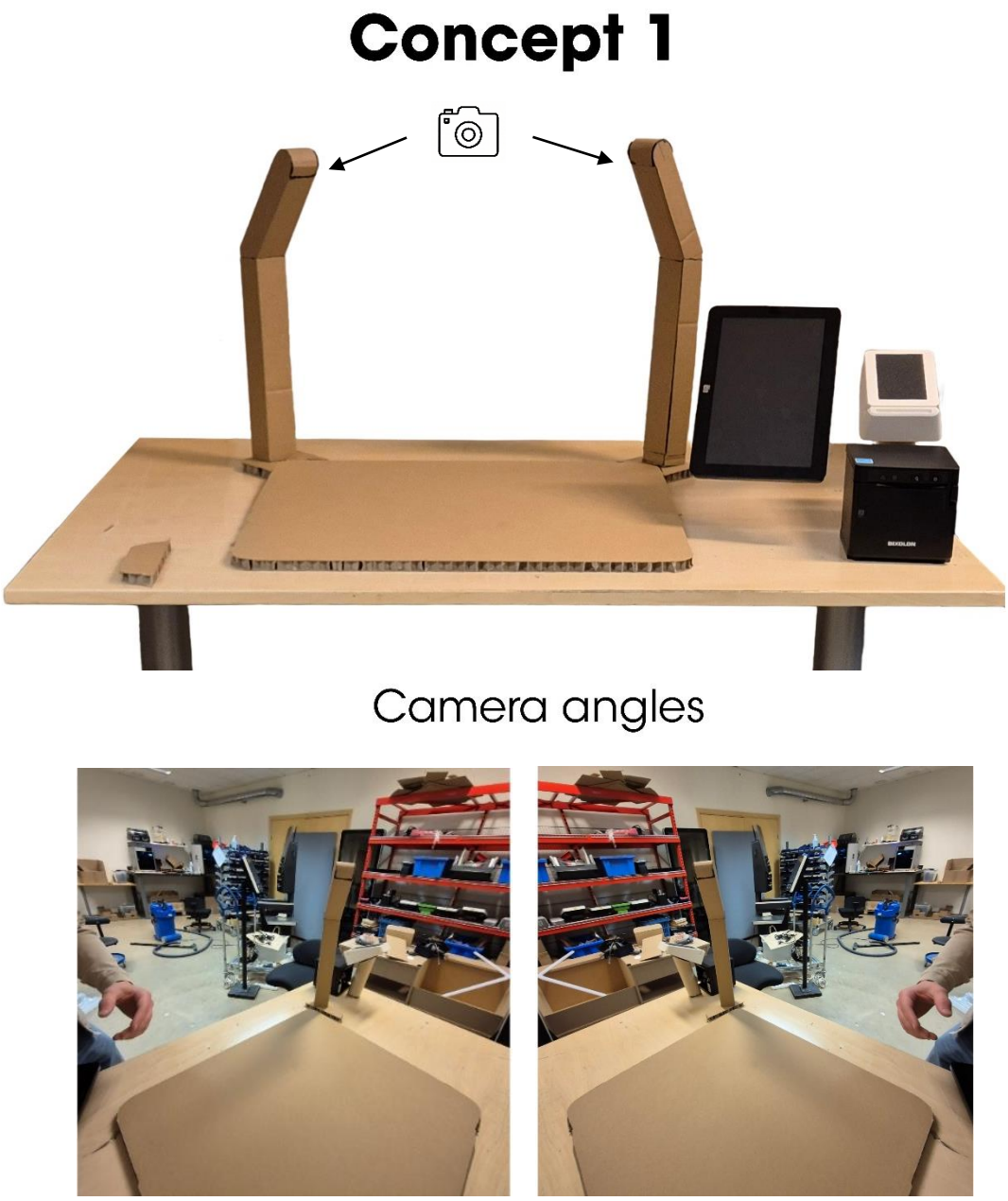


Figure 49: Concept mockup 2

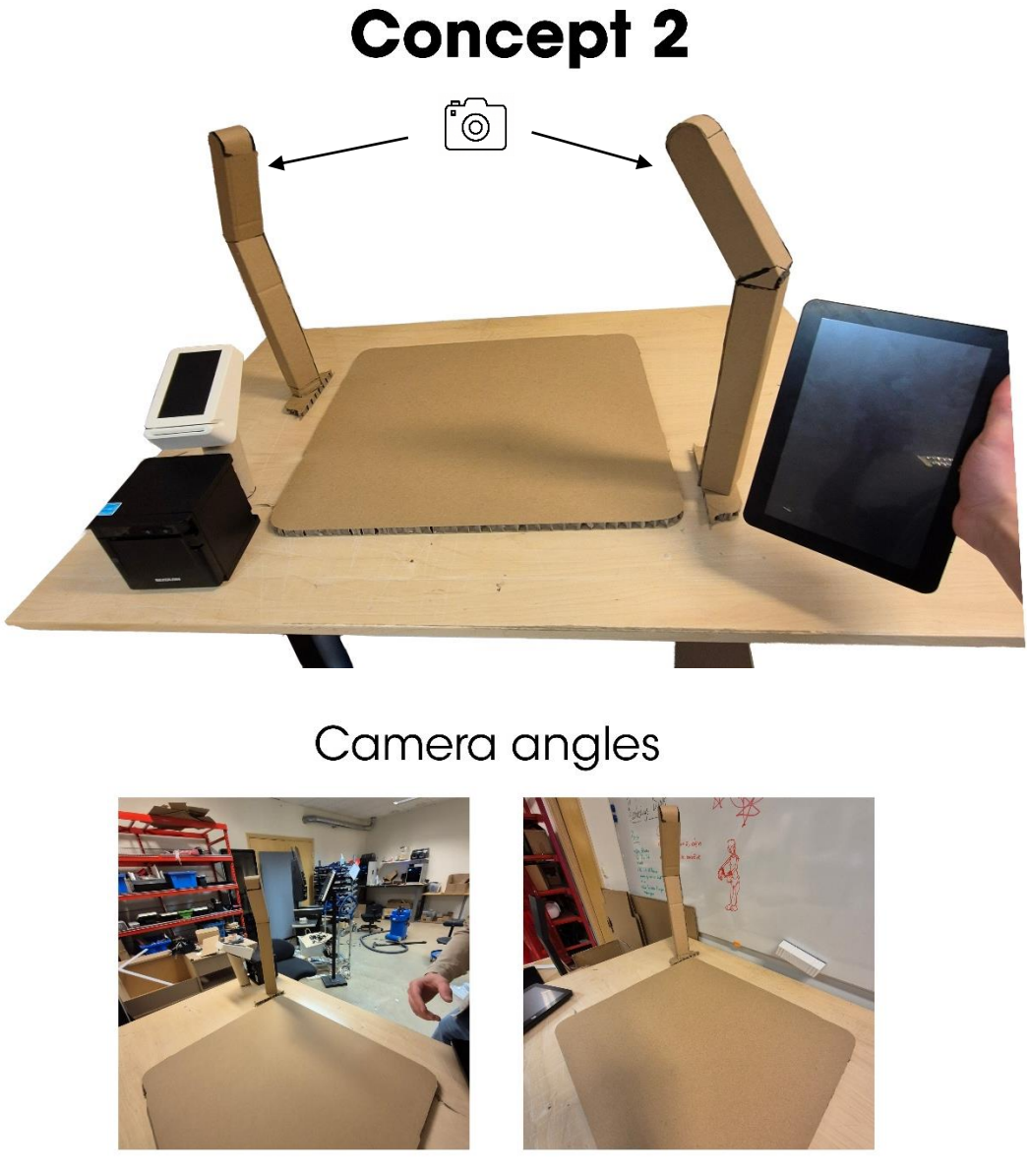


Figure 50: Concept mockup 3

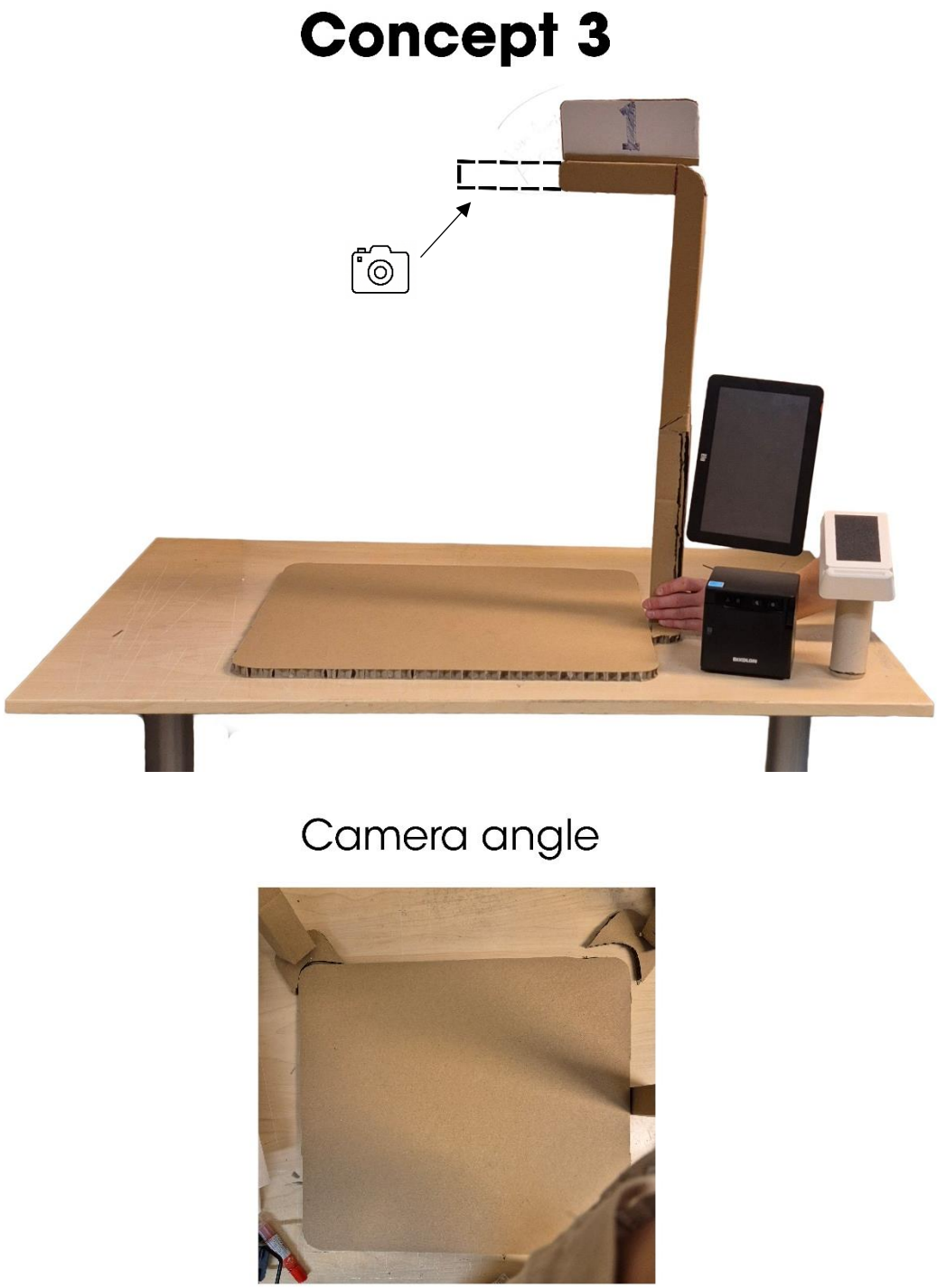


Figure 51: Concept mockup 4

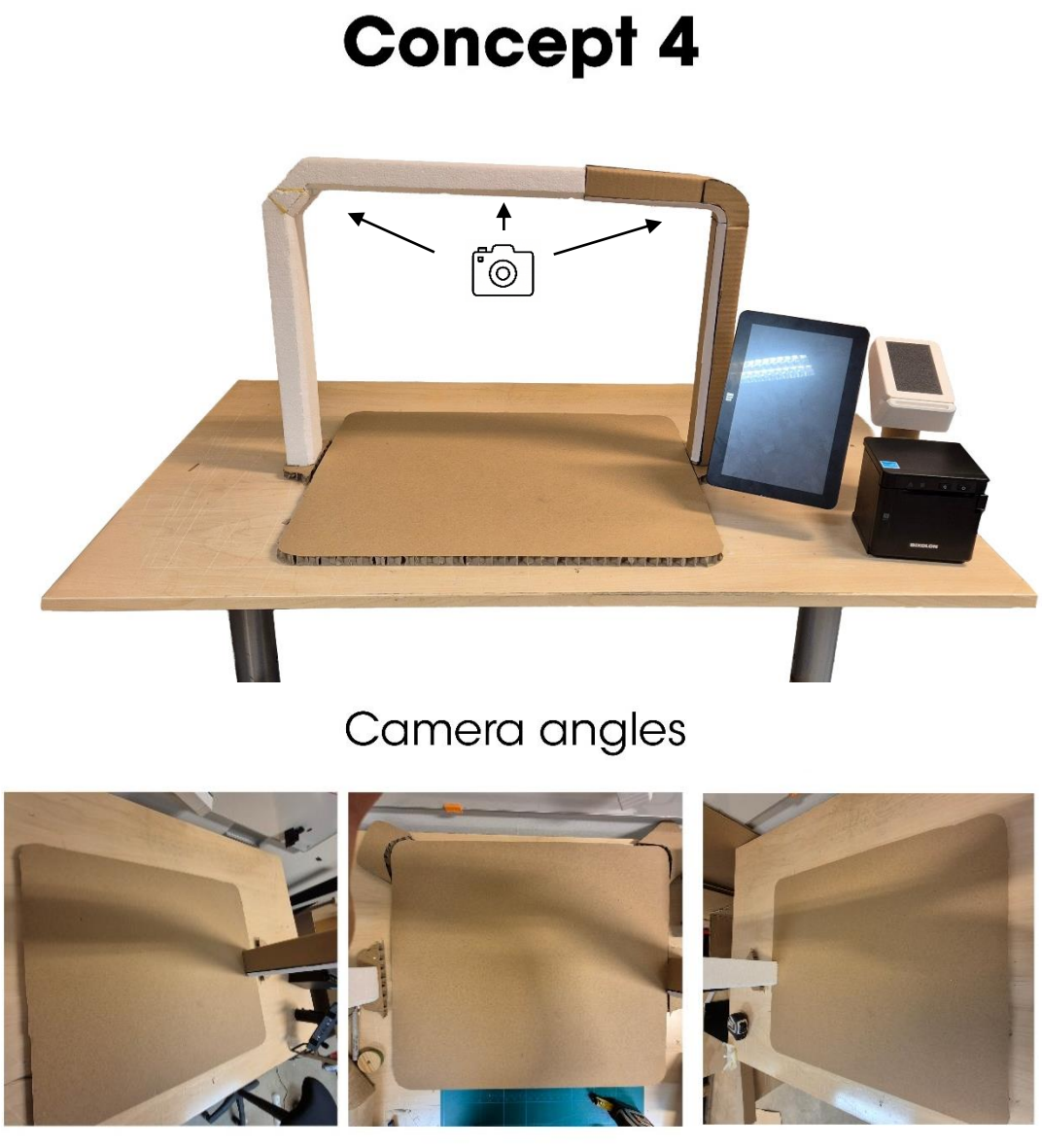
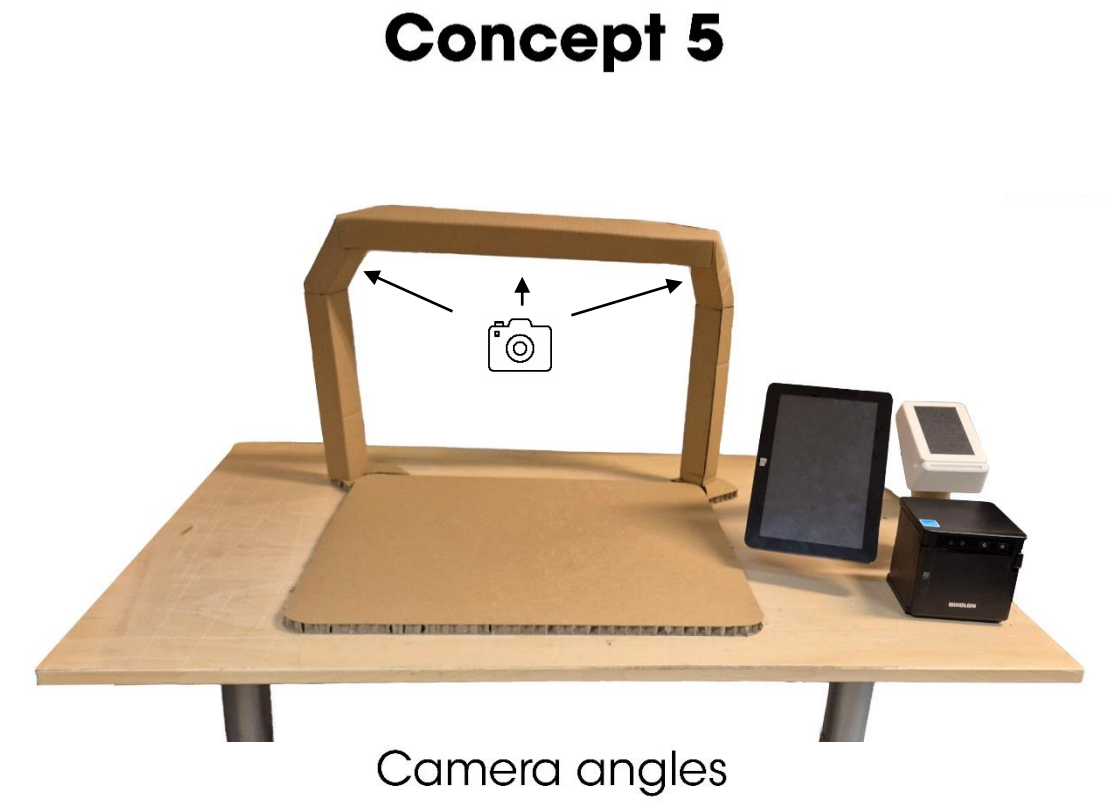


Figure 52: Concept mockup 5



Evaluated by the extended project team, these concepts were discussed from different aspects such as camera placement, camera count and possibilities for illumination. The most promising concepts that were chosen for further development were concept two and concept 4 because of their camera angles. Concept 2 would arguably have to be improved a lot more if it would be able to provide illumination and an interaction light, but it might have had an advantage in it's front camera placement compared to concept 4. Concept 4 utilized three cameras, providing illumination in its integrated arch design. Having 3 cameras might provide higher recognition accuracy than having 2 cameras. The design shape of the arch was mostly aligned with the ART series design language of the two concepts, as the two towers in concept two appeared a bit disconnected from one another, and possibly obstructing the user.

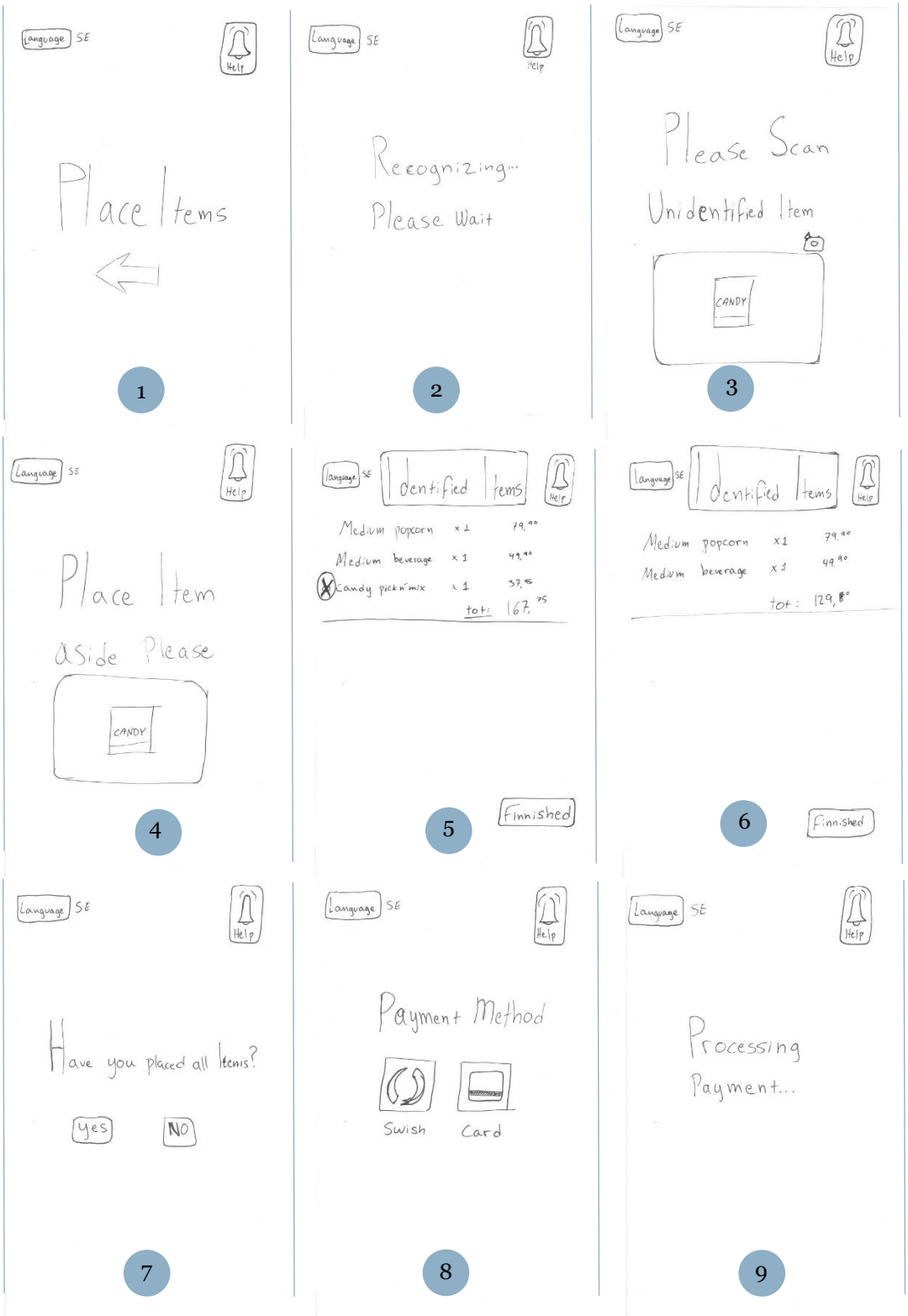
Before further developing any of the concepts, a user-based evaluation was conducted to understand how effectively each design communicated its intended purpose. Because the final project goal is an interactive model featuring a user interface, paper wireframes were created to simulate the digital experience along the physical model experience. The idea to test was two potential user scenarios. In the first scenario, users were asked to simulate purchasing a medium-sized popcorn and a medium beverage. In the second scenario, the participant was asked to process a medium popcorn, a medium beverage, and a bag of pick-and-mix candy.

After registering the candy, the user was instructed to change their mind and attempt to remove the item from the receipt. Based on insights from the try-it yourself session with the Mashgin smart checkout, and the ideation from the HMW method, The wireframes were inspired. See the wireframes for the scenarios are presented in Figure 53 and Figure 54.

Figure 53: Wireframe sequence for Scenario 1.



Figure 54: Wireframe sequence for Scenario 2.



4.3.6 Concept testing

To further evaluate the two concepts, 10 participants who were unfamiliar with the project were asked to perform two tasks. The user group consisted of six men and four women who were internal ITAB employees. While their employment at ITAB may have slightly influenced the evaluation due to their familiarity with the company's product line, selecting them was highly convenient. Furthermore, they still provided an objective perspective on the concepts since they were not involved in this specific project.

The concept testing took place at the ITAB office using physical mockups, supplemented by unintegrated electronic devices managed by the test conductor. A portion of the testing involved paper prototypes—sketched wireframes held by the facilitator to simulate a live screen for the two user tasks. These interactions were documented via audio recordings and digital field notes taken immediately after each session. Finally, the audio data was transcribed using Klang.ai, and the transcripts were cross-referenced and analyzed alongside the field notes.

The tasks included the participants to imagine they were at a cinema buying popcorn and a drink. They were instructed to "think out loud," sharing their thoughts and impressions during their interaction with each setup. For the first task, participants completed a standard purchase of a medium-sized popcorn and a drink. For the second task, they were asked to buy the same items but add a bag of candy, which they then had to attempt to delete from their virtual receipt. After completing the tasks, a set of open questions were asked regarding what they perceived about the design. For example, they were asked, “what was the first thing you looked at when you arrived at the checkout?” See an illustration of the user evaluations in Figure 55.

Figure 55: Shows photos from user evaluations.



4.3.6.1 Takeaways

User testing indicated that Concept 4 felt unobtrusive to most participants when placed in the middle or back of the item placement area. However, when positioned at the front, users felt it invaded their personal space. This front placement also created the misconception that items had to be passed through the arch rather than simply placed down, which could cause confusion. In contrast, Concept 2 felt intrusive to personal space and was less visually appealing at first glance than Concept 4.

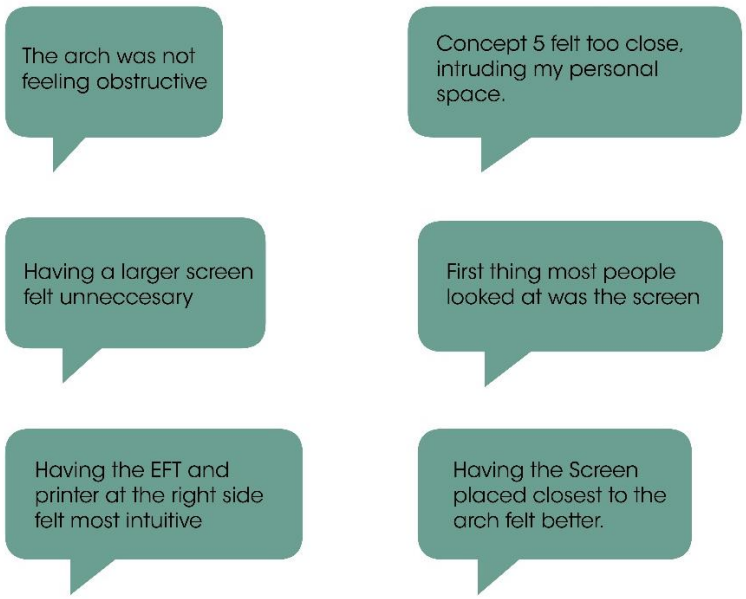
Regarding screen size, participants felt a smaller screen was more appropriate in relation to the overall design, noting that a larger screen seemed unnecessary. Interestingly, most users instinctively looked at the screen first rather than the placement area, despite the placement area being the largest component. This unexpected behavior suggests that the screen is the most effective location for user instructions and information.

Placing the EFT and printer on the right side felt the most intuitive; positioning either device on the left, especially the EFT, felt unnatural to users. Participants also preferred having the screen close to the placement area. Because the screen is the primary point of interaction before the EFT and printer, keeping it nearest to the placement area was noted as a key design priority.

The scanner type were also tested in two variants, one stationary in built within a screen, and on other handheld scanner. Most users perceived a hand scanner as 'old-school' and mismatched with the modernity of the rest of the solution; therefore, it was decided not to proceed with this option. This choice also eliminated cables from a hand scanner, which could otherwise introduce a risk of tilting open items.

The UI faced some problems in communicating its purpose to the users. Especially the "finish" button before making the transaction. Users were confused about what it meant and suggested changing it to "pay." Another suggestion was to add functionality to remove all items on the user interface, not just the candy that had to be added manually. A third suggestion was to not ask the user to remove the candy from the item placement area once the item was removed from the screen, since that system made customers confused. However, the rest of the interface appeared quite clear for the test users. See a figure of user citations from the user testing in Figure 56.

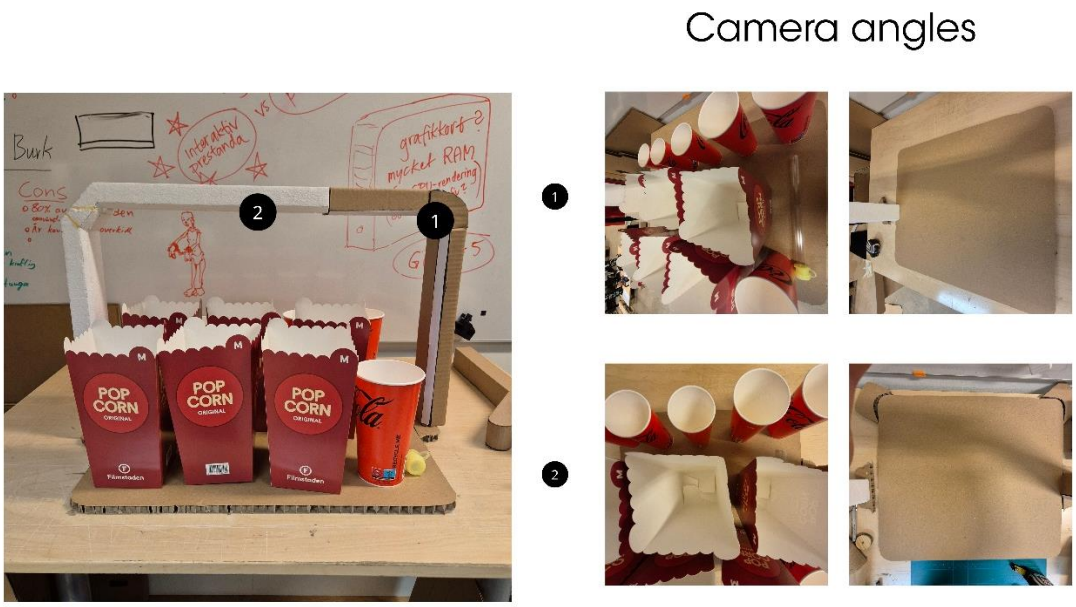
Figure 56: Citations from participants during user interaction.



4.4 Concept selection

Based on concept testing and discussions with the project team, Concept 4 was selected for further development. However, several critical design challenges still needed to be addressed. For instance, the original camera lens setup was insufficient, as the photo captures only half of some items within the placement area. See an illustration in Figure 57. While a 0.6 fisheye lens was initially used, achieving better product recognition across the entire placement area with this type of items requires a wider-angle (smaller focal length) fisheye lens.

Figure 57: Item placement area as captured through a 0.6 fisheye camera lens



Another area for future evaluation is how the arch connects to the main structure, as this project only explored attaching it directly to the baseplate. Other design alternatives might offer better aesthetics or easier installation.

The initial form development involved offsetting the arch pillars from the base and increasing the height of the arch to ensure the camera angle could capture all items within the scanning area. The shape was iterated with minor variations to explore different forms, design languages, and attachment points. Specifically, variations were tested with the arch mounted either on top of the placement area or to its sides. Both of these configurations allowed for internal cable routing. However, none of the generated concepts successfully reflected the "ART series" design language. Consequently, the team decided to retain the arch's original shape.

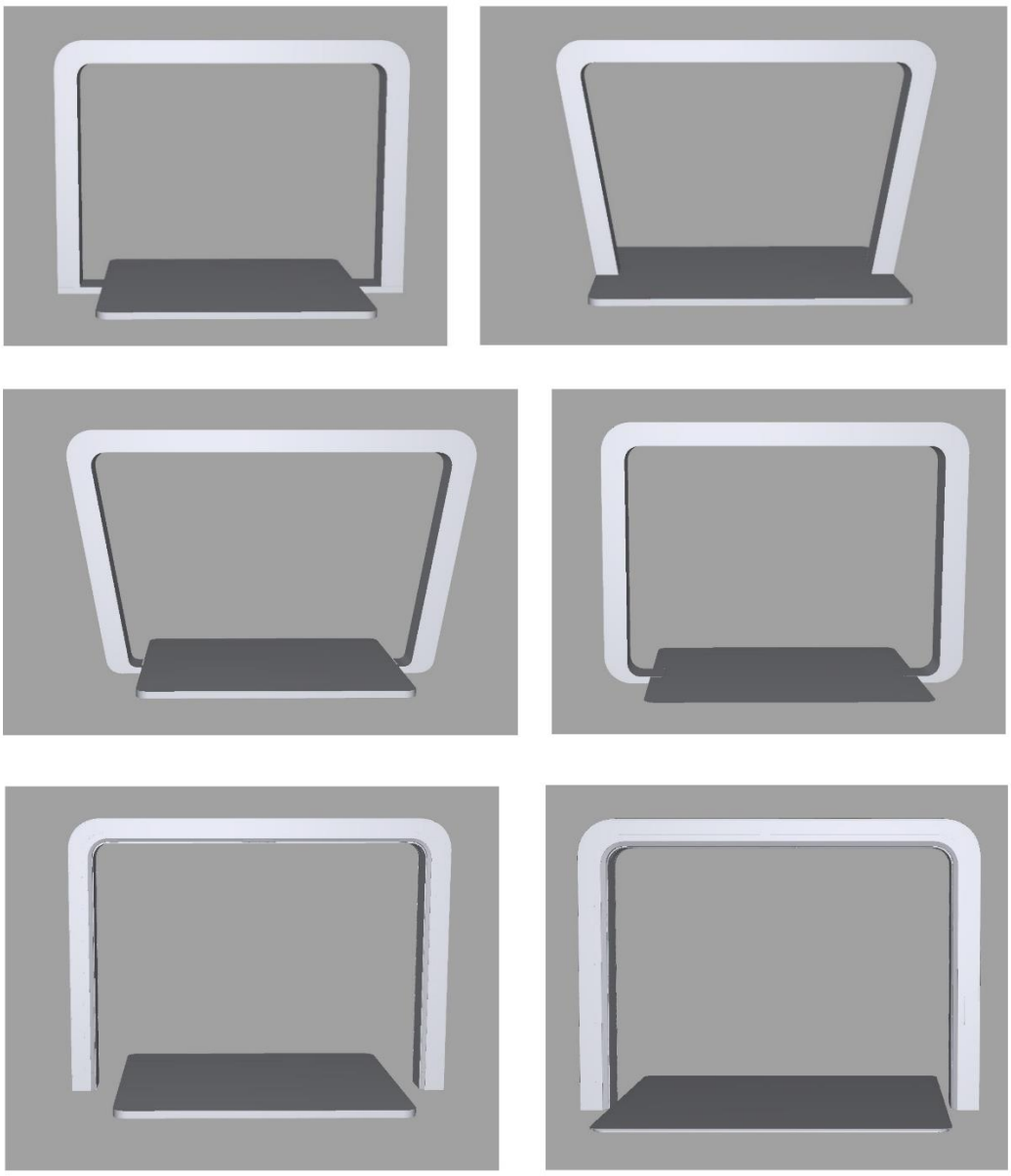
To give both the arch and the placement area a cleaner look, the connection point between them was eliminated. Instead, the arch was positioned on the side of the placement area, designed to mount straight onto the counter or table surface. This allowed cable management to be handled by routing wires through a hole drilled into the table. Because cables no longer needed to pass through the placement area, its component thickness could be significantly reduced.

While a scale could potentially be integrated into the design for weight based items, it was excluded to minimize production costs, aligning with the project's focus on developing an MVP. For the target context of cinema stores, pick-and-mix candy was the only item requiring weight measurement. Observations showed that these cinemas already utilize separate scales and barcode printers. According to the project's CV and ML ML engineer, a smart checkout system can process these items by using OCR-text to read the printed barcode labels.

The inward-pointing chamfer and interaction light were designed on the inside of the arch to give users a clear visual cue that the system recognized the items under the arch. Because this placement was widely accepted and not questioned by participants during user testing or by the extended project team, the original positioning was maintained.

Following discussions with the involved CV and ML engineer regarding camera constraints, it was decided to keep the prototype's original size. Instead of making the physical hardware larger, it was suggested to modify the camera system. Using a wider-angle lens than the initial 0.6 fisheye will expand the field of view and could potentially recognize all items on the placement area. See the design iterations in Figure 58.

Figure 58: Shows the form development inside the software Autodesk Alias.



4.4.1 Appearance model

To present the final concept and provide an interactive model, an appearance model was created. The model was created in the Computer-Aided Design (CAD) software Alias, where all files were exported to enable laser cutting and 3D printing. The model was built in the scale 1:1 to provide test users a the real product size. The structure was partly made of laser-cut MDF wood and partly 3D-printed PLA filament. For long, straight surfaces, it was easier to use laser cutting, while 3D printing was more appropriate for curved and more complicated shapes. To create an illuminated interaction light, a semi-transparent PET-filament had to be used to let the light through. With materials provided by ITAB, it was possible to create the illusion of acrylic plastic in the model. To cast light on the 3D printed pieces, some LED components had to be installed, and cables were connected to battery holders to power up the light source.

Additionally, the appearance model had to stand on a base to make it easy to transport and possible to exhibit. Therefore, a table constructed from plywood was created to place the model on top of. Moreover, the model was designed to have cables running inside the table, as that would be part of the final concept solution. The solution was also meant to include a screen and a scanner. The decision was made to utilize a small screen with an integrated scanner—a product that ITAB already have in their product portfolio: Scanmaster Media. The idea was to place it on a fixture on top of the table, alongside a 3D-printed EFT and a receipt terminal. See the workshop process in Figure 59 and Figure 60.

Figure 59: Illustrates the individual parts for the appearance model.

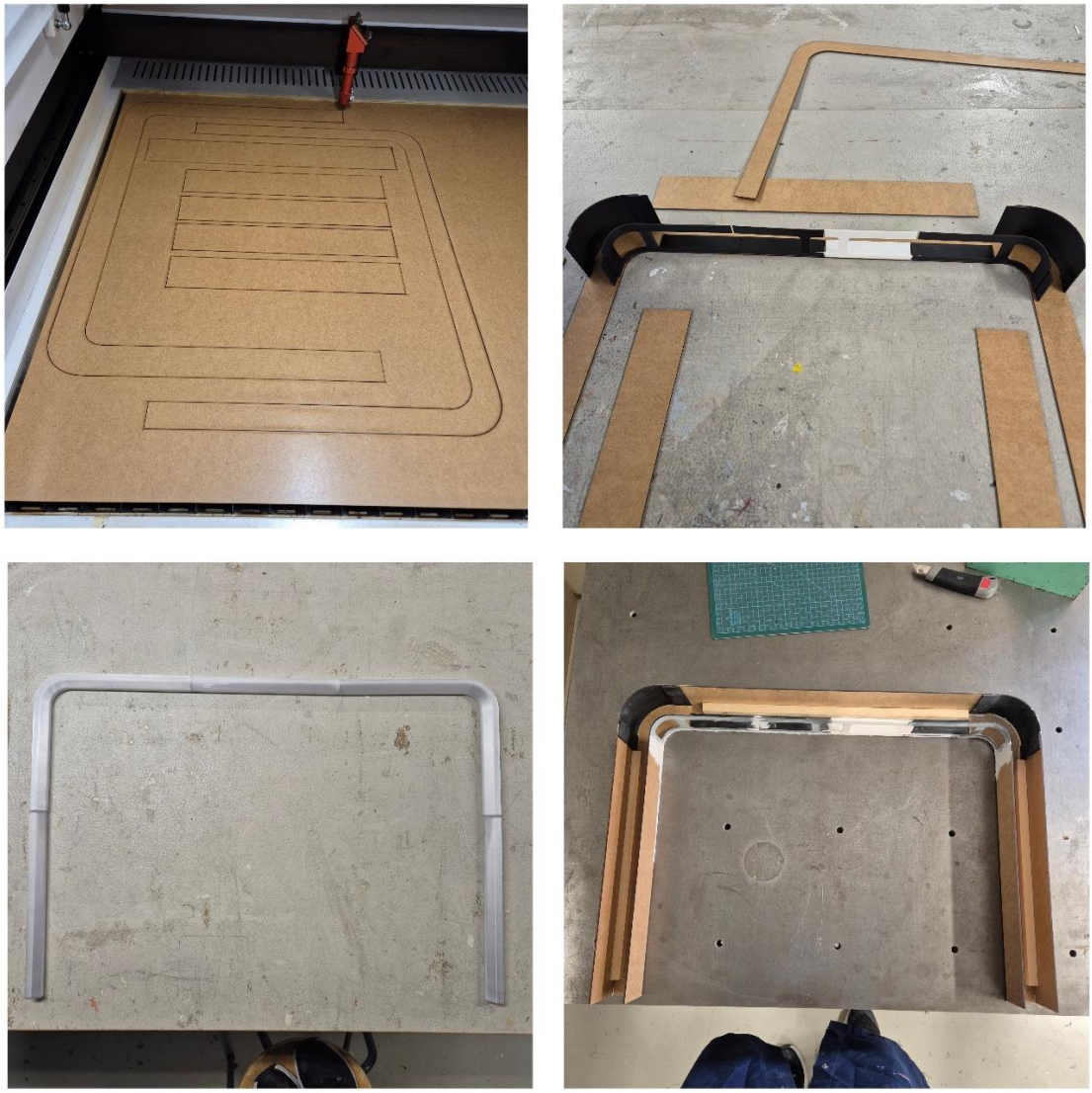


Figure 60: Illustrates the assembly process of the appearance model.



4.5 Solution delivery

This section presents the last part of the double diamond methodology and the final concept delivery for this thesis.

4.5.1 Appearance model

The final concept features a composition of the refined arch design with round edges, minimalistic shapes and clean surfaces to reflect the ART series. The dimensions of the arch were 80 x 50 mm to fit three cameras of 30 x 30 x 40 mm and a rotatable camera holder to provide the desired camera angle. The arch also has space for a cloudy white chamfer detail of 20 x 20 mm, plus an LED strip and a connection bar. Also containing light sources for illuminating the placement area.

The product composition features the item placement area together with a screen and integrated bar-code scanner from one of ITAB's current products, Scan master Media. On the side is the intended location for an EFT and a receipt printer. See an illustration of the final concept for this thesis in Figure 61.

Figure 61: Shows a hero shot of the assembled appearance model



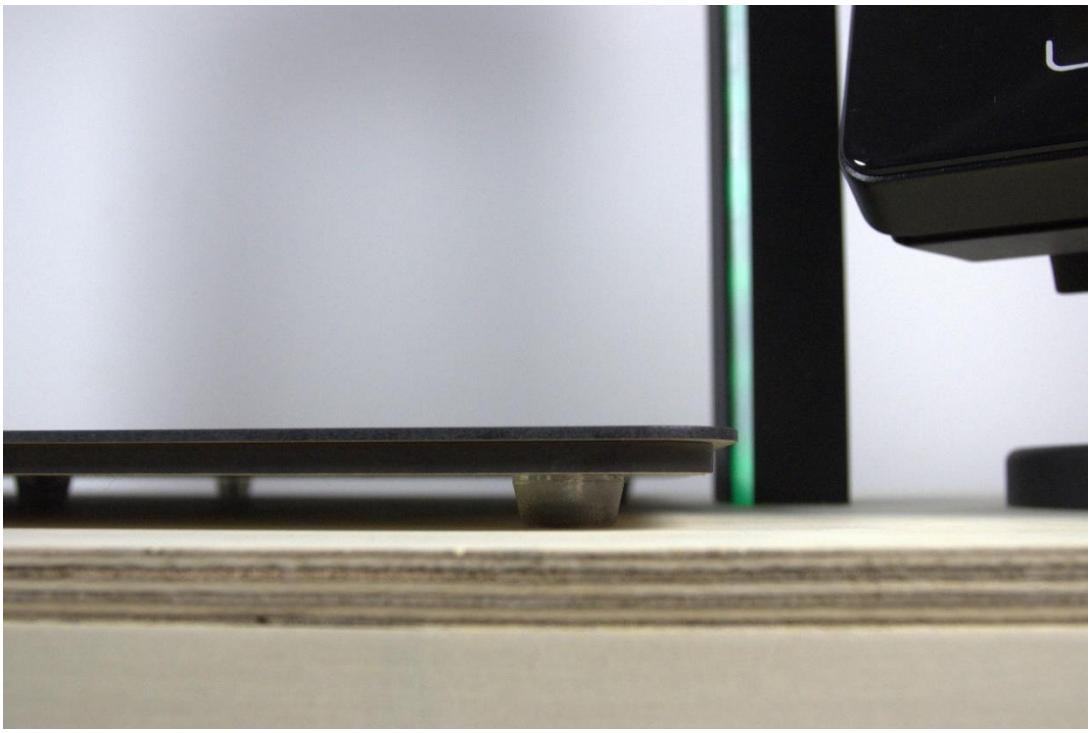
The cameras were designed to be placed behind a glass to prevent dirt from coming onto the cameras and to make them easy to clean. The camera angle were suggested to be 40 degrees as that seemed to be an angle that can capture a tall popcorn, even on the edges. See a closer illustration of the camera and illumination bar in Figure 62.

Figure 62 : Shows a closeup of the camera and illumination bar



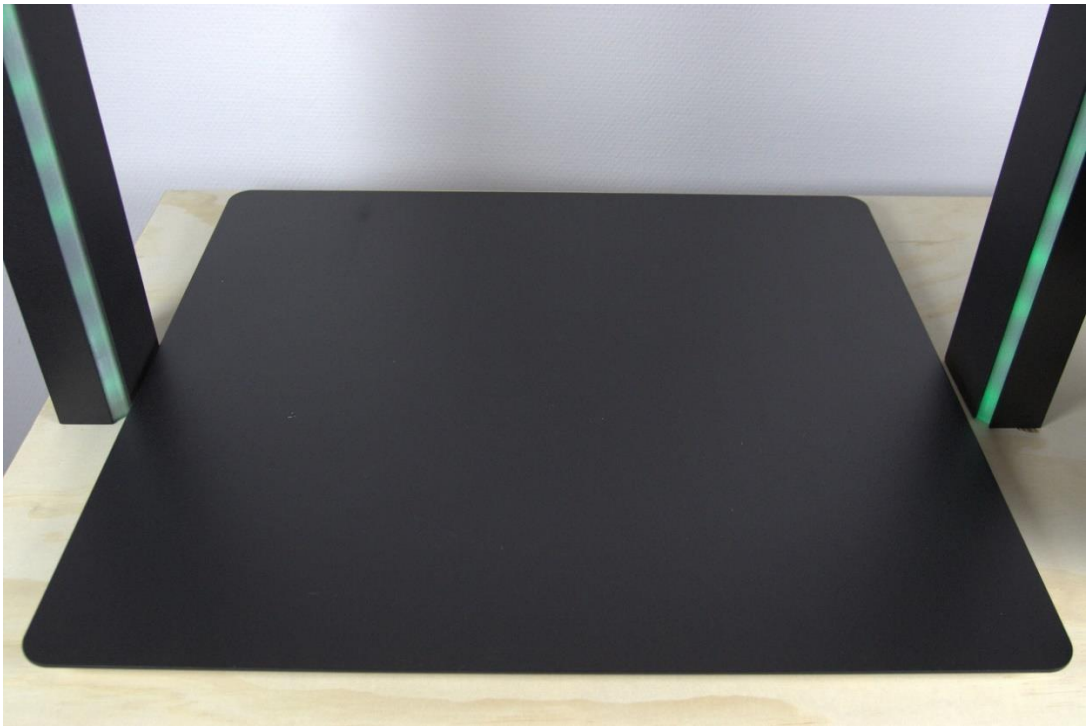
To prevent the placement are from slipping or deflecting when applying pressure, the design was supplemented with rubber dampers. Due to its thin construction, that was a necessary detail. See an illustration in Figure 63.

Figure 63: Shows a closeup of a rubber detail



The dimensions of the placement area was kept at 580 x 520 mm due to its practical size, enabling maximum 10 items to fit. It was designed with smooth rounds and a thin top surface to provide a feeling that it hovers above the table. See an illustration in Figure 64.

Figure 64: Shows a closeup of the placement area.



The interaction light was designed and built to enable color switches. Green color could indicating availability, yellow could indicate an ongoing customer process, and red could signal that assistance is needed. See Figure 65-67.

Figure 65: Shows the green interaction light.



Figure 66: Shows the yellow interaction light.

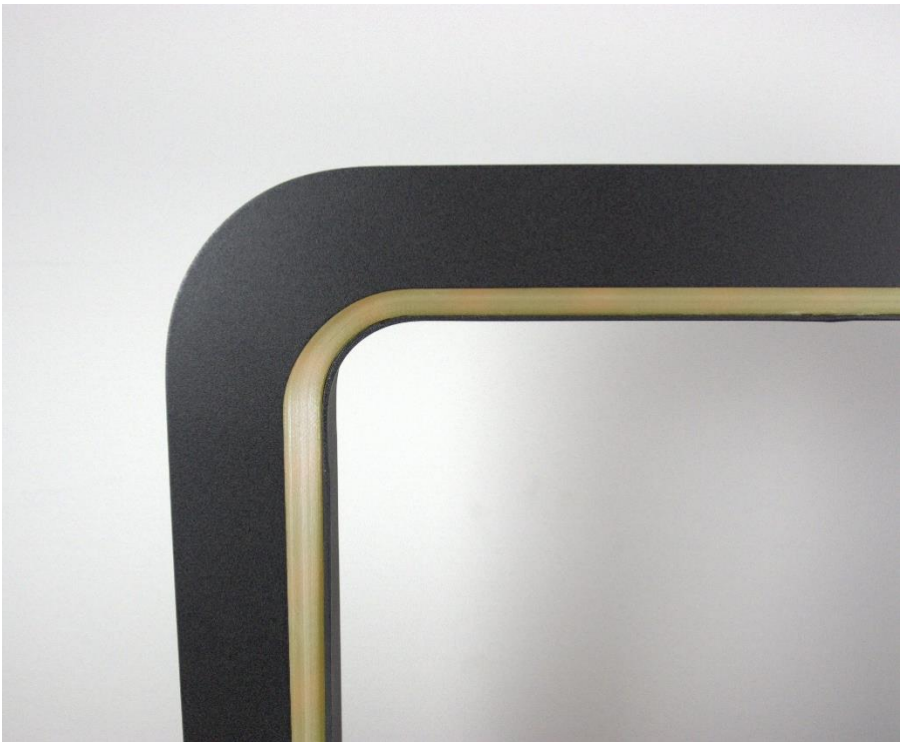


Figure 67: Shows the red interaction light.

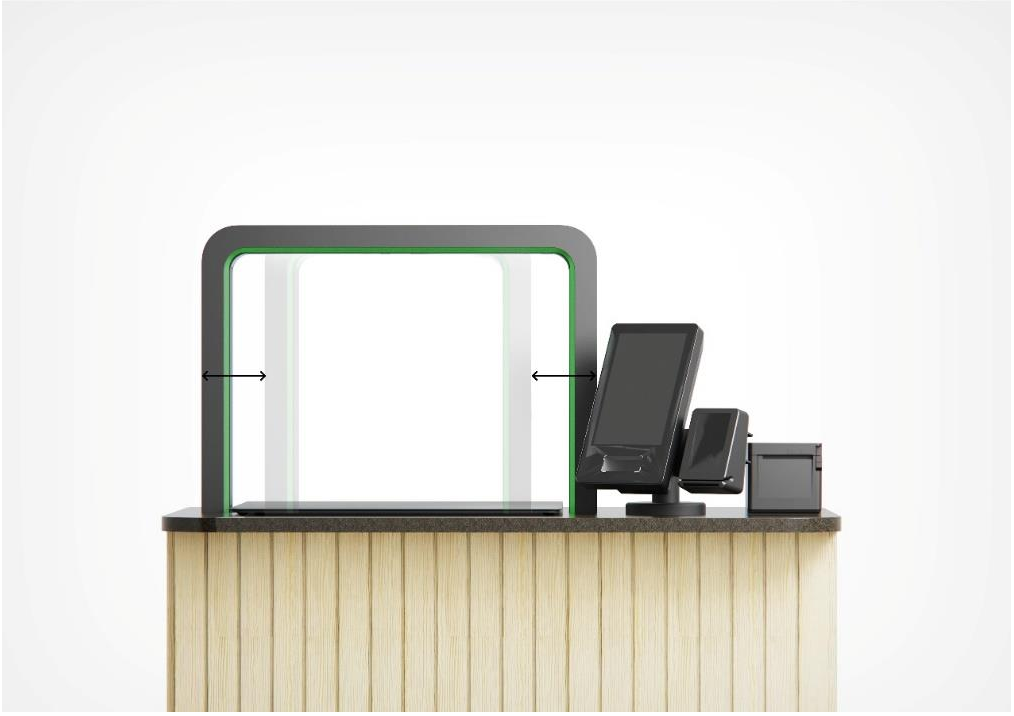


4.5.1 Renders

The solution is aimed at providing flexibility to be sold in multiple sizes, depending on the retailers' space requirements for a checkout. With the arch design, the pillars could potentially be adjusted down to one-third of their original size without changing too much of the design or compromising the technology. According to discussions with the ML and CV engineers, having two different sizes of the arch and slightly different camera placements would probably not require the vision system to be relearned to recognize items in new orientations. See the modularity of the arch in Figure 68.

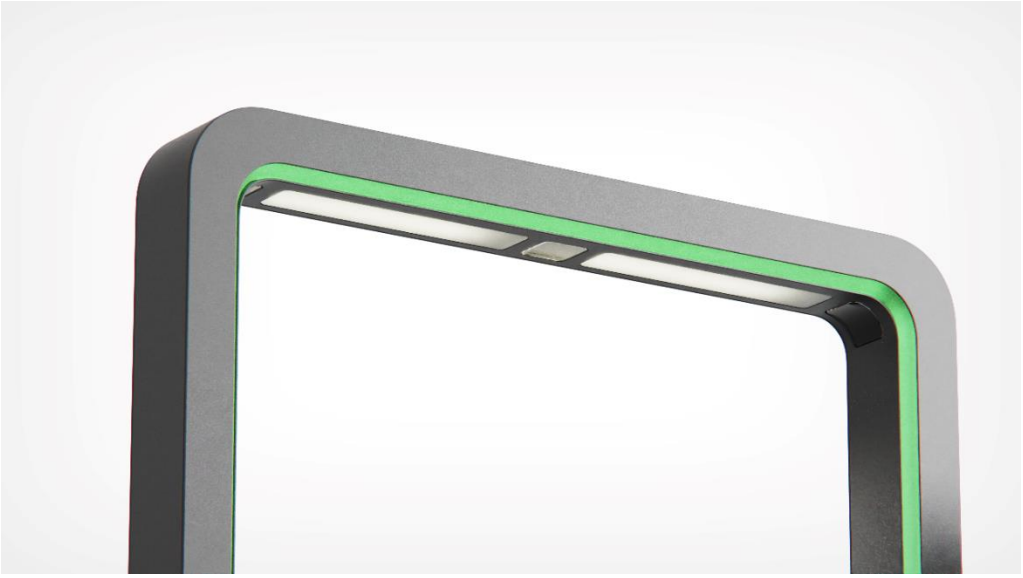
Moreover, the final design aimed at providing a modular POS hardware integration which is why the design focused on the arch and placement area. As highlighted by the interviews, these retailers seemed to have a preference of choosing their own POS system, which this design aims to enable. A suggestion is to have a screen integrated with a bar-code scanner for a backup solution if the vision system would fail. The retailer can chose if they want to have a larger screen and in a different orientation, another EFT and another printer, but this concept is presented like this based on the research findings. See a closeup of the illumination bar in Figure 69, and a hero shot of the final concept in Figure 70.

Figure 68: Design variation illustrating modular scaling of the arch and placement surface



Comment. Rendered in Keyshot and edited in Photoshop.

Figure 69: Close-up of the illumination bar



Comment. Rendered in Keyshot and edited in Photoshop.

Figure 70: Shows a hero shot of the final concept

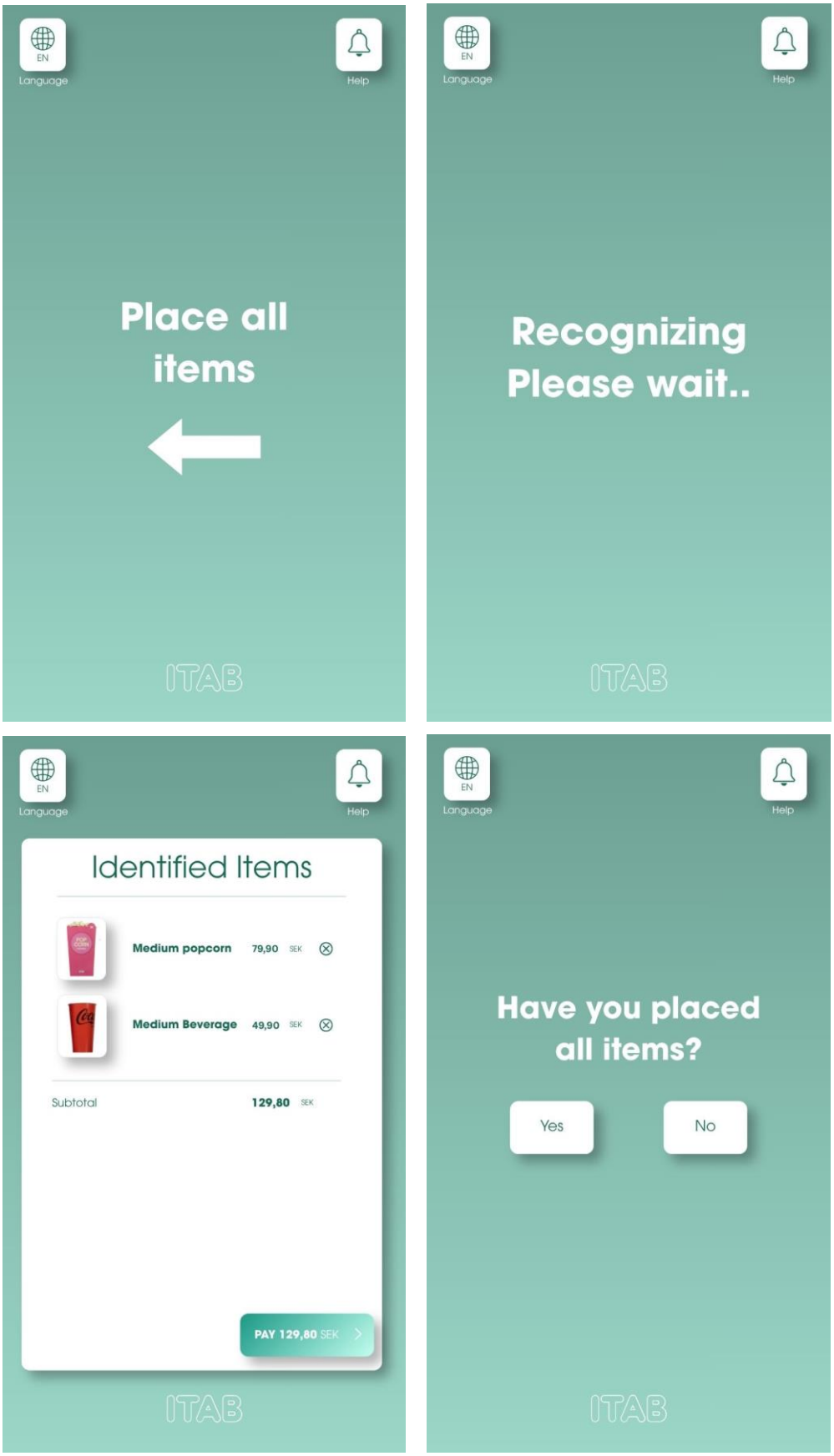


Comment. Rendered in Keyshot and edited with Weavy.ai

4.5.2 User interface

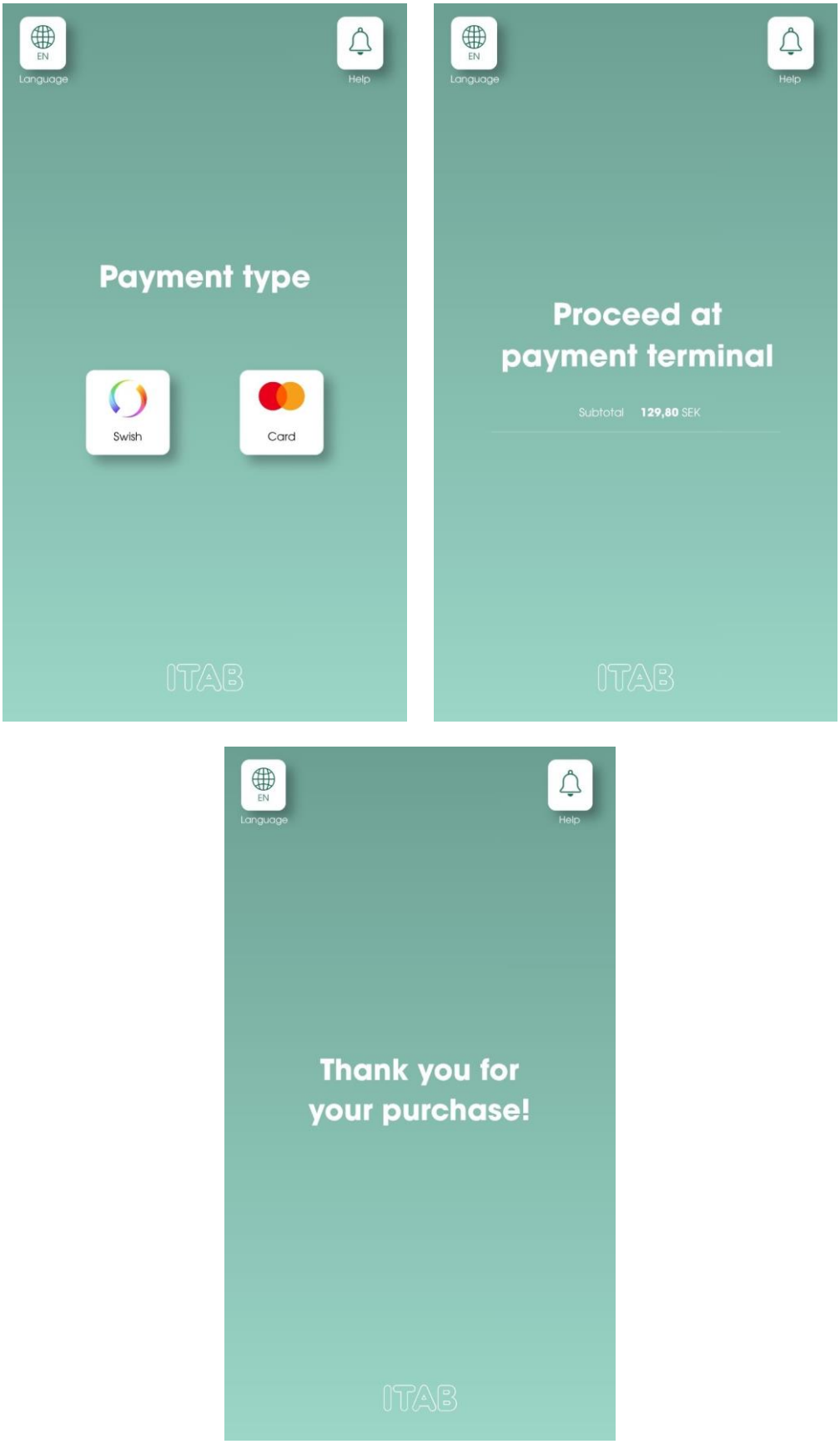
Based on the takeaways from the concept testing, the next step was to create a simplified UI for the interactive appearance model. The wireframes were based on the scenarios from the user evaluations: one for purchasing a beverage and popcorn, and another where the vision system fails to recognize an item. The starting screen was updated to instruct users to "place all items." This change was made because the concept evaluation revealed that some users tried to place items one by one when the screen simply read "place items." Additionally, the "Finish" button was changed to "Pay," as this more logically describes the next step in the purchase process. To provide visual clarity, symbols of the identified items were added to the shopping cart to accompany the text. Furthermore, the interface was updated so that all items could be removed, rather than only allowing the removal of manually scanned items. For the interactive prototype, animations were added between screens to make the interface feel more dynamic and engaging. Sound effects were also introduced, such as when the shopping cart appears or when manual scanning is required, to signal that user interaction is needed. Finally, a unique sound was added upon completion to indicate a successful transaction. See the UI wireframes for the interactive prototype in. Figure 71-74.

Figure 71: First segment of the UI sequence for user Scenario 1.



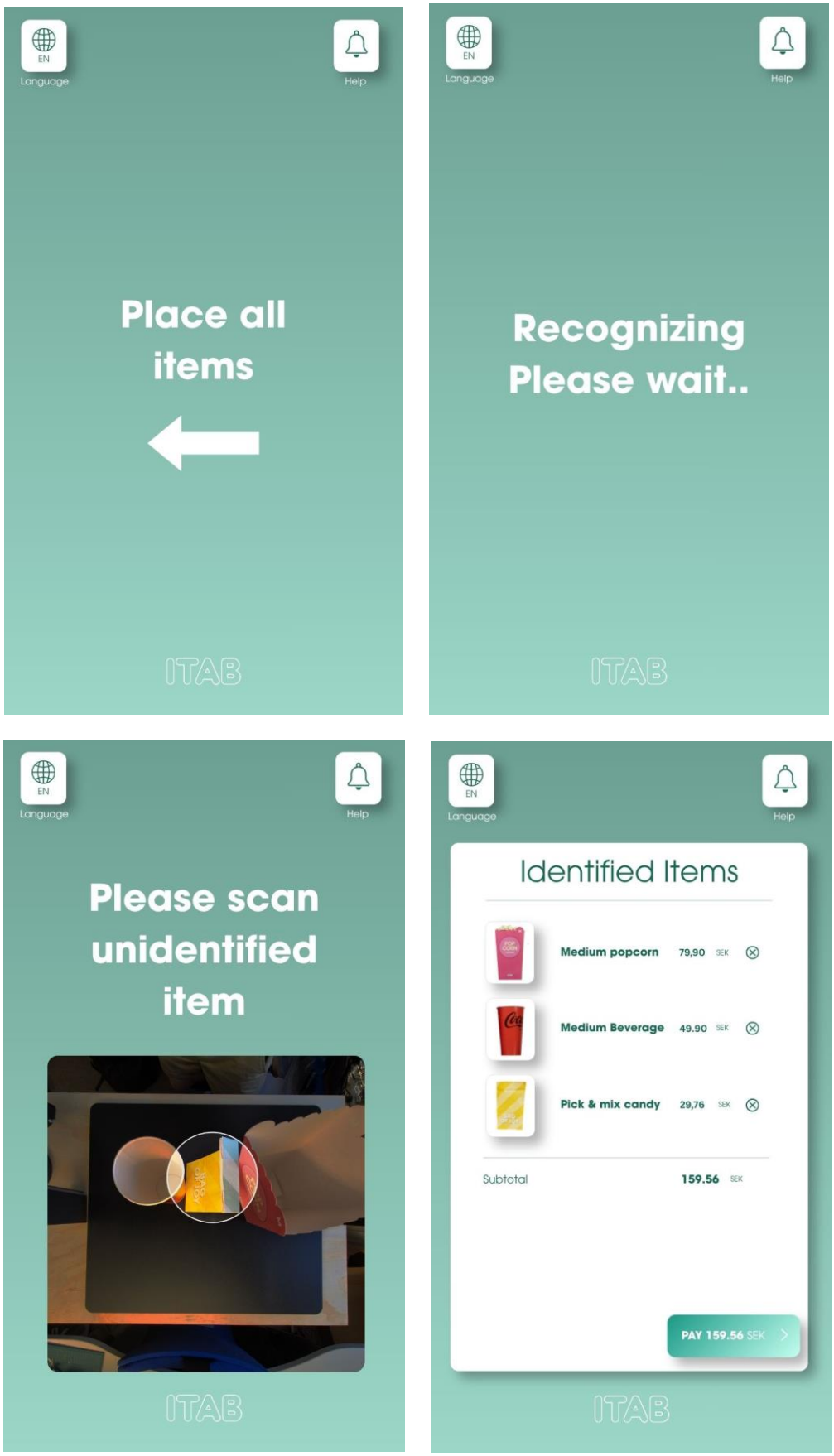
Comment. Created in Canva

Figure 72: Second segment of the UI sequence for user Scenario 1



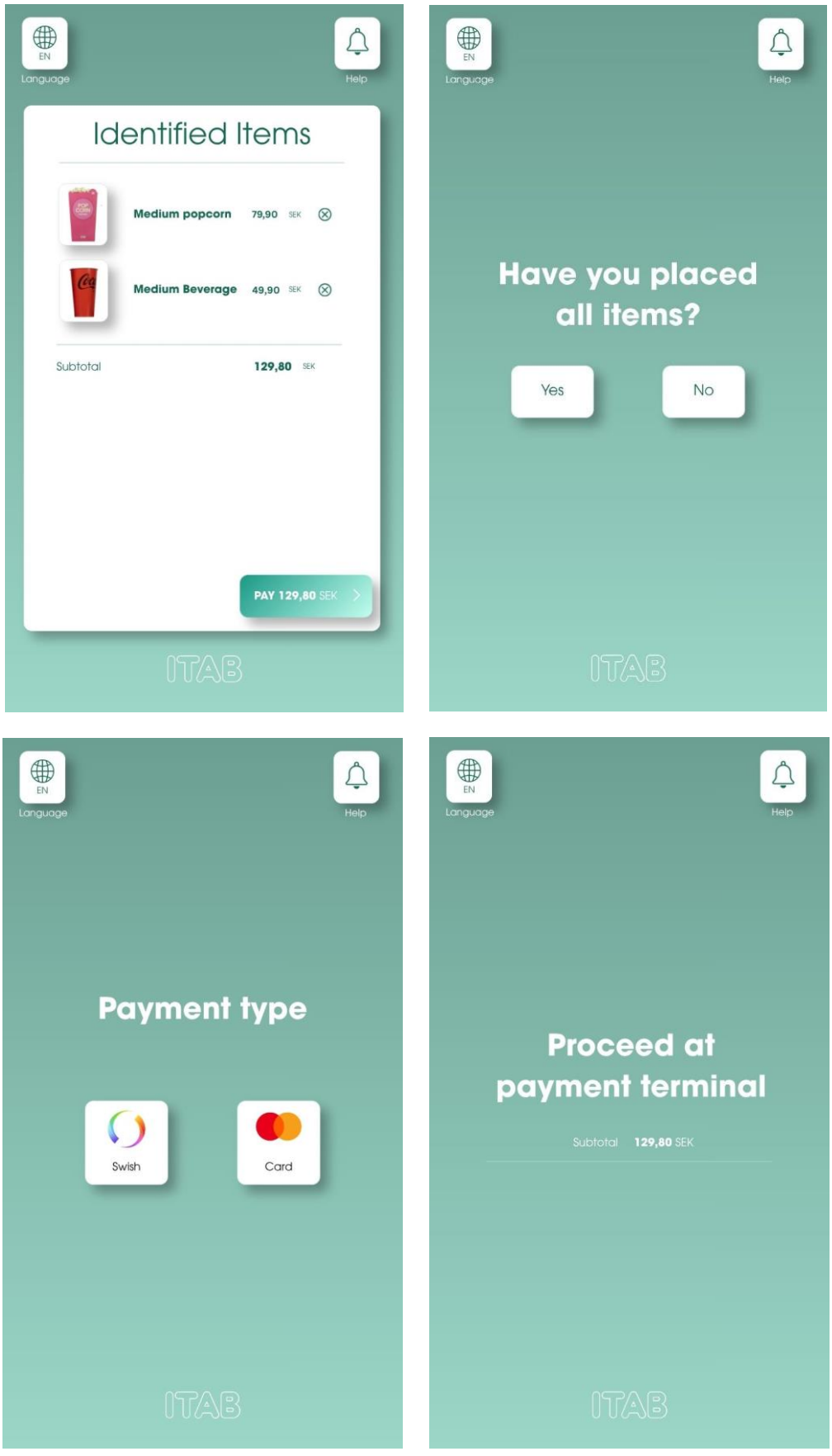
Comment. Created in Canva

Figure 73: First segment of the UI sequence for user Scenario 2



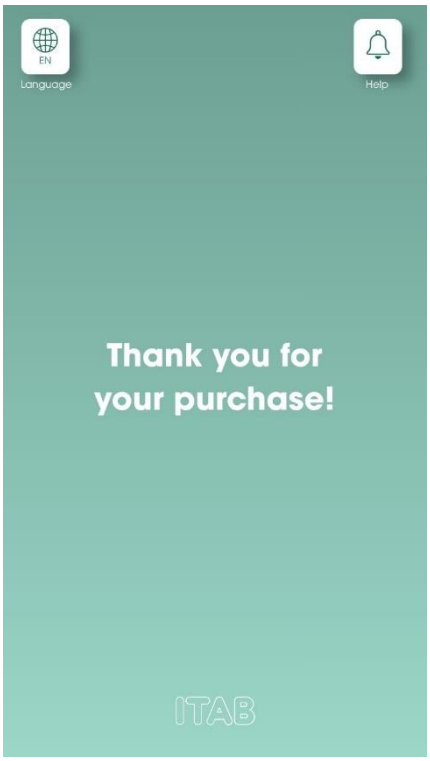
Comment. Created in Canva

Figure 74: Second segment of the UI sequence for user Scenario 2



Comment. Created in Canva

Figure 75: last segment of the UI sequence for user Scenario 2



Comment. Created in Canva.

4.6 User scenarios

To demonstrate how the final product concept can be utilized effectively, two scenarios were created covering both consumer and retailer perspectives. The first scenario was generated based on interactions with parents who had previously experienced the stress of getting children to the cinema on time. Furthermore, it was supplemented by insights from staff regarding stressed customers, as well as observations of checkout queues. The second scenario was developed from conversations with cinema staff, direct observations, and discussions with the extended project team. The latter parts of these scenarios leverage data from Mashgin studies highlighting checkout speed. Additionally, interactions with staff at gas stations currently operating Mashgin smart checkouts served as a foundation for understanding how the system can relieve stress and free up time for other tasks during staff shortages. Finally, consumer reactions within the scenarios are based on real customer interactions observed during field visits to cinemas, gas stations, and convenience stores.

Scenario 1

A family arrives late at the cinema after struggling to find parking, forcing them to skip buying cheaper snacks beforehand. With only five minutes until the movie starts, the children are excited but anxious for snacks, while the father is highly stressed by the long queues at the regular counters. To save time, the mother spots a faster lane leading to a Smart Checkout. The father doesn't understand how to use it, but the mother asks the service staff for help. They instruct them to place the items on the scanning area, and the system instantly generates the total price. As the kids place the items, the price appears on the screen. The father is surprised once the payment succeeds, and he is relieved to head to their seats just one minute before showtime. See an illustration of the scenario in Figure 76.

Figure 76: Shows the first scenario about a family going to the cinema



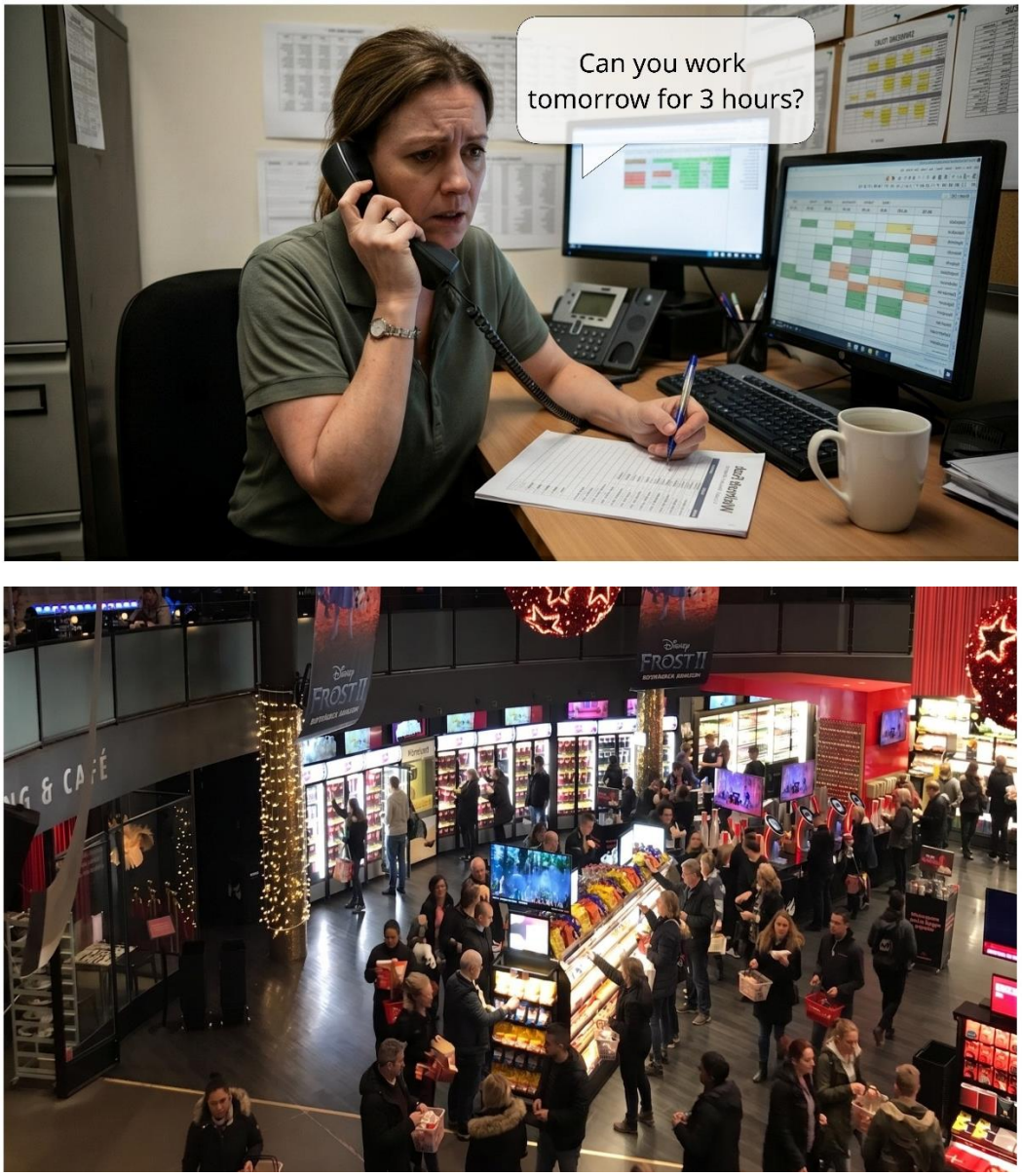
Comment: the images were generated with google Gemeni and edited in Photoshop

Scenario 2

In the cinema industry, it was mentioned difficult to predict which movies will sell more tickets than others. During a regular weekend, the schedule is already set for the employees, but at short notice, extra staff is needed due to sold-out theatres. The employer gets stressed and has to call around, but it is difficult to find staff who can step in, so they must manage with the staffing they have.

When the evening comes, the staff gets stressed at first because there are many customers queuing for buying snacks. However, shortly the queues reduce since some customers head to the smart checkouts instead. The customers gets an opportunity to skip the queues, and the employees get relieved. One staff can assist some customers at multiple smart checkouts simultaneously, instead of handling customers at one conventional checkout. The customer flow moves smoothly for the night. Se an illustration of the scenario in Figure 77-77.

Figure 77: Shows the first part of the second scenario



Comment: The images were generated using Google Gemini

Figure 78: Shows the second part of the second scenario



Comment: The images were generated using Google Gemini, utilizing reference images from photos and imported renders from KeyShot. Lastly, they were edited in Photoshop.

5. Discussion & conclusion

This section covers discussions related to the three research questions, the company's value proposition, conclusions, and suggested future research areas.

5.1 Overview

The objective of this thesis was to address the main challenges in customer journeys focusing on “less than 10 Item purchases”. Furthermore, the project aimed to identify a specific store type, suitable for developing a smart checkout MVP on Swedish market. To answer this, the first and second research question aimed at addressing these matters.

Moreover, the goal was to clearly describe the roles, jobs and interaction points of customers, store employees when it comes to purchase, deployment, support and training of a new product concept. Clearly state how the concept solves a problem that is valuable to the retailer and beneficial for the consumer. Ultimately, the goal was to create a prototype that incorporates solutions to the main challenges and show how the concept provides value-based outcomes.

This chapter aims to discuss and conclude whether the thesis has provided answers to the research questions and what could be addressed in future research. To address these questions the chapter is divided in sub chapters for each research question.

5.1.1 First research question

The first research question was formulated as follows:

RQ 1: What are the main challenges of a self-checkout customer journey that supports a less than 10 items purchase?

Checkout queues

Many different themes emerged in the thematical analysis that can be connected to challenges in customer journeys. The one thing that many main themes seem to have in common regarding challenges of customer journeys was the consequence; checkout queues. According to Finalpos, (2025), The benefits with SCO are usually shorter waiting times in line to the manned checkout and increased convenience sales. During some observations in time-sensitive environments with many customers, it seemed to cause longer checkout queues and increased SCO usage. In environments where the SCOs were more accessible and visible to customers, they were more utilized than when the SCO was more hidden or placed in cluttered spaces. As mentioned by some customers, they did not notice the SCO stations in the visited convenience store, however, they would prefer using it if they knew about it. This suggests that in some cases, checkout queues are caused by the poor visibility of the SCOs themselves.

Item accessibility and SCO placement

During some customer interactions, it was also noted that certain users specifically chose manned checkouts due to limited item accessibility. Products such as hot food served by staff, unprescribed medication, and tobacco are often restricted to manned counters. When multiple customers attempt to purchase these items simultaneously in a time-sensitive environment, long queues have been observed. When users who prefer self-service fail to see the SCO kiosks, they end up in the same line as customers who want a manned counter, creating a clear collision in customer needs.

Perceived Complexity and Labor Shortages

Based on customer interactions, a segment of shoppers prefers to queue for the manned counter due to perceived friction at the SCO. This friction includes age-verification checks, general checkout complexity, and a belief that it is more time-consuming than using a manned counter.

Because these findings align directly with a study by Softpay.io (2025) which identified perceived SCO complexity as a primary reason for underutilization, it could be a main challenge in user journeys resulting in queue friction. Another challenge highlighted by customer interactions, which also aligns with the Softpay.io report, is that some customers prefer interacting with service personnel rather than using an SCO. Together with that it was mentioned from the convenience store CIO that mentioned that high staff turnover is evident in their business, it might not always be possible to man up the manned counters during staff shortages. However, it could be possible to provide service in other ways for the customers who prefer a fast SCO experience in order to make way for the customers who really want to go to the manned counter faster.

The barriers identified regarding perceived SCO complexity can be deeply analyzed through the theoretical framework of UX Design. According to Kuniavsky et al., (2012) a good UX is defined through functionality, efficiency, and desirability. During customer interactions, some shoppers expressed a fear of making mistakes, while others mentioned that they do not perceive the SCO as the fastest alternative. Furthermore, a convenience store staff member noted that he does not believe customers are prone to reading instructions these days. When a customer experiences these friction points, both the efficiency and the functionality of the system collapse from the user's perspective. This directly harms the desirability of the self-checkout terminal, causing customers to default to manned counters even during observed high-traffic peaks. Consequently, the proposed design utilizes short, clear on-screen instructions, visual cart icons, expanded deletion capabilities, and dynamic feedback with animations and audio to potentially create a more intuitive and engaging experience.

5.1.2 Second research question

The second research question was formulated as follows:

RQ 2: What store type could be suitable for developing an MVP smart checkout?

MVP Suitability and Modular Scaling

Based on main themes, store assortments, time sensitive environments and high service availability, cinema stores appeared as a potentially suitable store type. This does not mean the solution is strictly restricted to cinema environments. Because the design is modular in terms of size, POS hardware, and attachable to multiple tables, it has the potential to scale beyond cinema retail. However, expanding into other store types presents challenges based on their assortment and service models, such as over-the-counter sales versus item accessibility at shelves. In other store types, new and more complex use cases would need to be managed to provide a viable solution than in the potential MVP environment.

Restricted Items and Food Handling

For instance, handling hot food would have to be solved in a convenience store environment. In bars, alcohol sales would require the system to handle both age verification and checking how sober the customer is. This raises important questions: How does a store handle situations where items are not directly accessible to customers? Do they still prioritize keeping restricted items behind the service desk, or do they prioritize customer convenience? That being mentioned, smart checkouts are already deployed in settings like gas stations and sports arenas, demonstrating that a solution does not need to solve every edge case to be viable. It is worth noting, however, that the international markets where Mashgin cases have taken place have different alcohol laws than Sweden, implying additional regulatory challenges when deploying a similar solution locally.

5.1.3 Third research question

The third research question was formulated as follows:

RQ 3: How can a vision-based SCO be designed to support user problems and retailer needs?

Based on the research and development conducted throughout this project, several design directions were explored to address the three perspectives stated in the ITAB methodology; Technology, User and Market. The final concept demonstrates that a vision-based SCO can be designed by understanding self-service product architecture, technical constraints, and brand identity.

Product architecture and user perception

During the initial ideation phase, sketching and mockups were used to explore how the positioning of components impacts user understanding. While a central screen layout was observed common for traditional SCOs, physical mockup testing showed that this configuration is impractical for a cinema store context. Taller items, such as a large popcorn box or a large drink cup, obstructed the screen and prevented a good interaction with the UI. To solve this problem, a detached layout was chosen where the screen and payment devices are positioned on the right side of the item placement area. According to the concept testing, this placement felt most intuitive for the participants and clearly separated the scanning process from the payment process.

Balancing technical constraints and environment

As highlighted during the interview with the ML and CV engineer, vision-based systems require consistent conditions to achieve high product recognition accuracy. While some competitor checkouts do not utilize an integrated light source, field visits to cinema stores revealed that the observed lobbies feature low, diffused ambient lighting to maintain a moody atmosphere. To handle this environmental challenge, the arch design (Concept 4) was developed with an integrated downward-facing illumination bar to ensure sufficient light reaches the cameras.

To avoid a bulky design, the arch uses three cameras to capture the cinema items: two angled at 40 degrees and one facing vertically downward. Discussions with the ML and CV engineer confirmed that using three cameras might provide a relevant balance between high recognition accuracy and cost-effectiveness for an MVP.

Meeting retailer and brand demands

To align with ITAB's ART-series design language, the physical appearance of the arch was designed with clean surfaces, soft radiuses, and hard materials to remain robust yet unobtrusive in the retail environment. Retailer demands were also addressed through hardware modularity. Interviews with store management indicated a strong preference for maintaining their own POS systems and components. By focusing the core design on the arch and the placement area, the concept allows retailers to choose their preferred screen size and EFT terminal, facilitating easier maintenance and deployment.

According to product development literature, a modular product architecture isolates distinct functions into separate physical components, allowing for greater system flexibility and easier integration into existing environments (Ulrich & Eppinger, 2012). In this project, the physical architecture was consciously designed as a modular system by isolating the retail technology within the arch and maintaining an unintegrated screen, payment terminal, and receipt printer. This architectural decision was partly to enhance the user experience because they know the devices signaling that it is a self-service station. In observed fully integrated SCO kiosks, it was often less clear where to scan items and receive receipts. To address this, the developed arch features an integrated interaction light designed to function as a status indicator that provides visual communication for both the user and store employees. Furthermore, while concept

evaluations showed that the isolated screen caused users to focus on it more than the placement area, the UI was intentionally designed to focus strictly on guiding the transaction flow through action-oriented prompts and visual icons.

5.1.4 ITAB's value proposition

In this section, the final product concept is evaluated against its alignment with ITAB's value propositions.

The desired consumer brand experience

The final design can elevate the consumer brand experience by eliminating the traditional friction of manual barcode scanning for most cases, allowing for a "grab-and-go" interaction. Furthermore, the minimalist and robust form language of the concept, which features clean surfaces and soft radiuses, adheres closely to the ART-series design principles. This aesthetic restraint ensures that the hardware remains visually unobtrusive, successfully allowing the unique identity of the retailer's brand to speak.

Increased conversion & sales

Based on that a smart checkout system might processes transactions significantly faster than conventional counters, a larger volume of customers can be handled simultaneously. An increased throughput speed could prevent potential customers from abandoning their purchases due to long queue lines, thereby boosting conversion rates and concession sales for the retailer. By increasing the speed of the checkout process, as demonstrated by case studies of existing smart checkouts, the concept could enhance overall customer satisfaction and increase the likelihood of repeat store visits. Based on the concept evaluation, the implementation of an intuitive layout combined with clear color-switching interaction lights effectively signals the system's purpose to users. By making the terminal easily recognizable as a self-service station, this visual clarity can encourage a higher adoption rate among customers.

Improved operational efficiency & services

The potential deployment of the smart checkout concept could introduce a highly scalable service model that might optimize daily store operations and mitigate the challenges of high staff turnover. By balancing the transaction workload with an smart checkout, a single staff member could monitor and assist four smart checkouts at the same time. The shift of work roles can drastically reduce the labor cost required to man multiple conventional checkout lanes during peak hours. While some store personnel can still stand at conventional checkouts, others can focus dynamically on other services such as shelf stocking, cleaning, and direct customer interactions.

Reduced operational cost

Integrating a modular smart checkout as a supplement could allow retailers to maintain service availability while lowering overall operational costs. By focusing the core checkout technology and the placement area, the concept might be integratable with preferred POS infrastructure. This hardware modularity eliminates the need for full-system replacements and simplifies long-term maintenance routines, as individual external components can be updated independently. While the system introduces upfront operational costs related to the initial training of new items, a financial balance could be evaluated in future research.

5.1.5 Future research

Many themes from the thematic analysis could not be addressed due to time constraints or complexity beyond industrial design, leaving several opportunities for future research. Many things have been discussed with the extended project team, but highlighting the most important takeaways is the main focus of this chapter.

Developing a UI prototype that covers more use cases than the ones covered in this study would be necessary to evaluate how customers might interact in different situations. Moreover, it would be necessary to research how to design a UI to teach the CV system new items, and also research how that could be implemented. Who would teach the system these items? As it was initially a report goal to examine how items can be learned for a potentially new smart checkout concept, limited contact with the right experts made it difficult to do a proper ideation of how that could be done. However, it was noted during field observations at gas stations that use smart checkouts that they do not train the system at the gas station itself; it gets done from somewhere else. Therefore, it was discussed with the extended project team that it would be promising if that could be done remotely. That way, all stores utilizing the same system and assortment can use the same dataset, instead of individual stores having to train the system separately.

As fraud detection was frequently mentioned during field visits, and that it has been mentioned as a major problem for self-service in retail business, it would be valuable to consider in future research. One possibility could be to utilize the cameras used for product recognition to understand fraudulent behavior, however, that needs to be further investigated. It could also be possible to implement some of ITAB's current anti-theft and fraud detection systems to address some problems. Moreover, that might impact the proposed design. How the smart checkout can provide sales data and interact with other technologies inside a store to be more beneficial for the retailer would be interesting to research in the future.

Mentioned during the interview with the CIO of the retail management company, it was noted that there are a few reasons why they have chosen not to implement smart checkouts in the past. One reason was their physical size. Another was that they found difficulties in how competitor systems handled different campaigns. The campaigns given as examples involved overlapping promotional prices. For instance, if there is a campaign to buy one cinnamon roll and one coffee for a cheaper price, and another campaign for a coffee and a hot dog for another cheap price, how would the system recognize which campaign to associate the coffee with to give the customer the best deal? From the software side, it would be an important factor to research deeper to find out if and how a vision system could handle that logic.

As the upselling problem was never handled due to lack of time, future industrial design research could focus on creating more value layers and ideating modular design alternatives, for instance, by exploring smart upselling concepts.

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7. Annexes

Annex A – Interview guide with the sales manager over a large cinema company

Annex B – Interview summary 1

Annex A - Interview guide with the sales manager over a large cinema company

1. Introduction of topic, project and highlighting cooperation with ITAB

2. Asking for consent to use the interview material in a report and mentioning the participant's right to adjust information after summary has been written of content.

3. What is your working title?

4. What are you currently working on?

5. What solutions are you looking for to increase sales and conversion in your stores today?

6. Are there any self-checkouts in the cinema stores today (why / why not?)

7. What are the main reasons why you have chosen not to use self-checkouts?

8. Have you evaluated whether to use Vision-based product recognition? (like Mashgin's)

9. Do you think there is potential to increase sales using vision based self-service stations in your store?

10. Why do you think Swedish cinema industry has not chosen to implement any smart checkout?

11. Why do you think Circle K has chosen to use smart checkouts?

12. What are the most common purchases at your cinema stores?

13. What number of customers do you have at a large cinema during weekends and or high season?

Annex B – Interview summary 1

Date: 2026/27/02
Time: 13:00-13:30
Location: Teams meeting
Respondent: Sales manager over a large swedish cinema company
Structure: Semi structured

An interview with a sales manager at a major Swedish cinema company confirmed that their cinema stores do not currently utilize SCO systems. The manager noted that this absence reflects a strategic decision to deprioritize self-service in the past. The current state of Swedish cinema retail might appear to be rooted in traditional organizational perspectives, specifically regarding the role of IT management and the deployment of technology outside the theatres themselves.

However, the interview also revealed that both the represented company and its parent organization are currently evaluating modernization strategies for their POS system. A reason for it was mentioned is that current technical constraints have begun to limit desired operational development. A key component of this modernization involves the exploration of potential integration of self-service technologies to facilitate a more seamless transaction process. Specifically, vision-based product recognition was subjectively expressed to be a technology that could further boost sales, speed up service, and improve the customer experience. The motivation was that self-service generally increase the average transaction value, and if the vision technology works smoother than traditional self-service, it could be an interesting alternative.

In discussing the cinema's value proposition, the interviewee highlighted the staff as a core element of the consumer experience. Moreover, this informed the view that automating transactions is not an alternative to replace the human element, but it is rather interesting to redirect staff focus on more service interactions. The manager clarified that unstaffed locations with fully automated checkout solutions would be unattractive for future cinema stores, as maintaining high service levels remains paramount. Instead, the focus could be utilizing self-service technology to free up personnel, thereby enhancing the overall customer experience through more attentive service.

The interview also touched upon the significant shift in queuing dynamics at manned counters following the COVID-19 pandemic. Previously, customers could reserve tickets online and collect them in-person up to fifteen minutes before showtime. However, the transition to a purchase-only model has effectively eliminated many queues.

Consequently, staff requirements at the traditional counters have been drastically reduced, as the remaining customer traffic is almost exclusively dedicated to purchasing concessions such as popcorn and snacks. While streamlining the checkout process may not have been the primary driver behind this policy change, it represents a substantial step toward operational efficiency. This reduction in transaction friction at the main counter may, in turn, influence the organization’s perception of whether self-checkout systems are a necessary or compelling investment.

In further discussions addressing POS integration, the interviewee highlighted that accelerated implementation timelines is attractive. Moreover, it was mentioned that swedish POS solutions, for instance, stand as particularly interesting. While a seamless deployment is vital and cost-effectiveness remains a significant consideration, the overarching priority is the stability of daily operations. Beyond the initial rollout, the organization must prioritize operational resilience, specifically concerning how technical disruptions are addressed and what formal contingency protocols are established to mitigate risks when problems arise.